# Probability

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## Contents

**1 Probability Models and Axioms** **6**
1.1 Sample spaces: what can happen? . . . 6
1.2 Events and probability axioms . . . 7
1.3 A finite probability law: two die rolls . . . 7
1.4 The discrete uniform law . . . 8
1.5 The continuous uniform law . . . 9
1.6 A countably infinite law and countable additivity . . . 9

**2 Conditioning and Bayes' Rule** **11**
2.1 Start with a probability model . . . 11
2.2 Conditional probability: restrict and renormalize . . . 11
2.3 Building a model from conditional probabilities . . . 13
2.4 The multiplication rule . . . 14
2.5 Total probability: combine disjoint cases . . . 14
2.6 Bayes' rule: reverse the conditioning . . . 15

**3 Independence** **17**
3.1 Review: probabilities along a coin-toss tree . . . 17
3.2 Independence of two events . . . 18
3.3 Conditioning can change independence . . . 18
3.4 Independence of a collection of events . . . 20
3.5 Pairwise independence is not enough . . . 21
3.6 The king's sibling: specify the experiment . . . 21

**4 Counting** **23**
4.1 The discrete uniform law . . . 23
4.2 The basic counting principle . . . 23
4.3 From ordered choices to combinations . . . 24
4.4 Binomial probabilities . . . 25
4.5 Conditioning on the number of heads . . . 26
4.6 Partitions: dealing a deck to four players . . . 27

**5 Discrete Random Variables, PMFs, and Expectations** **28**
5.1 Random variables: numbers assigned to outcomes . . . 28
5.2 Probability mass functions . . . 28
5.3 Computing a PMF by grouping outcomes . . . 29
5.4 The binomial PMF . . . 30
5.5 Expectation: a probability-weighted average . . . 31
5.6 Expectations of functions and affine expressions . . . 31
5.7 Variance: squared distance from the mean . . . 32

**6 Discrete Random Variable Examples and Joint PMFs** **34**
6.1 Review: means, deviations, and standard deviation . . . 34

6.2 Random speed and average travel time . . . 34
6.3 Conditional PMFs and conditional expectations . . . 35
6.4 The geometric PMF and memorylessness . . . 36
6.5 Total expectation and the geometric mean . . . 37
6.6 Joint PMFs, marginals, and conditioning . . . 38

**7 Multiple Discrete Random Variables: Expectations, Conditioning, and Independence 40**
7.1 Joint PMFs and the multiplication rule . . . 40
7.2 Independence and the effect of conditioning . . . 40
7.3 Expectations of functions, sums, and products . . . 42
7.4 Variances of sums . . . 43
7.5 Binomial mean and variance using indicators . . . 44
7.6 The hat problem: an expected number of matches . . . 45
7.7 The hat problem: variance and pair probabilities . . . 46

**8 Continuous Random Variables 48**
8.1 Probability as area under a density . . . 48
8.2 Means, variances, and the continuous uniform law . . . 49
8.3 Cumulative distribution functions . . . 49
8.4 Mixed distributions: area and point mass . . . 51
8.5 Gaussian densities and their moments . . . 52
8.6 Normal probabilities by standardization . . . 54
8.7 How the descriptions fit together . . . 55

**9 Multiple Continuous Random Variables 56**
9.1 Review: densities and expectations . . . 56
9.2 Joint densities, marginals, and independence . . . 56
9.3 Buffon's needle . . . 58
9.4 Conditional densities: normalize a slice . . . 59
9.5 Stick breaking: construct the joint density . . . 60
9.6 Stick breaking: marginal density and mean . . . 61

**10 Continuous Bayes' Rule; Derived Distributions 63**
10.1 Review and the discrete Bayes rule . . . 63
10.2 Continuous and mixed Bayes rules . . . 64
10.3 Derived distributions and discrete transformations . . . 65
10.4 Continuous transformations: start with the CDF . . . 66
10.5 Driving time: a decreasing transformation . . . 67
10.6 Affine transformations and normal variables . . . 68

**11 Derived Distributions; Convolution; Covariance and Correlation 70**
11.1 A ratio from the unit square . . . 70
11.2 A density formula for a monotone transformation . . . 71
11.3 Discrete convolution: the distribution of a sum . . . 72
11.4 Continuous convolution . . . 73
11.5 The joint density of independent normal variables . . . 74
11.6 The sum of independent normal variables . . . 74
11.7 Covariance and the variance of a sum . . . 75
11.8 Correlation: covariance on a common scale . . . 76

**12 Iterated Expectations; Sum of a Random Number of Random Variables 78**
12.1 Conditional expectation as a random variable . . . 78
12.2 The law of iterated expectations . . . 78
12.3 Conditional variance and total variance . . . 79
12.4 Variability within and between class sections . . . 80

12.5 A two-part density example . . . 81
12.6 The mean of a sum with a random number of terms . . . 82
12.7 The variance of a sum with a random number of terms . . . 83

**13 Bernoulli Process** **84**
13.1 The Bernoulli process . . . 84
13.2 Two views of a random process . . . 84
13.3 The number of successes in a fixed interval . . . 85
13.4 The waiting time to the first arrival . . . 85
13.5 Memorylessness and a string of lottery losses . . . 86
13.6 Interarrival times and the time of the kth success . . . 87
13.7 Splitting a Bernoulli process . . . 88
13.8 Merging independent Bernoulli processes . . . 89

**14 Poisson Process - I** **91**
14.1 From Bernoulli trials to continuous arrivals . . . 91
14.2 Defining a Poisson process . . . 91
14.3 The distribution of an arrival count . . . 92
14.4 Count moments and the email example . . . 93
14.5 The time of the kth arrival . . . 94
14.6 Exponential waiting and memorylessness . . . 95
14.7 The Bernoulli–Poisson relation . . . 95
14.8 Sums and merged Poisson processes . . . 96
14.9 Which process supplies the next arrival? . . . 97

**15 Poisson Process - II** **98**
15.1 The count and waiting-time laws . . . 98
15.2 Poisson fishing: the stopping rule and probabilities . . . 99
15.3 Poisson fishing: expected catches and time . . . 99
15.4 Merging processes and identifying the next arrival . . . 100
15.5 The lifetime of the last of three bulbs . . . 101
15.6 Splitting a Poisson process . . . 101
15.7 Random incidence in a Poisson process . . . 102
15.8 Renewal processes and the bus example . . . 103

**16 Markov Chains - I** **105**
16.1 The checkout counter as a state model . . . 105
16.2 Finite-state Markov chains . . . 106
16.3 Transition probabilities over several steps . . . 106
16.4 A two-state calculation . . . 107
16.5 The limit in the two-state example . . . 108
16.6 Transition probabilities can oscillate . . . 109
16.7 Limits can depend on the initial state . . . 109
16.8 Recurrent states, transient states, and classes . . . 110
16.9 Why transient states disappear . . . 111

**17 Markov Chains - II** **112**
17.1 Review and the warmup chain . . . 112
17.2 Periodic recurrent classes . . . 113
17.3 A periodic class with two return cycles . . . 113
17.4 Steady-state convergence and balance equations . . . 114
17.5 Solving the two-state balance equations . . . 115
17.6 Visit and transition frequencies . . . 116
17.7 Birth-death chains and flow across a cut . . . 116
17.8 Constant probabilities and the load factor . . . 117

**18 Markov Chains - III** **119**
18.1 Review of steady-state behavior . . . 119
18.2 Joint events in the two-state chain . . . 119
18.3 The phone company model . . . 120
18.4 Steady-state occupancy and blocking . . . 120
18.5 Absorption probabilities . . . 121
18.6 Expected time to absorption . . . 122
18.7 Mean first passage and recurrence times . . . 123

**19 Weak Law of Large Numbers** **125**
19.1 Chebyshev's inequality . . . 125
19.2 Deterministic limits and convergence in probability . . . 126
19.3 Convergence of the sample mean . . . 127
19.4 The pollster's problem . . . 128
19.5 Different scalings of the sum . . . 128
19.6 Standardization and the central limit theorem . . . 129

**20 Central Limit Theorem** **131**
20.1 Standardizing the sum . . . 131
20.2 What converges, and how quickly? . . . 132
20.3 The pollster's problem with the CLT . . . 133
20.4 Normal approximation to a binomial CDF . . . 133
20.5 Approximating a binomial probability mass . . . 134
20.6 Poisson versus normal approximation . . . 135

**21 Bayesian Statistical Inference - I** **137**
21.1 Models and inference questions . . . 137
21.2 Classical and Bayesian approaches . . . 137
21.3 Bayes' rule for inference . . . 138
21.4 A continuous unknown with discrete data . . . 139
21.5 Posterior distributions and point estimates . . . 140
21.6 Least mean squares without observations . . . 141
21.7 Least mean squares with observations . . . 141

**22 Bayesian Statistical Inference - II** **143**
22.1 LMS estimation and a uniform-noise model . . . 143
22.2 The posterior in the uniform-noise example . . . 144
22.3 Conditional mean squared error . . . 145
22.4 Properties of the LMS error . . . 146
22.5 Linear LMS with one observation . . . 146
22.6 Linear LMS with several observations . . . 148
22.7 Independent noisy measurements . . . 148
22.8 Choosing inputs and comparing methods . . . 149

**23 Classical Statistical Inference - I** **151**
23.1 A fixed unknown parameter . . . 151
23.2 Maximum likelihood estimation . . . 151
23.3 Bias, consistency, and mean squared error . . . 152
23.4 Estimating a mean . . . 153
23.5 What a confidence interval means . . . 154
23.6 A confidence interval for the mean . . . 155
23.7 When the standard deviation is unknown . . . 156
23.8 Estimating the variance from the sample . . . 157

**24 Classical Inference - II** **159**
24.1 Likelihood and confidence: a review . . . 159
24.2 Regression, residuals, and least squares . . . 160
24.3 Deriving the fitted intercept and slope . . . 161
24.4 The covariance interpretation . . . 161
24.5 Several predictors and transformed variables . . . 162
24.6 Regression reports and model concerns . . . 163
24.7 Binary tests, rejection regions, and errors . . . 164
24.8 From MAP to a likelihood-ratio test . . . 165
24.9 Choosing the rejection threshold . . . 165

**25 Classical Inference - III; Course Overview** **167**
25.1 Simple binary tests: a review . . . 167
25.2 Testing a normal mean . . . 167
25.3 Testing a normal variance . . . 168
25.4 Composite hypotheses: is a coin fair? . . . 169
25.5 Is a die fair? . . . 170
25.6 Checking a proposed distribution with bins . . . 171
25.7 Comparing the model and empirical CDFs . . . 172
25.8 Further statistical questions . . . 173

# Lecture 1: Probability Models and Axioms

A probability model pairs a sample space of possible outcomes with a probability law on events.

## 1.1 Sample spaces: what can happen?

Before assigning probabilities, decide what one complete outcome records. The detail must be enough to answer the questions of interest.

**Definition 1.1 [Sample space].** The sample space, denoted by $\Omega$, is the set of all possible outcomes of an experiment. Its outcomes must be mutually exclusive: only one outcome occurs in a run. They must also be collectively exhaustive: every possible result is included.

**Example:** *Two rolls of a tetrahedral die.* Each roll produces one of $1, 2, 3, 4$. Write $X$ for the first result and $Y$ for the second. One complete outcome is the ordered pair $(X, Y)$, so

$$\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4\}\}.$$

There are $4 \times 4 = 16$ outcomes. The order matters: $(1, 2)$ and $(2, 1)$ describe different results. We can display the pairs in a grid or describe the two rolls sequentially with a tree. Each full path through the tree ends at one of the same 16 outcomes.

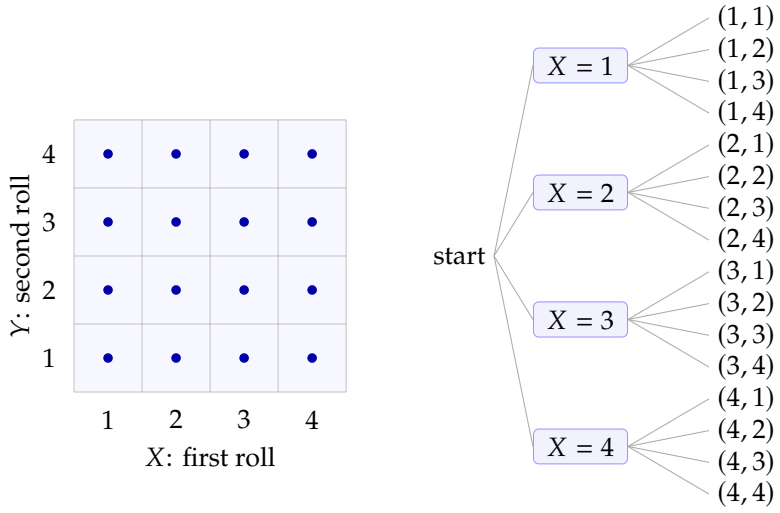



**Figure 1:** The grid and the tree describe the same sample space. A cell or a complete path records both rolls.

**Example:** *Two numbers in the unit interval.* If each result can be any real number from 0 to 1, the sample space is

$$\Omega = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le 1\} = [0, 1]^2.$$

It is the entire unit square, including its boundary. The die space is finite; this space is continuous. Both still need a probability law.

## 1.2 Events and probability axioms

We often ask whether a result satisfies a condition, rather than whether one particular outcome occurs.

**Definition 1.2 [Event and probability law].** An event $A$ is a subset of $\Omega$: it occurs when the outcome belongs to $A$. A probability law assigns a number $P(A)$ to each event in its domain, subject to the probability axioms below.

For a finite or countably infinite sample space, every subset can be an event. On a continuous space, the domain of the law must be chosen with care; we use ordinary geometric regions here.

**Definition 1.3 [Probability axioms].** For events in the domain of $P$:

1. Nonnegativity: $P(A) \geq 0$ for every event $A$.
2. Normalization: $P(\Omega) = 1$.
3. Additivity: for disjoint events $A$ and $B$, meaning $A \cap B = \varnothing$,

$$P(A \cup B) = P(A) + P(B).$$

For overlapping events, adding probabilities counts their intersection twice.

Distinct individual outcomes give disjoint singleton events. Repeatedly applying additivity therefore gives

$$P(\{s_1, \ldots, s_k\}) = P(\{s_1\}) + \cdots + P(\{s_k\}).$$

We may abbreviate $P(\{s\})$ to $P(s)$. The two-event axiom handles finite unions; an infinite union will require a stronger form of additivity.

## 1.3 A finite probability law: two die rolls

Return to the 16 ordered die outcomes and assign probability $1/16$ to each. The probabilities are nonnegative and sum to $16(1/16) = 1$. An event containing $k$ outcomes has probability $k/16$.

**Example:** *Either* $(1,1)$ *or* $(1,2)$. These are two distinct outcomes, so

$$P\big(\{(1,1),(1,2)\}\big) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}.$$

**Example:** *The first roll is* 1. The second roll can be $1, 2, 3$, or $4$. Thus

$$\{X = 1\} = \{(1,1),(1,2),(1,3),(1,4)\}, \qquad P(X = 1) = \frac{4}{16} = \frac{1}{4}.$$

The notation $\{X = 1\}$ names the subset of outcomes satisfying the condition.

**Example:** *The sum is odd.* One roll must be odd and the other even. There are two odd values and two even values, so there are $2 \times 2 = 4$ pairs of each order. Hence

$$P(X + Y \text{ is odd}) = \frac{4+4}{16} = \frac{1}{2}.$$

**Example:** *The smaller roll is* 2. Both rolls must be at least 2, and at least one must equal 2. The event is

$$\{\min(X, Y) = 2\} = \{(2,2),(2,3),(2,4),(3,2),(4,2)\}.$$

There are five outcomes, giving

$$P(\min(X, Y) = 2) = \frac{5}{16}.$$

Counting three pairs with $X = 2$ and three with $Y = 2$ would count $(2,2)$ twice, so it must be included only once.

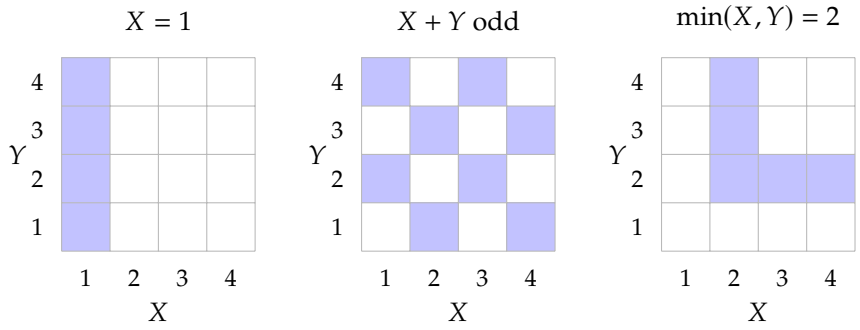



**Figure 2:** Highlighted cells show the last three die events. Each cell has probability $1/16$. In the last panel, the corner $(2,2)$ is counted once.

## 1.4 The discrete uniform law

When a finite sample space has $N$ equally likely outcomes, normalization forces each outcome to have probability $1/N$. This is the DISCRETE UNIFORM LAW. Additivity then gives

$$\boxed{P(A) = \frac{|A|}{|\Omega|}},$$

where $|A|$ is the number of outcomes in $A$ and $|\Omega| = N$. Computing a probability becomes a counting problem. Fair coins, fair dice, and well-shuffled decks are modeled this way. The formula relies on the outcomes being equally likely.

## 1.5 The continuous uniform law

Now give the unit square a CONTINUOUS UNIFORM LAW: the probability of a region is its area. The square's total area is 1, so the law is normalized. Equal-area regions have equal probabilities, and areas add for disjoint regions.

**Example:** *The two numbers sum to at most* $1/2$. The event is

$$A = \{(x, y) \in [0, 1]^2 : x + y \leq \tfrac{1}{2}\}.$$

It is the triangle bounded by the two coordinate axes and the line $x + y = 1/2$. Both perpendicular sides have length $1/2$, so

$$P(X + Y \leq \tfrac{1}{2}) = \text{area}(A) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

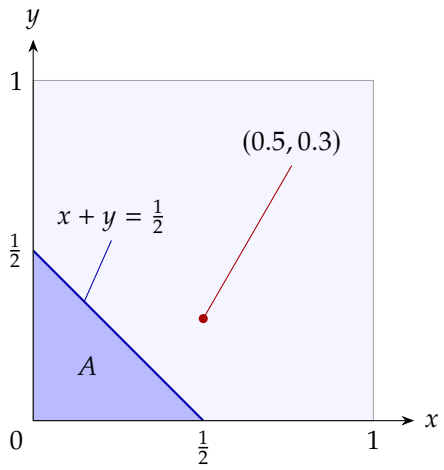



**Figure 3:** The shaded triangle has area $1/8$. The point $(0.5, 0.3)$ is a possible outcome, but its singleton has zero area.

**Example:** *The exact pair* $(0.5, 0.3)$. A single point has zero area, so

$$P\big((X, Y) = (0.5, 0.3)\big) = 0.$$

The point is still in the sample space. A probability-zero event need not be impossible: in this model every individual point has probability zero, while the whole square has probability one.

## 1.6 A countably infinite law and countable additivity

Consider the sample space $\Omega = \{1, 2, 3, \ldots\}$ with

$$P(n) = 2^{-n}, \qquad n = 1, 2, \ldots.$$

Each probability is positive, and the geometric series checks normalization:

$$\sum_{n=1}^{\infty} 2^{-n} = \frac{1/2}{1-1/2} = 1.$$

**Example:** *The outcome is even.* The event $\{2, 4, 6, \ldots\}$ contains infinitely many disjoint singleton events. Adding their probabilities gives

$$\begin{aligned} P(\{2,4,6,\ldots\}) &= \sum_{k=1}^{\infty} 2^{-2k} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots \\ &= \frac{1/4}{1-1/4} = \frac{1}{3}. \end{aligned}$$

This calculation uses an infinite sum, which finite additivity alone does not justify.

**Definition 1.4 [**Countable additivity**].** If $A_1, A_2, \ldots$ are pairwise disjoint events, the countable additivity axiom requires

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Countable additivity includes finite unions. It does not justify summing singleton probabilities over an uncountable space such as the unit square.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 1, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 2: Conditioning and Bayes' Rule

Conditioning restricts a probability model to an observed event and renormalizes it. Multiplication, total probability, and Bayes' rule follow from this update.

## 2.1 Start with a probability model

The sample space $\Omega$ lists mutually exclusive, collectively exhaustive outcomes at the detail needed for the question. Events are subsets of $\Omega$. Their probabilities are nonnegative, satisfy $P(\Omega) = 1$, and add over disjoint events. For a disjoint sequence $A_1, A_2, \ldots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

To solve a problem, specify the sample space, define its probability law, identify the event of interest, and then calculate. A calculation is only meaningful once these ingredients are clear.

## 2.2 Conditional probability: restrict and renormalize

Suppose we learn that $B$ occurred. Outcomes outside $B$ can be discarded. Among the outcomes in $B$, the event $A$ occurs exactly on $A \cap B$. We compare the probability of this intersection with the probability of the new universe $B$.

**Definition 2.1 [Conditional probability].** For events $A$ and $B$ with $P(B) > 0$, the conditional probability of $A$ given $B$ is

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}.$$

This definition does not assign a value when $P(B) = 0$.

Dividing by $P(B)$ normalizes the restricted law: $P(B \mid B) = 1$.

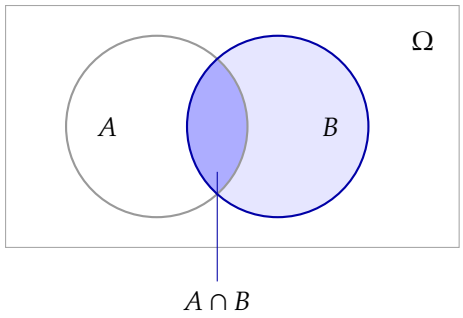



**Figure 4:** Conditioning keeps $B$ and compares $A \cap B$ with all of $B$. This is a schematic set diagram; region sizes do not represent probabilities.

**Example:** *The minimum of two die rolls is* 2. Let $X$ and $Y$ be the two results of a tetrahedral die, with each of the 16 ordered pairs assigned probability $1/16$. Suppose we know

$$B = \{\min(X, Y) = 2\} = \{(2,2), (2,3), (2,4), (3,2), (4,2)\}.$$

Then $P(B) = 5/16$. Each remaining pair has conditional probability $(1/16)/(5/16) = 1/5$.

Let $M = \max(X, Y)$. The event $M = 1$ has no outcome in common with $B$: if the minimum is 2, the maximum cannot be 1. Therefore

$$P(M = 1 \mid B) = \frac{0}{5/16} = 0.$$

For $M = 2$, the only compatible pair is $(2, 2)$, giving

$$P(M = 2 \mid B) = \frac{1/16}{5/16} = \frac{1}{5}.$$

The denominator counts the five outcomes allowed by the information, not all 16 outcomes in the original space.

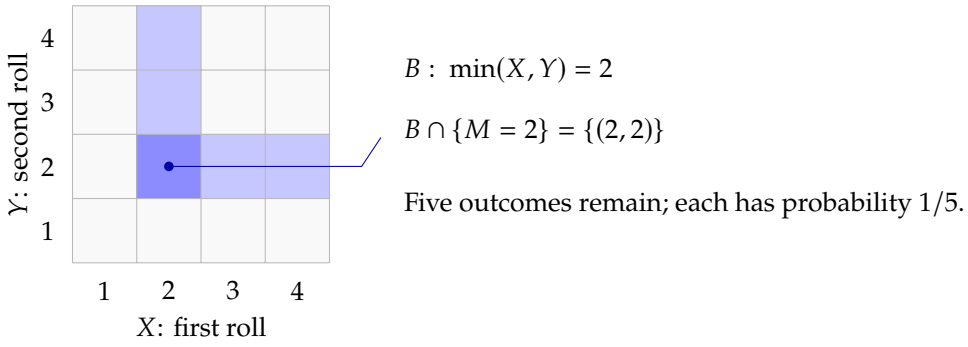



**Figure 5:** The five colored cells form $B$. The darker cell $(2, 2)$ is the only one where $M = 2$. Each colored cell has probability $1/5$ given $B$.

## 2.3 Building a model from conditional probabilities

A model can be specified in stages. First give the probability of one event, then give the probabilities of later events under each possible earlier result. Write $A^c$ for the COMPLEMENT of $A$, the event that $A$ does not occur.

**Example:** *An airplane and a radar reading.* Let $A$ mean that an airplane is flying above, and let $B$ mean that something registers on the radar screen. The model is

$$P(A) = 0.05, \qquad P(A^c) = 0.95,$$

with the following conditional probabilities:

$$\begin{aligned} P(B \mid A) &= 0.99 \qquad & P(B^c \mid A) &= 0.01, \\ P(B \mid A^c) &= 0.10 \qquad & P(B^c \mid A^c) &= 0.90. \end{aligned}$$

Thus a registration is likely when a plane is present, but registrations also occur when no plane is present. Within each condition, the two radar probabilities sum to one.

Rearranging the conditional-probability definition gives

$$P(A \cap B) = P(A)P(B \mid A).$$

Multiply the probability of the first stage by the conditional probability of the second stage to obtain the probability of a complete path.

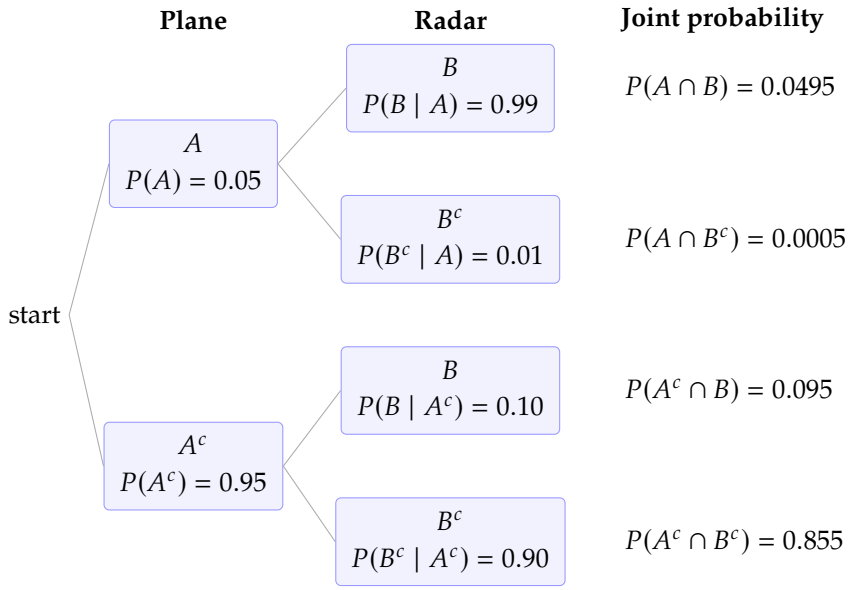



**Figure 6:** Each path specifies both plane presence and radar status. Multiply the two factors on a path to obtain the joint probability on the right.

For a plane and a registration,

$$P(A \cap B) = 0.05(0.99) = 0.0495.$$

A registration can come from either $A \cap B$ or $A^c \cap B$. These events are disjoint, so

$$\begin{aligned} P(B) &= P(A \cap B) + P(A^c \cap B) \\ &= 0.05(0.99) + 0.95(0.10) = 0.1445. \end{aligned}$$

Once a registration is observed, the probability of a plane is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.0495}{0.1445} = \frac{99}{289} \approx 0.3426.$$

The two directions answer different questions: $P(B \mid A) = 0.99$ describes the radar given a plane; $P(A \mid B) \approx 0.3426$ describes plane presence given a registration. The latter must account for how often planes are present and how often the radar registers without one.

## 2.4 The multiplication rule

For three events, a complete path specifies $A$, then $B$ given $A$, then $C$ given both earlier events. Each factor must use all the information accumulated so far.

**Proposition 2.2 [Multiplication rule].** *If $P(A) > 0$ and $P(A \cap B) > 0$, then*

$$P(A \cap B \cap C) = P(A)P(B \mid A)P(C \mid A \cap B).$$

*The positivity assumptions ensure that the conditional probabilities on the right are defined.*

*Proof.* Apply the conditional-probability definition twice: first condition on the whole event $A \cap B$, then express $P(A \cap B)$ by conditioning on $A$.

$$\begin{aligned} P(A \cap B \cap C) &= P(A \cap B)P(C \mid A \cap B) \\ &= P(A)P(B \mid A)P(C \mid A \cap B). \end{aligned}$$

□

## 2.5 Total probability: combine disjoint cases

Sometimes the probability of $B$ is easier to calculate after dividing the sample space into cases. Compute the contribution from each case, then add the contributions.

**Definition 2.3 [Partition].** Events $A_1, A_2, A_3$ form a partition of $\Omega$ if they are pairwise disjoint and cover the whole space:

$$A_i \cap A_j = \varnothing \quad (i \neq j), \qquad A_1 \cup A_2 \cup A_3 = \Omega.$$

Every outcome belongs to exactly one of the three cases.

**Theorem 2.4 [Total probability].** *For a partition $A_1, A_2, A_3$ with $P(A_i) > 0$ for each $i$,*

$$P(B) = P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2) + P(A_3)P(B \mid A_3).$$

*Proof.* Each outcome of $B$ belongs to exactly one part of the partition, so $B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B)$ is a disjoint union. Additivity gives

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B).$$

For each piece, use $P(A_i \cap B) = P(A_i)P(B \mid A_i)$. Substituting these three products gives the formula. □

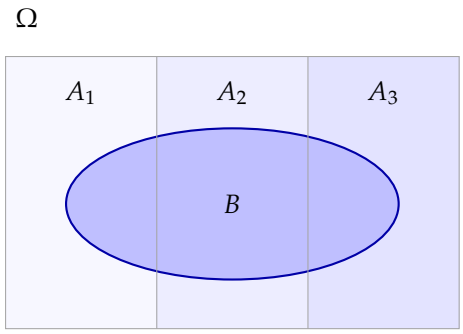



**Figure 7:** The partition cuts $B$ into three disjoint pieces. Their probabilities add to $P(B)$. The geometry is schematic.

Each term weights a conditional probability by the probability of its case. In the radar example, the two cases are $A$ and $A^c$, giving the calculation $P(B) = 0.05(0.99) + 0.95(0.10)$.

## 2.6 Bayes' rule: reverse the conditioning

Suppose we know how likely $B$ is under each case $A_i$, but observe $B$ and want to infer which case occurred. The prior probabilities $P(A_i)$ describe the cases before using this observation. The updated probabilities are $P(A_i \mid B)$.

**Theorem 2.5 [Bayes' rule].** *For the partition above, if $P(A_i) > 0$ for each $i$ and $P(B) > 0$, then*

$$\boxed{P(A_i \mid B) = \frac{P(A_i)P(B \mid A_i)}{\sum_{j=1}^{3} P(A_j)P(B \mid A_j)}}, \qquad i = 1, 2, 3.$$

*Proof.* The conditional-probability definition gives

$$P(A_i \mid B) = \frac{P(A_i \cap B)}{P(B)}.$$

The multiplication rule supplies the numerator, $P(A_i \cap B) = P(A_i)P(B \mid A_i)$. The total probability theorem supplies the denominator by adding the contributions from all three cases. Substituting both expressions yields Bayes' rule. □

In figure 7, the numerator is the probability of the piece $A_i \cap B$ and the denominator is the probability of all of $B$. Thus Bayes' rule compares one case's contribution with the total probability of the observation. In the radar model, it updates $P(A) = 0.05$ to $P(A \mid B) \approx 0.3426$ after a registration.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 2, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 3: Independence

Independent events do not change one another's probabilities. Independence depends on the probability law: conditioning can remove it or create it.

## 3.1 Review: probabilities along a coin-toss tree

For $P(B) > 0$, conditioning gives $P(A \mid B) = P(A \cap B)/P(B)$. The multiplication rule, total probability, and Bayes' rule then give

$$\begin{aligned} P(A \cap B) &= P(A)P(B \mid A) = P(B)P(A \mid B), \\ P(B) &= P(A)P(B \mid A) + P(A^c)P(B \mid A^c), \\ P(A \mid B) &= \frac{P(A)P(B \mid A)}{P(B)}. \end{aligned}$$

Each conditional factor requires a conditioning event of positive probability. Multiply along a path and add disjoint paths.

**Example:** *Three tosses of a biased coin.* At each stage, the probability of heads is $p$ and of tails is $1 - p$, regardless of earlier results. The sample space consists of the eight strings $HHH, HHT, HTH, HTT, THH, THT, TTH, TTT$.

The path $THT$ has probability

$$P(THT) = (1 - p)p(1 - p) = p(1 - p)^2.$$

Exactly one head occurs on the three disjoint paths $HTT, THT, TTH$. Each has the same probability, so

$$P(\text{exactly one head}) = 3p(1 - p)^2.$$

Given exactly one head, the first toss is heads only on $HTT$. For $0 < p < 1$,

$$P(\text{first toss is } H \mid \text{exactly one head}) = \frac{p(1 - p)^2}{3p(1 - p)^2} = \frac{1}{3}.$$

The bias cancels because all three remaining paths have equal probability. At $p = 0$ or $p = 1$, the conditioning event has probability zero, so this conditional probability is undefined.

## 3.2 Independence of two events

If learning that $A$ occurred gives no information about $B$, then $P(B \mid A) = P(B)$, provided $P(A) > 0$. Substitution in the multiplication rule gives a definition that also works for probability-zero events.

**Definition 3.1 [Independence].** Events $A$ and $B$ are INDEPENDENT if

$$\boxed{P(A \cap B) = P(A)P(B)}.$$

The definition is symmetric in $A$ and $B$. When $P(A) > 0$, it is equivalent to $P(B \mid A) = P(B)$; when $P(B) > 0$, it is equivalent to $P(A \mid B) = P(A)$.

*Proof.* If $P(A) > 0$, divide $P(A \cap B) = P(A)P(B)$ by $P(A)$ to obtain $P(B \mid A) = P(B)$. Conversely, multiplying that conditional equality by $P(A)$ gives the defining product. Interchanging $A$ and $B$ proves the other equivalence. □

If $P(A) = 0$, then $A \cap B \subseteq A$ also has probability zero. Thus $P(A \cap B) = 0 = P(A)P(B)$ for every $B$, even though conditioning on $A$ is undefined. Independence is a numerical property of the probability law; the shape of a set diagram alone does not establish it.

## 3.3 Conditioning can change independence

After learning that $C$ occurred, use the conditional probability law $P(\cdot \mid C)$ throughout. Independence in this new universe is tested using conditional probabilities.

**Definition 3.2 [Conditional independence].** For $P(C) > 0$, events $A$ and $B$ are CONDITIONALLY INDEPENDENT given $C$ if

$$P(A \cap B \mid C) = P(A \mid C)P(B \mid C).$$

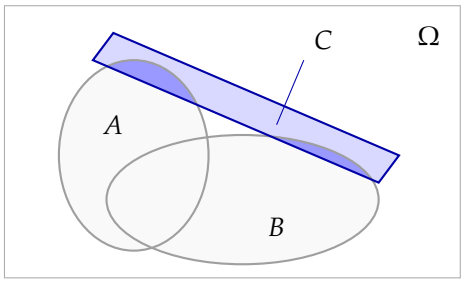



**Figure 8:** Within $C$, the pieces $A \cap C$ and $B \cap C$ are disjoint. The diagram is schematic; areas do not specify probabilities.

Suppose $A$ and $B$ are independent before conditioning, but $C$ meets each of them while avoiding $A \cap B$. If both $P(A \mid C)$ and $P(B \mid C)$ are positive, then

$$P(A \cap B \mid C) = 0 \quad \text{while} \quad P(A \mid C)P(B \mid C) > 0.$$

They are therefore dependent given $C$. Unconditional independence does not guarantee conditional independence.

**Example:** *A coin chosen once, then tossed repeatedly.* Choose Coin A or Coin B with probability 1/2 each. Coin A has head probability 0.9; Coin B has head probability 0.1. Toss the chosen coin repeatedly, with independent tosses conditional on its identity. Let $G$ be the event that Coin A was chosen, and let $H_i$ mean that toss $i$ is heads. Then

$$P(H_i \mid G) = 0.9, \qquad P(H_i \mid G^c) = 0.1.$$

Knowing the identity fixes the head probability. Within either conditional model, earlier tosses give no further information about later tosses.

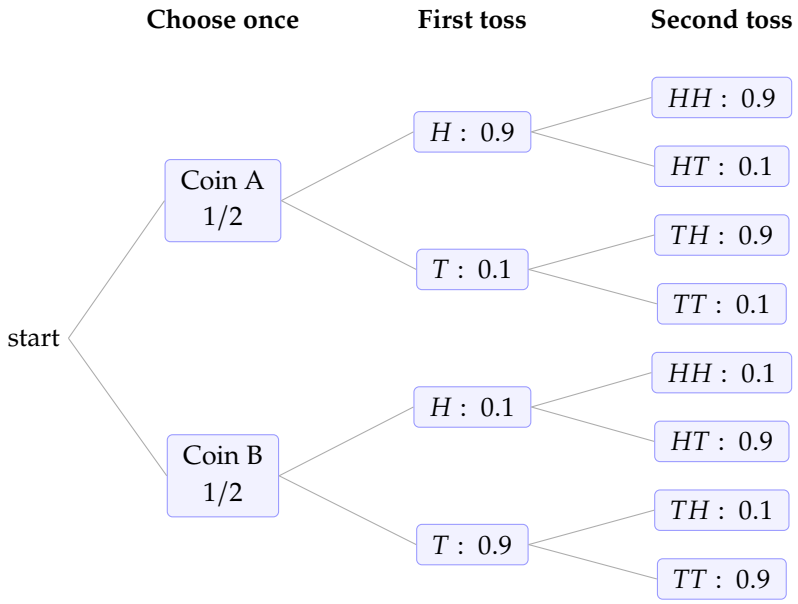



**Figure 9:** The first choice selects one coin for all tosses. The boxes give the probability of each new step conditional on the preceding history. The two tosses shown are independent given the chosen coin.

Without knowing the coin, total probability gives

$$P(H_{11}) = \tfrac{1}{2}(0.9) + \tfrac{1}{2}(0.1) = \tfrac{1}{2}.$$

Now suppose the first ten tosses were heads. Write $D = H_1 \cap \cdots \cap H_{10}$. Conditional independence within each coin model gives $P(D \mid G) = 0.9^{10}$ and $P(D \mid G^c) = 0.1^{10}$. The observation is far more

likely with Coin A. Bayes' rule gives

$$P(G \mid D) = \frac{0.9^{10}}{0.9^{10} + 0.1^{10}}.$$

For the next toss, divide the probability of eleven heads by that of ten:

$$\begin{aligned} P(H_{11} \mid D) &= \frac{\frac{1}{2}(0.9^{11}) + \frac{1}{2}(0.1^{11})}{\frac{1}{2}(0.9^{10}) + \frac{1}{2}(0.1^{10})} \\ &= \frac{0.9^{11} + 0.1^{11}}{0.9^{10} + 0.1^{10}} \approx 0.9. \end{aligned}$$

This differs from $P(H_{11}) = 1/2$. Earlier results reveal information about the shared coin and change the probability of later results. Thus conditional independence does not guarantee unconditional independence.

### 3.4 Independence of a collection of events

For a collection, the idea is stronger than checking two events at a time: information about any group of events must give no information about the remaining events.

**Definition 3.3 [Independence of a collection].** Events $A_1, \ldots, A_n$ are mutually independent if, for every nonempty subset $S \subseteq \{1, \ldots, n\}$,

$$\boxed{P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i)}.$$

Here $\prod$ means multiplication over the indicated indices. The condition must hold for every subcollection, not just all $n$ events together.

For three events, this requires the three pairwise products and also

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3).$$

For a larger independent collection, even combinations of events and their complements involving disjoint groups of indices give no information about one another. For example, when $n \geq 6$ and $P(A_5 \cap A_6^c) > 0$,

$$P\big(A_1 \cap (A_2^c \cup A_3) \mid A_5 \cap A_6^c\big) = P\big(A_1 \cap (A_2^c \cup A_3)\big).$$

The condition concerns indices $5, 6$, while the event of interest concerns indices $1, 2, 3$.

### 3.5 Pairwise independence is not enough

Pairwise independence checks each pair; mutual independence requires every subcollection.

**Example:** *Two fair coin tosses and whether they match.* Consider two independent fair tosses, with the four equally likely outcomes $HH, HT, TH, TT$. Define

$$A = \{HH, HT\}, \qquad B = \{HH, TH\}, \qquad C = \{HH, TT\}.$$

Thus $A$ means the first toss is heads, $B$ the second is heads, and $C$ that both results are the same.

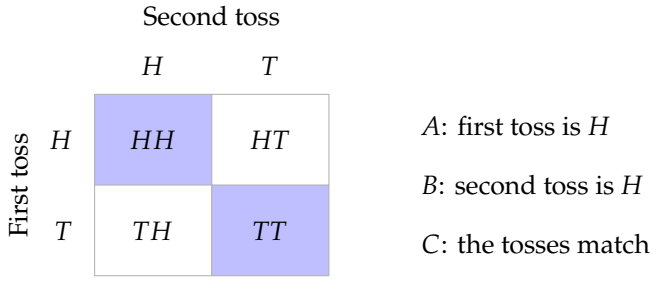



**Figure 10:** Every cell has probability $1/4$. The two colored cells form $C$. Knowing both $A$ and $B$ identifies $HH$, which is in $C$.

Each event contains two outcomes, so $P(A) = P(B) = P(C) = 1/2$. Each pairwise intersection is the singleton $\{HH\}$:

$$P(A \cap B) = P(A \cap C) = P(B \cap C) = \tfrac{1}{4} = \tfrac{1}{2} \cdot \tfrac{1}{2}.$$

All three pairs are independent. But the triple intersection is also $\{HH\}$, giving

$$P(A \cap B \cap C) = \tfrac{1}{4} \neq \tfrac{1}{8} = P(A)P(B)P(C).$$

Equivalently,

$$P(C \mid A \cap B) = \frac{1/4}{1/4} = 1 \neq \tfrac{1}{2} = P(C).$$

Neither toss alone changes the probability of matching; knowing both tosses determines whether they match. Pairwise independence misses this dependence on combined information.

### 3.6 The king's sibling: specify the experiment

**Example:** *A king comes from a family with two children. What is the probability that his sibling is female?* Assume the parents decided in advance to have exactly two children. Each child is a boy or girl with probability $1/2$, independently of the other. Assume that a boy takes precedence in succession, so a king exists exactly when at least one child is a boy.

Order the children by birth. The four outcomes $BB, BG, GB, GG$ each have probability $1/4$, where $B$ means boy and $G$ girl. The information that there is a king is the event

$$K = \{BB, BG, GB\}, \qquad P(K) = \tfrac{3}{4}.$$

His sibling is female in $BG$ or $GB$. Writing $F$ for this event within $K$ gives

$$P(F \mid K) = \frac{P(F \cap K)}{P(K)} = \frac{2/4}{3/4} = \frac{2}{3}.$$

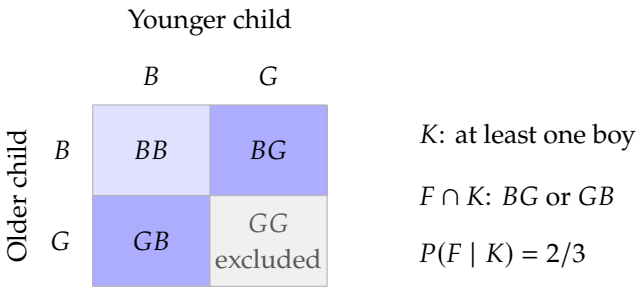



**Figure 11:** Under the fixed two-child model, conditioning on $K$ removes $GG$. The other three outcomes are equally likely; two give a female sibling. The darker cells are $F \cap K$.

The answer depends on the experiment. If the parents instead have children until the first boy and then stop, having exactly two children forces the outcome $GB$. Under that rule, the king's sibling is female with probability one. The phrase "a family with two children" alone does not specify how the family came to have two children.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 3, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 4: Counting

Under a uniform law, probability reduces to counting outcomes. Count each outcome once, distinguishing order, repetition, and labeled groups.

## 4.1 The discrete uniform law

If a finite sample space $\Omega$ has equally likely outcomes, each outcome has probability $1/|\Omega|$. For an event $A \subseteq \Omega$,

$$\boxed{P(A) = \frac{|A|}{|\Omega|}}.$$

Here $|A|$ is the number of outcomes in $A$. Count the numerator and denominator using the same kind of outcome. The formula requires equal likelihood; counting alone does not establish that assumption.

## 4.2 The basic counting principle

Multiply the choices at successive stages when every partial outcome has the same number of extensions and each full outcome has one description.

**Proposition 4.1 [Basic counting principle].** *Suppose an outcome is constructed in $r$ stages. At stage $i$, there are $n_i$ choices for every permitted history of earlier stages. If each complete sequence of choices describes exactly one outcome, and each outcome has exactly one such description, the number of outcomes is*

$$n_1 n_2 \cdots n_r.$$

*Proof.* After the first stage there are $n_1$ partial outcomes. Each has $n_2$ extensions, giving $n_1 n_2$ partial outcomes after stage two. Repeating this reasoning at each stage gives the product. The unique-description assumption prevents counting the same outcome more than once. □

**Example:** *License plates with three letters followed by four digits.* Use 26 letters and the ten digits $0, \ldots, 9$ in these fixed positions. With repetition allowed, the number of plates is

$$26^3\, 10^4 = 175{,}760{,}000.$$

If letters cannot repeat and digits cannot repeat, the choices decrease as positions are filled:

$$(26 \cdot 25 \cdot 24)(10 \cdot 9 \cdot 8 \cdot 7) = 78{,}624{,}000.$$

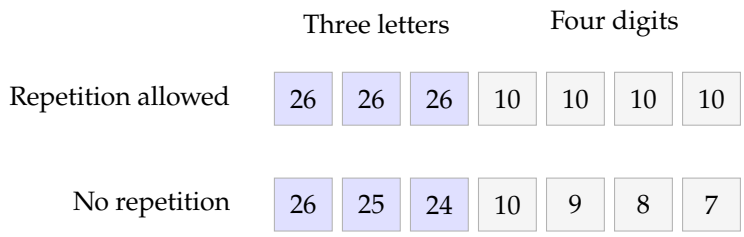



**Figure 12:** Numbers in the slots count the available choices. Multiply across a row. Without repetition, used symbols are unavailable later.

A PERMUTATION of $n$ distinct elements is an ordering of all of them. There are $n$ choices for the first position, $n-1$ for the second, and so on, giving

$$n(n-1)\cdots 2\cdot 1 = n!.$$

The notation $n!$ is the FACTORIAL of $n$; by convention, $0! = 1$.

To construct a subset of $\{1,\dots,n\}$, decide separately for each element whether to include it. There are two choices at each of $n$ stages, so the number of subsets, including the empty set, is

$$2^n.$$

**Example:** *Six die rolls, all different.* Roll a fair six-sided die independently six times. An outcome is an ordered sequence of six results. All $6^6$ sequences are equally likely. For the results to be different, the numbers of choices are $6, 5, 4, 3, 2, 1$, giving $6!$ favorable sequences. Thus

$$P(\text{all six different}) = \frac{6!}{6^6} = \frac{720}{46{,}656} = \frac{5}{324} \approx 0.01543.$$

The denominator allows repeated results; the numerator forbids them.

### 4.3 From ordered choices to combinations

Choosing $k$ distinct elements in order from an $n$-element set is a $k$-PERMUTATION. For integers $0 \le k \le n$, its count is

$$n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

For $k = 0$, there is one empty sequence. An unordered selection must identify all the different orderings of the same chosen elements.

**Definition 4.2 [COMBINATIONS].** The BINOMIAL COEFFICIENT $\binom{n}{k}$, read "$n$ choose $k$," is the number of $k$-element subsets of a given $n$-element set. These unordered selections are called COMBINATIONS.

**Proposition 4.3 [Counting combinations].** *For integers $0 \le k \le n$,*

$$\boxed{\binom{n}{k} = \frac{n!}{k!(n-k)!}}.$$

*Proof.* Count ordered selections of $k$ distinct elements in two ways. Choosing the elements one at a time gives $n!/(n-k)!$ selections. Alternatively, choose a $k$-element subset in $\binom{n}{k}$ ways, then order its elements in $k!$ ways. Both procedures describe the same ordered selections, so

$$\binom{n}{k} k! = \frac{n!}{(n-k)!}.$$

Dividing by $k!$ gives the formula. □

The factor $k!$ removes repeated counts of the same subset. Also, every subset has one of the sizes $0, 1, \ldots, n$. Counting subsets by size or by the include/exclude decisions above gives

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

## 4.4 Binomial probabilities

Consider $n$ independent tosses of a coin with head probability $p$. A specified sequence with $k$ heads and $n-k$ tails has probability $p^k(1-p)^{n-k}$. For the sequence $HTTHHH$, for example,

$$P(HTTHHH) = p(1-p)(1-p)ppp = p^4(1-p)^2.$$

The positions of the heads identify the sequence. Choosing $k$ of the $n$ positions therefore counts the sequences with exactly $k$ heads.

**Proposition 4.4 [Binomial probability].** *For $0 < p < 1$ and $k = 0, 1, \ldots, n$,*

$$\boxed{P(\textit{exactly } k \textit{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}}.$$

*Proof.* Exactly $k$ heads is a disjoint union of the $\binom{n}{k}$ sequences whose head positions form a $k$-element subset. Independence gives each sequence probability $p^k(1-p)^{n-k}$. Add their equal probabilities to obtain the formula. □

If $p = 0$, the all-tail sequence has probability one; if $p = 1$, the all-head sequence has probability one. These are the boundary cases of the same model. For a biased coin, sequences with different head

counts need not be equally likely. Equal likelihood is being used only among sequences with a fixed head count.

## 4.5 Conditioning on the number of heads

**Example:** *Three heads in ten tosses.* Toss a coin independently ten times, with $0 < p < 1$. Let $B$ be the event that exactly three tosses are heads, and let $A$ be the event that the first two tosses are heads. Find $P(A \mid B)$.

Every sequence in $B$ has probability $p^3(1-p)^7$. After conditioning on $B$, these sequences remain equally likely, even when the coin is biased. There are

$$|B| = \binom{10}{3} = 120$$

possible sets of head positions. To be in $A \cap B$, positions 1 and 2 must be heads, leaving exactly one head to place among positions 3 through 10. Hence

$$|A \cap B| = \binom{8}{1} = 8.$$

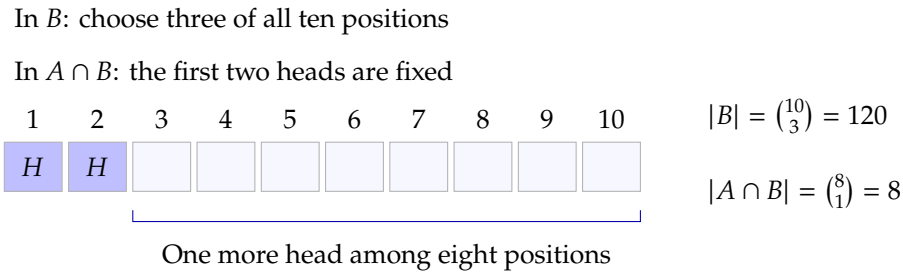



**Figure 13:** Fixing the first two heads leaves eight possible positions for the last head. Without those fixed heads, any three of the ten positions may be the head positions.

The conditional uniform law gives

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{\binom{8}{1}}{\binom{10}{3}} = \frac{8}{120} = \frac{1}{15}.$$

Equivalently, the common factor $p^3(1-p)^7$ cancels from the conditional-probability ratio. The denominator counts only sequences compatible with $B$.

### 4.6 Partitions: dealing a deck to four players

**Example:** *Each player receives an ace.* Deal a uniformly shuffled 52-card deck to four labeled players, with 13 cards per player. What is the probability that each receives one ace? An outcome records each player's hand, with order within a hand ignored. The four hands are disjoint and cover the deck, so they form a partition.

An ordering of the deck induces four labeled hands. Divide by the $(13!)^4$ orders within those hands:

$$|\Omega| = \frac{52!}{(13!)^4}.$$

Uniform shuffling makes these partitions equiprobable. Do not divide by 4!: exchanging labeled players' hands changes the outcome.

Assign the four aces to the players in 4! ways, then partition the 48 non-aces into four hands of 12:

$$|A| = 4!\,\frac{48!}{(12!)^4},$$

where $A$ is the event that each player has an ace.

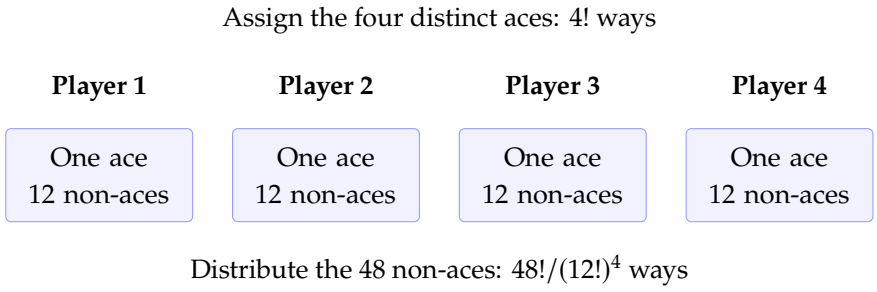



**Figure 14:** A favorable deal puts one distinct ace and 12 non-aces in each labeled hand. Choose the ace assignment, then the distribution of the remaining cards.

Taking the ratio gives

$$\begin{aligned} P(A) &= \frac{4!\,48!/(12!)^4}{52!/(13!)^4} \\ &= \frac{4!\,13^4}{52\cdot 51\cdot 50\cdot 49} = \frac{2197}{20825} \approx 0.10550. \end{aligned}$$

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 4, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 5: Discrete Random Variables, PMFs, and Expectations

A random variable assigns a number to each outcome. Its PMF describes the probabilities, its expectation the center, and its variance the spread.

## 5.1 Random variables: numbers assigned to outcomes

**Definition 5.1 [Random variable].** A random variable is a function $X : \Omega \to \mathbb{R}$. For each outcome $\omega$, the number $X(\omega)$ is the value recorded by $X$.

The rule $X$ is fixed; the outcome determines which value it takes. Use a capital letter $X$ for the random variable and a lower-case letter $x$ for a possible numerical value. Several random variables can record different numbers from the same outcome.

A random variable is discrete when its possible values form a finite or countably infinite set. Random variables can also take continuous values, but the probability mass functions below describe the discrete case.

## 5.2 Probability mass functions

To find the probability of a value $x$, collect the outcomes that produce that value. The event $\{X = x\}$ is the set $\{\omega \in \Omega : X(\omega) = x\}$.

**Definition 5.2 [Probability mass function].** The probability mass function, or PMF, of a discrete random variable $X$ is

$$p_X(x) = P(X = x) = P\big(\{\omega \in \Omega : X(\omega) = x\}\big).$$

It describes the probability distribution of $X$.

Each value has nonnegative probability, and the probabilities of all possible values add to one:

$$p_X(x) \geq 0, \qquad \sum_x p_X(x) = 1.$$

The sum runs over the possible values of $X$, rather than over individual outcomes. The events $\{X = x\}$ are disjoint and cover the sample space, which explains normalization.

**Example:** *Tosses until the first head.* Toss a coin independently, with head probability $p > 0$. Let $X$ be the number of tosses up to and including the first head. For $X = k$, the first $k - 1$ tosses must be tails and the $k$th must be heads. For $0 < p < 1$,

$$p_X(k) = P(\underbrace{TT \cdots T}_{k-1 \text{ tails}} H) = (1-p)^{k-1}p, \qquad k = 1, 2, \ldots.$$

This is the GEOMETRIC PMF. Its probabilities sum to $p \sum_{j=0}^{\infty}(1-p)^j = p/[1-(1-p)] = 1$. Equivalently, the probability of seeing no head in the first $m$ tosses is $(1-p)^m$, which tends to zero. A head therefore occurs eventually with probability one. If $p = 1$, then $X = 1$ with probability one.

## 5.3 Computing a PMF by grouping outcomes

Group outcomes by their value of $X$ and sum each group's probabilities.

**Example:** *The minimum of two die rolls.* Roll a fair tetrahedral die twice, independently. Let $F$ and $S$ be the first and second results, and set $X = \min(F, S)$. The 16 ordered pairs $(f, s)$ are equally likely, each with probability $1/16$.

For $X = 2$, both results must be at least 2 and at least one must equal 2. The five pairs are

$$(2,2),\ (2,3),\ (2,4),\ (3,2),\ (4,2),$$

so $p_X(2) = 5/16$. Count $(2,2)$ once, even though both rolls equal 2. Grouping the other pairs by their minimum gives the complete PMF:

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $p_X(x)$ | 7/16 | 5/16 | 3/16 | 1/16 |

These four probabilities sum to one. The values of $X$ are not equally likely, even though the underlying die outcomes are.

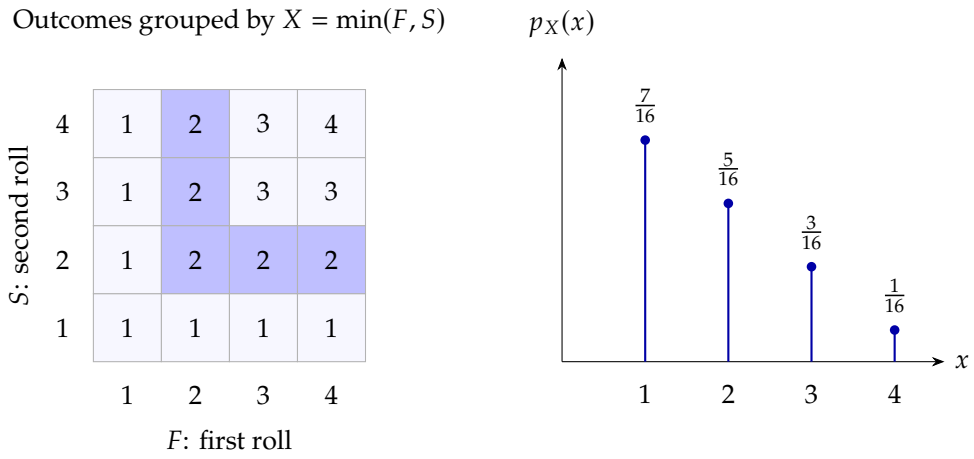



**Figure 15:** Each grid cell shows $X = \min(F, S)$ and has probability $1/16$. The highlighted cells give $X = 2$. Grouping cells with the same number produces the PMF on the right.

## 5.4 The binomial PMF

Let $X$ count the heads in $n$ independent coin tosses, with head probability $p$. A sequence with $k$ heads and $n - k$ tails has probability $p^k(1-p)^{n-k}$. There are $\binom{n}{k}$ such sequences, one for each choice of the $k$ head positions. Thus the BINOMIAL PMF is

$$\boxed{p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}}, \qquad k = 0, 1, \ldots, n, \quad 0 < p < 1.$$

The PMF is zero at other values. At $p = 0$, $X = 0$ with probability one; at $p = 1$, $X = n$ with probability one.

**Example:** *Two heads in four tosses.* For $n = 4$, the event $X = 2$ consists of six disjoint sequences:

$$HHTT,\ HTHT,\ HTTH,\ THHT,\ THTH,\ TTHH.$$

Each has probability $p^2(1-p)^2$. Adding them gives

$$p_X(2) = 6p^2(1-p)^2 = \binom{4}{2} p^2 (1-p)^2.$$

The coefficient counts sequences, and the remaining factor gives the probability of one such sequence.

## 5.5 Expectation: a probability-weighted average

Values with more probability receive more weight in the average.

**Definition 5.3 [Expectation].** For a discrete random variable $X$, its expectation, or mean, is

$$\boxed{\mathbb{E}[X] = \sum_x x\, p_X(x)},$$

provided $\sum_x |x| p_X(x) < \infty$.

For finitely many possible values, the sum is automatically finite. For infinite sets of values, the expectation identities below assume that the relevant sums are finite. Think of $\mathbb{E}[X]$ as the center of gravity when mass $p_X(x)$ is placed at position $x$. It also represents the average approached over many independent repetitions of the same experiment; this interpretation requires a limit result.

**Example:** *Uniform on* $0, 1, \ldots, n$. Let $n$ be a nonnegative integer. Each value has probability $1/(n+1)$, so

$$\mathbb{E}[X] = \frac{0 + 1 + \cdots + n}{n+1}.$$

To evaluate the numerator, write $S = 0 + 1 + \cdots + n$ and also $S = n + (n-1) + \cdots + 0$. Adding the two expressions gives $2S = n(n+1)$. Therefore

$$\mathbb{E}[X] = \frac{n(n+1)/2}{n+1} = \boxed{\frac{n}{2}}.$$

Equal probabilities at positions equally far from $n/2$ also explain why this is the center.

## 5.6 Expectations of functions and affine expressions

Suppose $Y = g(X)$. One way to calculate $\mathbb{E}[Y]$ is to find the PMF of $Y$ and sum $y\, p_Y(y)$. We can instead use the PMF of $X$ directly.

**Proposition 5.4 [Expectation of a function].** *If* $\sum_x |g(x)| p_X(x) < \infty$, *then*

$$\boxed{\mathbb{E}[g(X)] = \sum_x g(x) p_X(x)}.$$

*Proof.* Group the values of $X$ according to the common value they produce under $g$. For a possible value $y$ of $Y$,

$$p_Y(y) = \sum_{x : g(x) = y} p_X(x).$$

Substitute this into the definition of expectation:

$$\begin{aligned}\mathbb{E}[Y] &= \sum_y y \sum_{x:g(x)=y} p_X(x) \\ &= \sum_y \sum_{x:g(x)=y} g(x)p_X(x) = \sum_x g(x)p_X(x).\end{aligned}$$

Every value of $X$ belongs to one group. Absolute convergence justifies regrouping an infinite sum. □

In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$. Taking a function of an average need not give the average of that function. An AFFINE FUNCTION has the form $g(x) = \alpha x + \beta$, with constants $\alpha, \beta$. For these functions, the two operations agree.

**Proposition 5.5 [CONSTANTS AND AFFINE FUNCTIONS].** *For constants $\alpha, \beta$ and a random variable with finite mean,*

$$\mathbb{E}[\alpha] = \alpha, \qquad \mathbb{E}[\alpha X] = \alpha\mathbb{E}[X], \qquad \boxed{\mathbb{E}[\alpha X + \beta] = \alpha\mathbb{E}[X] + \beta}.$$

*Proof.* The PMF sums to one, so a constant contributes $\sum_x \alpha p_X(x) = \alpha$. Applying the expectation-of-a-function formula to $g(x) = \alpha x + \beta$ gives

$$\sum_x (\alpha x + \beta)p_X(x) = \alpha \sum_x x p_X(x) + \beta \sum_x p_X(x) = \alpha\mathbb{E}[X] + \beta.$$

Taking $\beta = 0$ gives the scaling identity. □

## 5.7 Variance: squared distance from the mean

The mean locates the center, but does not describe how far values lie from it. Squaring their deviations from the mean prevents positive and negative deviations from canceling.

**Definition 5.6 [SECOND MOMENT AND VARIANCE].** The SECOND MOMENT of $X$ is $\mathbb{E}[X^2] = \sum_x x^2 p_X(x)$. When this is finite, write $\mu = \mathbb{E}[X]$. The VARIANCE of $X$ is

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p_X(x).$$

The expectation-of-a-function formula applies with $g(x) = (x - \mu)^2$. It also gives a useful way to compute variance.

**Proposition 5.7 [VARIANCE IDENTITIES].** *For finite second moment,*

$$\boxed{\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2}, \qquad \operatorname{Var}(X) \geq 0.$$

*For constants $\alpha, \beta$,*

$$\boxed{\operatorname{Var}(\alpha X + \beta) = \alpha^2 \operatorname{Var}(X)}.$$

*Proof.* Expand the square inside the defining sum and use $\sum_x x p_X(x) = \mu$ and $\sum_x p_X(x) = 1$:

$$\begin{aligned}\operatorname{Var}(X) &= \sum_x (x^2 - 2\mu x + \mu^2) p_X(x) \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2.\end{aligned}$$

Nonnegativity follows from the defining sum of nonnegative squared deviations, not from treating the last difference as automatically nonnegative.
For $Y = \alpha X + \beta$, its mean is $\mathbb{E}[Y] = \alpha\mu + \beta$. Consequently

$$Y - \mathbb{E}[Y] = \alpha(X - \mu), \qquad \operatorname{Var}(Y) = \mathbb{E}[\alpha^2 (X - \mu)^2] = \alpha^2 \operatorname{Var}(X).$$

□

A shift $\beta$ moves the values and their mean by the same amount, leaving deviations unchanged. Scaling by $\alpha$ multiplies each deviation by $\alpha$, so squared deviations scale by $\alpha^2$.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 5, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 6: Discrete Random Variable Examples and Joint PMFs

Conditioning changes the weights in an expectation; a joint PMF describes two variables together. In general, averaging and applying a function do not commute.

## 6.1 Review: means, deviations, and standard deviation

A random variable $X$ assigns a real number to each outcome. For a discrete $X$, its PMF is $p_X(x) = P(X = x)$. With the relevant sums finite,

$$\mathbb{E}[X] = \sum_x x p_X(x), \qquad \mathbb{E}[g(X)] = \sum_x g(x) p_X(x), \qquad \mathbb{E}[\alpha X + \beta] = \alpha \mathbb{E}[X] + \beta.$$

Writing $\mu = \mathbb{E}[X]$, the deviations from the mean average to zero:

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mu = 0.$$

Their squares need not average to zero. They give the variance:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p_X(x) = \mathbb{E}[X^2] - \mu^2.$$

**Definition 6.1 [Standard deviation].** The standard deviation of $X$ is

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

Variance has squared units; standard deviation has the same units as $X$.

## 6.2 Random speed and average travel time

Travel a distance of 200 miles at a speed $V$ that is constant during the trip but random before it begins. The speed is either 1 or 200 miles per hour, each with probability $1/2$.

The mean speed is

$$\mathbb{E}[V] = \tfrac{1}{2}(1) + \tfrac{1}{2}(200) = \frac{201}{2} = 100.5 \quad \text{miles per hour.}$$

Both possible speeds are $199/2$ from this mean, so

$$\mathrm{Var}(V) = \tfrac{1}{2}\left(-\frac{199}{2}\right)^2 + \tfrac{1}{2}\left(\frac{199}{2}\right)^2 = \frac{39601}{4}, \qquad \sigma_V = \frac{199}{2} = 99.5 \quad \text{miles per hour.}$$

Travel time, in hours, is the function $T = t(V) = 200/V$:

| $V$ | 1 | 200 |
|---|---|---|
| $T = 200/V$ | 200 | 1 |
| Probability | 1/2 | 1/2 |

Average the times using their probabilities:

$$\mathbb{E}[T] = \sum_{v} \frac{200}{v} p_V(v) = \tfrac{1}{2}(200) + \tfrac{1}{2}(1) = \frac{201}{2} = 100.5 \quad \text{hours.}$$

Using the mean speed instead would give

$$\frac{200}{\mathbb{E}[V]} = \frac{400}{201} \quad \text{hours,} \qquad \mathbb{E}[200/V] \neq \frac{200}{\mathbb{E}[V]}.$$

The slow trip contributes 200 hours to the possible times, so it strongly affects their average. Taking a reciprocal does not commute with averaging.

Also, $TV = 200$ on every outcome, hence

$$\mathbb{E}[TV] = 200 \neq \mathbb{E}[T]\mathbb{E}[V] = \frac{40401}{4}.$$

The speed and time come from the same trip. Their product must be averaged using the values that occur together.

### 6.3 Conditional PMFs and conditional expectations

Once we know that an event $A$ occurred, use conditional probabilities as the new weights. Assume $P(A) > 0$.

**Definition 6.2 [Conditional PMF and expectation].** The conditional PMF of $X$ given $A$ is

$$p_{X|A}(x) = P(X = x \mid A) = \frac{P(\{X = x\} \cap A)}{P(A)}.$$

Its probabilities sum to one. The conditional expectation is

$$\mathbb{E}[X \mid A] = \sum_{x} x\, p_{X|A}(x),$$

when the corresponding absolute-value sum is finite.

**Example:** *A uniform variable after conditioning.* Let $X$ be uniform on $\{1, 2, 3, 4\}$, so $p_X(x) = 1/4$ at these four values. Take $A = \{X \geq 2\}$. Then $P(A) = 3/4$. Conditioning removes the value 1 and

divides each remaining probability by $3/4$:

$$p_{X|A}(x) = \begin{cases} 1/3, & x \in \{2,3,4\}, \\ 0, & \text{otherwise.} \end{cases} \qquad \mathbb{E}[X \mid A] = \frac{2+3+4}{3} = 3.$$

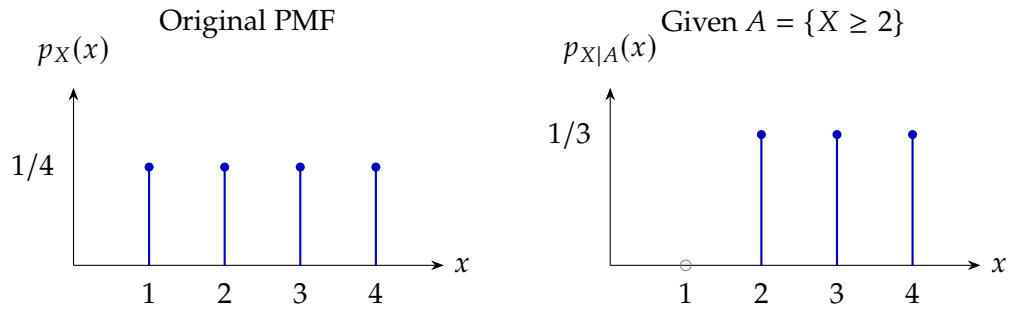



**Figure 16:** Conditioning on $X \geq 2$ removes the probability at 1. The retained probabilities are rescaled to sum to one.

## 6.4 The geometric PMF and memorylessness

Toss a coin independently until the first head, with head probability $0 < p < 1$. Let $X$ count all tosses, including the head, and write $q = 1 - p$. The first $k - 1$ tosses must be tails, giving

$$p_X(k) = q^{k-1}p, \quad k = 1, 2, \ldots, \qquad \mathbb{E}[X] = \sum_{k=1}^{\infty} kq^{k-1}p.$$

This expectation is finite: successive terms have ratio $q(k+1)/k$, eventually bounded strictly below one. We will evaluate the mean by conditioning.

Suppose we already know that $X > 2$. The first two tosses were tails, with probability $q^2$. For $k \geq 3$,

$$p_{X|X>2}(k) = \frac{q^{k-1}p}{q^2} = q^{k-3}p.$$

It is zero for $k \leq 2$. Let $R = X - 2$ be the number of *remaining* tosses, including the first head. For $j = 1, 2, \ldots$,

$$P(R = j \mid X > 2) = P(X = j + 2 \mid X > 2) = q^{j-1}p.$$

Thus, given that the first two tosses failed, the remaining waiting time has the original geometric PMF. This is the MEMORYLESS PROPERTY. The total waiting time still starts at 3 under this condition; it is the remaining waiting time that starts at 1.

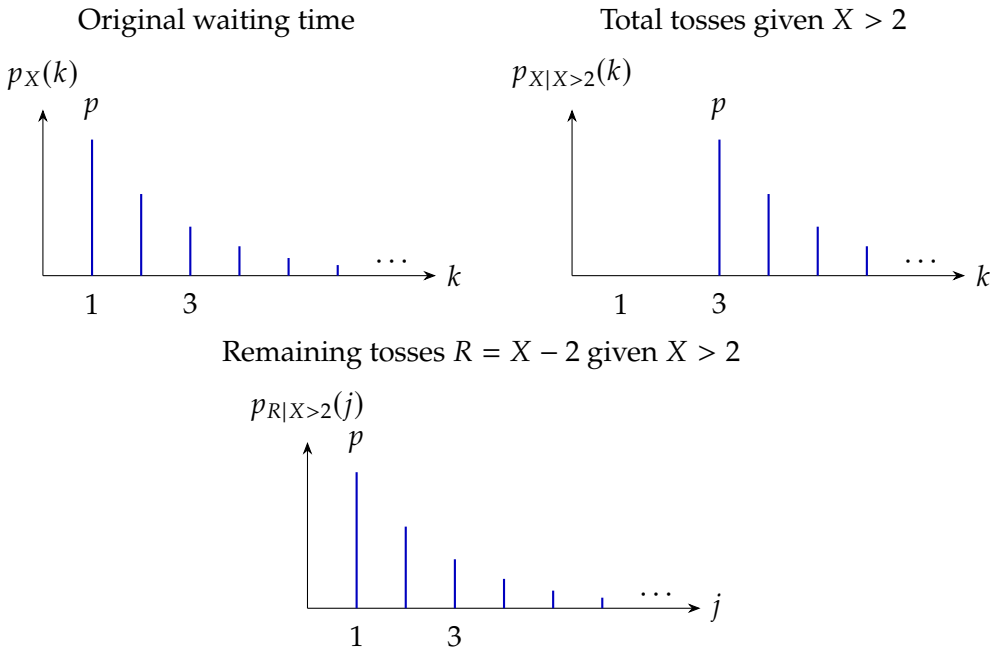



**Figure 17:** The geometric PMFs are schematic. Conditioning on two tails renormalizes the tail of the original PMF. Subtracting the two elapsed tosses restores its original shape and starting value.

## 6.5 Total expectation and the geometric mean

Average within each event of a partition, then weight by its probability. Omit probability-zero events.

For any event $B$, total probability gives

$$P(B) = \sum_{i=1}^{n} P(A_i)P(B \mid A_i).$$

Taking $B = \{X = x\}$ yields

$$p_X(x) = \sum_{i=1}^{n} P(A_i)p_{X|A_i}(x).$$

**Theorem 6.3 [Total expectation].** *For a discrete random variable with finite absolute expectation,*

$$\boxed{\mathbb{E}[X] = \sum_{i=1}^{n} P(A_i)\mathbb{E}[X \mid A_i]}.$$

*Proof.* Substitute the total-probability expression for the PMF into the expectation, then collect terms from each event:

$$\begin{aligned}\mathbb{E}[X] &= \sum_x x \sum_{i=1}^{n} P(A_i) p_{X|A_i}(x) \\ &= \sum_{i=1}^{n} P(A_i) \sum_x x p_{X|A_i}(x) = \sum_{i=1}^{n} P(A_i) \mathbb{E}[X \mid A_i].\end{aligned}$$

Absolute convergence justifies exchanging the sums. □

**Example:** *The geometric mean.* Return to the tosses until the first head. Write $m = \mathbb{E}[X]$ and partition by $A_1 = \{X = 1\}$ and $A_2 = \{X > 1\}$. Their probabilities are $p$ and $q$. On $A_1$, $X = 1$. On $A_2$, one tail has elapsed, and the remaining tosses start the same experiment. Thus $\mathbb{E}[X \mid X > 1] = 1 + m$. Total expectation gives

$$m = p \cdot 1 + q(1 + m) = 1 + qm.$$

Since $1 - q = p$, solving gives

$$\boxed{\mathbb{E}[X] = \frac{1}{p}}.$$

If $p = 1$, the first toss is a head with certainty and $\mathbb{E}[X] = 1$, which agrees with this formula.

## 6.6 Joint PMFs, marginals, and conditioning

Assign a joint probability to each pair of values.

**Definition 6.4 [Joint PMF].** The joint PMF of discrete random variables $X, Y$ is

$$p_{X,Y}(x, y) = P(\{X = x\} \cap \{Y = y\}).$$

It is nonnegative and satisfies

$$\sum_x \sum_y p_{X,Y}(x, y) = 1.$$

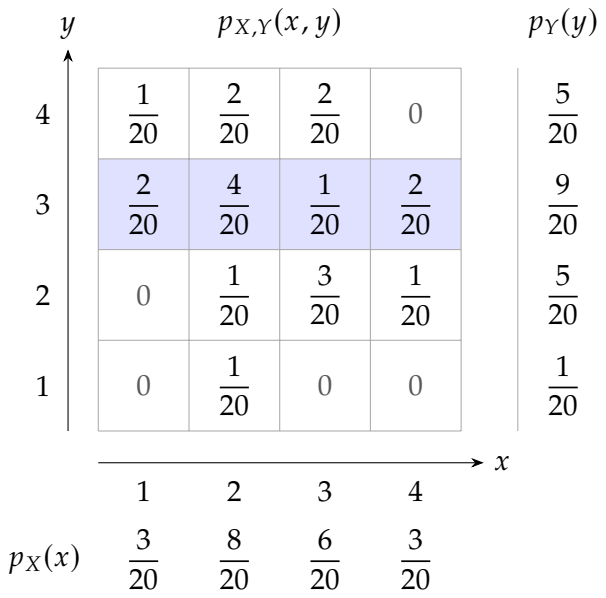



**Figure 18:** Each cell gives the probability of one pair $(x, y)$. Column sums give $p_X(x)$ and row sums give $p_Y(y)$. The highlighted row contains the pairs retained when conditioning on $Y = 3$.

To find the probability of $X = x$, allow every possible value of $Y$:

$$\boxed{p_X(x) = \sum_y p_{X,Y}(x,y)}, \qquad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

These are the MARGINAL PMFs. The displayed column totals are $(3, 8, 6, 3)/20$; for example, $p_X(3) = (0 + 3 + 1 + 2)/20 = 6/20$.

If $p_Y(y) > 0$, conditional probability gives

$$\boxed{p_{X|Y}(x \mid y) = P(X = x \mid Y = y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}}.$$

Here the denominator is the whole row's probability, so

$$\sum_x p_{X|Y}(x \mid y) = \frac{\sum_x p_{X,Y}(x,y)}{p_Y(y)} = 1.$$

For $Y = 3$, the row has probability $9/20$. Its conditional PMF is

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $p_{X\|Y}(x \mid 3)$ | 2/9 | 4/9 | 1/9 | 2/9 |

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 6, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 7: Multiple Discrete Random Variables: Expectations, Conditioning, and Independence

Independence factors a joint PMF into marginals. Indicator variables let us compute expected counts without deriving their full distributions.

## 7.1 Joint PMFs and the multiplication rule

For discrete random variables $X, Y$,

$$p_X(x) = P(X = x), \qquad p_{X,Y}(x, y) = P(X = x, Y = y).$$

Sum over the other variable to obtain a marginal PMF:

$$p_X(x) = \sum_y p_{X,Y}(x, y).$$

When $p_Y(y) > 0$, the conditional PMF is

$$p_{X|Y}(x \mid y) = P(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Rearranging the corresponding formula with $X$ as the condition gives

$$p_{X,Y}(x, y) = p_X(x)p_{Y|X}(y \mid x), \qquad p_X(x) > 0.$$

The first factor describes $X$; the second uses the information that $X = x$ to describe $Y$.

For three variables, apply the multiplication rule once more:

$$p_{X,Y,Z}(x, y, z) = p_X(x)p_{Y|X}(y \mid x)p_{Z|X,Y}(z \mid x, y).$$

The conditional factors require $p_X(x) > 0$ and $p_{X,Y}(x, y) > 0$. If a conditioning event has probability zero, the joint probability is zero, but the conditional expression is undefined.

## 7.2 Independence and the effect of conditioning

Under independence, knowing the values of some variables does not change the probabilities for the others.

**Definition 7.1 [Independent random variables].** Discrete random variables $X, Y$ are independent random variables if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y.$$

Three variables $X, Y, Z$ are independent if

$$p_{X,Y,Z}(x, y, z) = p_X(x)p_Y(y)p_Z(z) \quad \text{for every } x, y, z.$$

For two independent variables and $p_X(x) > 0$, division gives $p_{Y|X}(y \mid x) = p_Y(y)$. The product condition itself also applies at values of probability zero. It must hold for *every* pair, rather than just one selected pair.

**Example:** *A joint table before and after conditioning.* Consider the joint PMF on $\{1, 2, 3, 4\}^2$ shown below; all other pairs have probability zero. Its marginals give $p_X(1) = 3/20$ and $p_Y(1) = 1/20$. Yet $p_{X,Y}(1, 1) = 0$, whereas $p_X(1)p_Y(1) = 3/400 > 0$. Thus $X, Y$ are not independent.

Now condition on $A = \{X \leq 2,\ Y \geq 3\}$. The retained probabilities sum to $(1 + 2 + 2 + 4)/20 = 9/20$. Dividing by $P(A)$ gives

| | $x = 1$ | $x = 2$ |
|---|---|---|
| $y = 4$ | $1/9$ | $2/9$ |
| $y = 3$ | $2/9$ | $4/9$ |

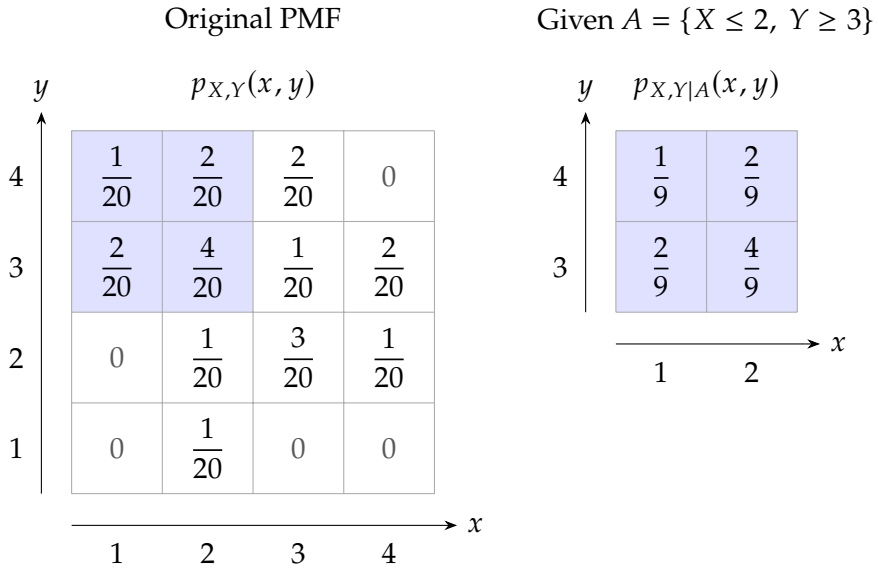



**Figure 19:** The highlighted cells are the event $A$. Conditioning keeps these four probabilities and divides them by $9/20$. The resulting two-by-two table factors into its conditional marginals.

The conditional marginals are

$$p_{X|A}(1) = 1/3, \quad p_{X|A}(2) = 2/3,$$
$$p_{Y|A}(3) = 2/3, \quad p_{Y|A}(4) = 1/3.$$

Their four products are precisely the four entries above; all other entries and the corresponding products are zero. Therefore

$$p_{X,Y|A}(x, y) = p_{X|A}(x)p_{Y|A}(y) \quad \text{for every } x, y.$$

The variables are independent under the conditional probability law, even though they are dependent under the original law.

## 7.3 Expectations of functions, sums, and products

Weight each function value by its joint probability. Assume the expectation sums converge absolutely.

**Proposition 7.2 [Expectation of a function of two variables].** *For $W = g(X, Y)$,*

$$\boxed{\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y)p_{X,Y}(x, y)}.$$

*Proof.* Collect the pairs that give the same value $w$ of $W$:

$$p_W(w) = \sum_{(x,y):g(x,y)=w} p_{X,Y}(x, y).$$

Substitute into $\mathbb{E}[W] = \sum_w w p_W(w)$. Each pair belongs to one group, and within that group $w = g(x, y)$. Regrouping gives the stated double sum. □

In general, $\mathbb{E}[g(X, Y)]$ need not equal $g(\mathbb{E}[X], \mathbb{E}[Y])$. Affine functions satisfy $\mathbb{E}[\alpha X + \beta] = \alpha\mathbb{E}[X] + \beta$. Sums of variables have a further useful property.

**Proposition 7.3 [Linearity of expectation].** *For variables with finite absolute expectations, no independence assumption is needed for*

$$\boxed{\mathbb{E}[X + Y + Z] = \mathbb{E}[X] + \mathbb{E}[Y] + \mathbb{E}[Z]}.$$

*Proof.* For two variables, split the joint-PMF sum and then use the marginals:

$$\begin{aligned}\mathbb{E}[X+Y] &= \sum_x \sum_y (x+y)p_{X,Y}(x,y) \\ &= \sum_x x \sum_y p_{X,Y}(x,y) + \sum_y y \sum_x p_{X,Y}(x,y) \\ &= \mathbb{E}[X] + \mathbb{E}[Y].\end{aligned}$$

Apply this identity first to $X+Y$ and $Z$, then to $X$ and $Y$. □

Products require more care: values that occur together matter. Independence makes the double sum factor.

**Proposition 7.4 [Products under independence].** *If $X, Y$ are independent and $g(X), h(Y)$ have finite absolute expectations, then*

$$\boxed{\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]}.$$

*In particular, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ when $X, Y$ have finite absolute expectations.*

*Proof.* Replace the joint PMF by the product of its marginals:

$$\begin{aligned}\mathbb{E}[g(X)h(Y)] &= \sum_x \sum_y g(x)h(y)p_X(x)p_Y(y) \\ &= \left(\sum_x g(x)p_X(x)\right)\left(\sum_y h(y)p_Y(y)\right).\end{aligned}$$

The corresponding absolute-value sum is the product of two finite sums, so the factorization is justified. □

## 7.4 Variances of sums

A constant shift leaves deviations from the mean unchanged; scaling by $a$ scales their squares by $a^2$. Thus

$$\operatorname{Var}(X+a) = \operatorname{Var}(X), \qquad \operatorname{Var}(aX) = a^2 \operatorname{Var}(X).$$

For a sum, expand its squared deviation. This reveals where independence enters.

**Proposition 7.5 [Variance of an independent sum].** *If $X, Y$ are independent with finite second moments, then*

$$\boxed{\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)}.$$

*Proof.* Write $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. Linearity gives the mean of $X + Y$, so expanding its squared deviation yields

$$\begin{aligned}\operatorname{Var}(X+Y) &= \mathbb{E}[((X-\mu_X)+(Y-\mu_Y))^2] \\ &= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\mathbb{E}[(X-\mu_X)(Y-\mu_Y)].\end{aligned}$$

Independence factors the last expectation into $\mathbb{E}[X-\mu_X]\mathbb{E}[Y-\mu_Y] = 0$. This removes the cross term. □

The independence assumption cannot simply be dropped:

- If $X = Y$, then $X + Y = 2X$, so $\operatorname{Var}(X+Y) = 4\operatorname{Var}(X)$.
- If $X = -Y$, then $X + Y = 0$, so $\operatorname{Var}(X+Y) = 0$.
- If $X, Y$ are independent and $Z = X - 3Y$, scaling $Y$ and applying the independent-sum rule gives $\operatorname{Var}(Z) = \operatorname{Var}(X) + 9\operatorname{Var}(Y)$.

These cases distinguish adding means, which always works for finite absolute expectations, from adding variances.

### 7.5 Binomial mean and variance using indicators

Let $X$ count the successes in $n$ independent trials, each with success probability $p$. For $0 < p < 1$, a direct PMF calculation gives

$$\mathbb{E}[X] = \sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k}.$$

We can compute the mean through the individual trials. Define the INDICATOR RANDOM VARIABLE

$$X_i = \begin{cases} 1, & \text{trial } i \text{ succeeds,} \\ 0, & \text{otherwise.} \end{cases} \qquad X = \sum_{i=1}^{n} X_i.$$

An indicator turns an event into a number. Its expectation is the event's probability:

$$\mathbb{E}[X_i] = 1 \cdot p + 0 \cdot (1-p) = p.$$

Apply linearity repeatedly:

$$\boxed{\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = np}.$$

Since $X_i^2 = X_i$, its second moment is also $p$. Therefore

$$\mathrm{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = p - p^2 = p(1-p).$$

The independent trials make the indicators independent. In the squared deviation of their sum, every cross term has expectation zero by the same product argument used for two variables. Hence

$$\boxed{\mathrm{Var}(X) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = np(1-p)}.$$

At $p = 0$ or $p = 1$, the success count is constant, and these mean and variance formulas still hold. They also give zero mean and variance when $n = 0$.

## 7.6 The hat problem: an expected number of matches

There are $n \geq 1$ people, each with a distinct hat. Put the hats in a box and return them at random, without replacement. Every assignment of the $n$ hats to the $n$ people is equally likely. Let $X$ count the people who receive their own hats.

Define indicators for the individual matches:

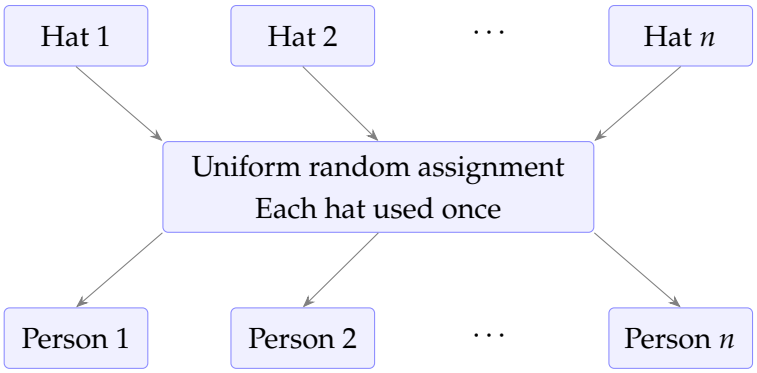
$$X_i = \begin{cases} 1, & \text{person } i \text{ receives their own hat,} \\ 0, & \text{otherwise.} \end{cases} \qquad X = X_1 + \cdots + X_n.$$

Each person is equally likely to receive any of the $n$ hats, so

$$P(X_i = 1) = \frac{1}{n}, \qquad \mathbb{E}[X_i] = \frac{1}{n}.$$

For $n \geq 2$, these indicators are not independent. Given that person 1 receives their own hat, person 2 has probability $1/(n-1)$ of receiving theirs, rather than $1/n$. Nevertheless, linearity gives

$$\boxed{\mathbb{E}[X] = \sum_{i=1}^{n} \frac{1}{n} = 1}.$$




$X_i = 1$ exactly when person $i$ receives hat $i$

**Figure 20:** All hats are used once in a uniformly random assignment. The match indicators share this assignment, so they are dependent for $n \geq 2$. Their means can still be added.

### 7.7 The hat problem: variance and pair probabilities

To find the variance, compute the second moment of the total. Assume $n \geq 2$. Since $\mathbb{E}[X] = 1$,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - 1.$$

Expand the square before taking expectations:

$$X^2 = \sum_{i=1}^{n} X_i^2 + \sum_{i \neq j} X_i X_j.$$

The second sum uses *ordered* pairs: both $(i, j)$ and $(j, i)$ appear. There are $n(n-1)$ terms.

An indicator satisfies $X_i^2 = X_i$, so $\mathbb{E}[X_i^2] = 1/n$. For two distinct people, $X_i X_j$ is itself the indicator that both receive their own hats. Conditioning on the first match gives

$$\begin{aligned} \mathbb{E}[X_i X_j] &= P(X_i = 1, X_j = 1) \\ &= P(X_i = 1) P(X_j = 1 \mid X_i = 1) \\ &= \frac{1}{n} \frac{1}{n-1} = \frac{1}{n(n-1)}. \end{aligned}$$

After fixing one correct hat, the remaining $n - 1$ hats are still uniformly assigned to the remaining people. In particular, this joint probability is not $1/n^2$.

Now sum the individual and pair contributions:

$$\mathbb{E}[X^2] = n \frac{1}{n} + n(n-1) \frac{1}{n(n-1)} = 2.$$

Consequently,

$$\boxed{\text{Var}(X) = 2 - 1 = 1 \quad (n \geq 2)}.$$

When $n = 1$, the sole person always receives their own hat: $X = 1$ and $\text{Var}(X) = 0$. The pair calculation does not apply in this case.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 7, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 8: Continuous Random Variables

A density describes probability per unit length; integration gives interval probabilities. A CDF records the accumulated probability up to a value.

## 8.1 Probability as area under a density

**Definition 8.1 [Continuous random variable and density].** A random variable $X$ is continuous if its probabilities can be described by a probability density function, or PDF, $f_X$, with

$$f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1,$$

such that, for $a \leq b$,

$$\boxed{P(a \leq X \leq b) = \int_a^b f_X(x)\,dx}.$$

The total area is one. The area over a particular interval is the probability that the value of $X$ belongs to that interval. More generally, for a set $B$ over which the integral is defined,

$$P(X \in B) = \int_B f_X(x)\,dx.$$

For a union of disjoint intervals, add their areas.

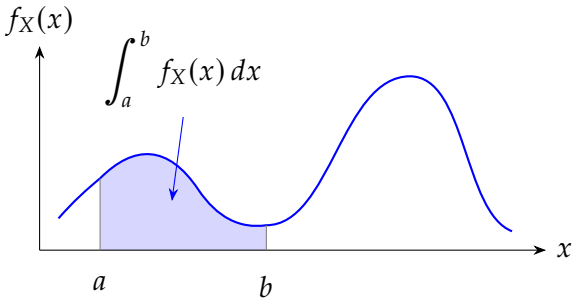



**Figure 21:** The shaded area represents $P(a \leq X \leq b)$. The curve is schematic: its height describes density, while its area describes probability.

A PDF value is a density, not a point probability: $P(X = x) = 0$, so interval endpoints do not matter. Density may exceed one; its integral must equal one.

For $\delta > 0$,

$$P(x \le X \le x + \delta) = \int_x^{x+\delta} f_X(s)\, ds \approx f_X(x)\delta.$$

The approximation replaces a short strip of area by a rectangle. It is accurate when the density varies little across the interval, for example near a point where it is continuous as $\delta$ becomes small.

## 8.2 Means, variances, and the continuous uniform law

The expectation formulas use the same weighting idea as for a PMF, with an integral in place of a sum:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx, \qquad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx.$$

These formulas assume that the corresponding absolute-value integrals are finite. With $\mu = \mathbb{E}[X]$ and finite second moment,

$$\mathrm{Var}(X) = \sigma_X^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f_X(x)\, dx.$$

Thus variance still averages squared distances from the mean.

**Example:** *Uniform on* $[a, b]$. Suppose $a < b$ and the density is constant on this interval and zero outside it. This is the CONTINUOUS UNIFORM DISTRIBUTION. Normalization gives the height: if it is $c$, then $c(b - a) = 1$. Hence

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

The mean is the midpoint:

$$\mathbb{E}[X] = \frac{1}{b-a} \int_a^b x\, dx = \frac{b^2 - a^2}{2(b-a)} = \boxed{\frac{a+b}{2}}.$$

To compute variance, write $m = (a + b)/2$ and $L = b - a$. The substitution $u = x - m$ centers the interval:

$$\mathrm{Var}(X) = \frac{1}{L} \int_{-L/2}^{L/2} u^2\, du = \frac{1}{L} \left[ \frac{u^3}{3} \right]_{-L/2}^{L/2}$$
$$= \frac{L^2}{12} = \boxed{\frac{(b-a)^2}{12}}.$$

## 8.3 Cumulative distribution functions

A CDF describes the probability to the left of a value, and works for both continuous and discrete variables.

**Definition 8.2 [Cumulative distribution function].** The cumulative distribution function, or CDF, of $X$ is

$$\boxed{F_X(x) = P(X \le x)}.$$

For a continuous variable with density $f_X$,

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

As $x$ increases, the event $\{X \le x\}$ grows, so the CDF cannot decrease. Its limits at $-\infty$ and $\infty$ are zero and one. For $a < b$,

$$P(a < X \le b) = F_X(b) - F_X(a).$$

This subtracts the probability already accumulated by $a$. For a continuous variable, endpoints do not matter. Where the density is continuous, differentiation recovers it: $F_X'(x) = f_X(x)$.

For the uniform distribution on $[a, b]$, the accumulated area gives

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$

The CDF is flat before the interval, increases linearly through it, and is flat again once all the probability has been included.

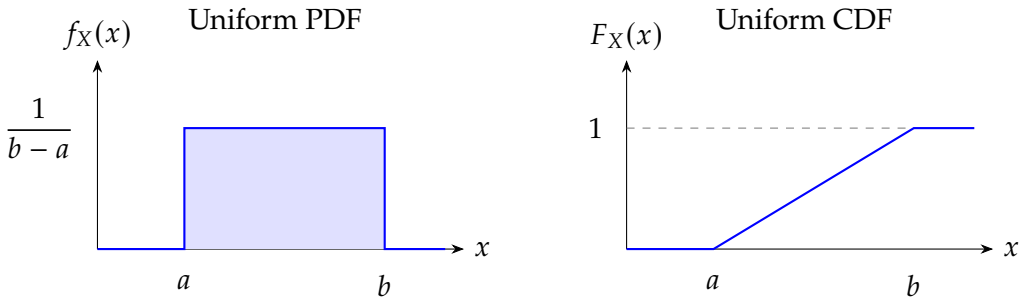



**Figure 22:** A constant uniform density produces a linear increase in the CDF. The density has total area one; the CDF reaches one at $b$. Horizontal spacing is schematic.

For a discrete variable, accumulate masses instead of areas:

$$F_X(x) = \sum_{k \le x} p_X(k),$$

where the sum runs over its possible values. Each positive mass produces a jump at that value.

**Example:** *Masses at 1, 2, and 4.* Let $p_X(1) = 1/6$, $p_X(2) = 3/6$, and $p_X(4) = 2/6$. These masses sum to one. The CDF is

$$F_X(x) = \begin{cases} 0, & x < 1, \\ 1/6, & 1 \le x < 2, \\ 4/6, & 2 \le x < 4, \\ 1, & x \ge 4. \end{cases}$$

At a mass location, $F_X$ includes that mass because its defining event uses $X \le x$.

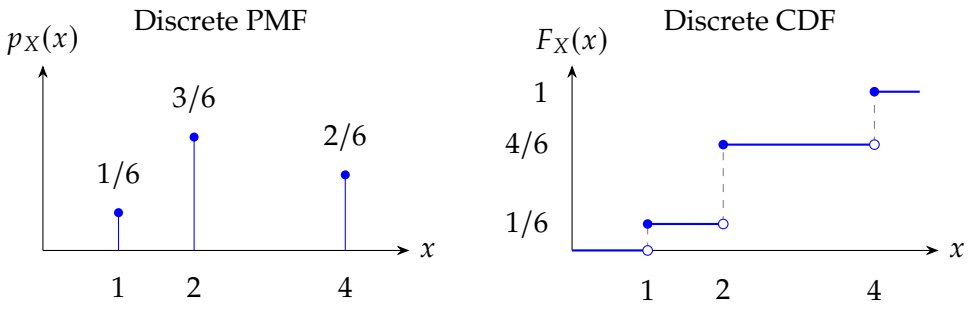



**Figure 23:** The jump sizes are $1/6$, $3/6$, and $2/6$. Filled circles show the CDF's value at each jump; open circles show the value approached from the left.

## 8.4 Mixed distributions: area and point mass

A distribution can have both a continuous part and point masses. Such a MIXED DISTRIBUTION is described by adding the two kinds of probability. Its CDF has continuous increases as well as jumps.

Consider probability $1/2$ spread uniformly over $[0, 1]$, with the other $1/2$ concentrated at $x = 1/2$. The continuous part has density height $1/2$ on $[0, 1]$, so its area is $1/2$. For a set $B$,

$$P(X \in B) = \frac{1}{2} \int_{B \cap [0,1]} 1 \, dx + \frac{1}{2} \mathbf{1}_{\{1/2 \in B\}},$$

where the indicator is one if $B$ contains $1/2$, and zero otherwise. The point mass is not an ordinary density height.

For $0 \le x < 1/2$, the CDF accumulates only the continuous area $x/2$. At $1/2$, it also includes the point mass. Thus

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x/2, & 0 \le x < 1/2, \\ x/2 + 1/2, & 1/2 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

Immediately before $1/2$, the accumulated probability approaches $1/4$. At $1/2$, it is $3/4$. The jump of $1/2$ is exactly the point mass. The remaining continuous area brings the CDF to one.

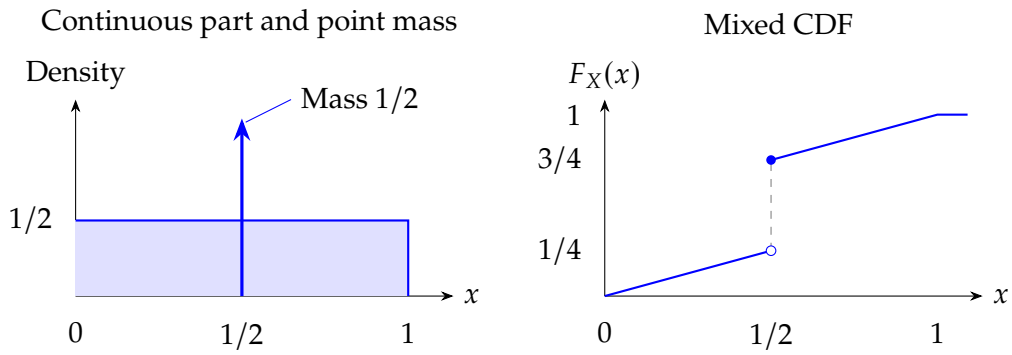



**Figure 24:** The arrow marks a point mass of $1/2$; its drawn height has no density meaning. The continuous part contributes area $1/2$. The CDF combines a linear increase with a jump of $1/2$.

## 8.5 Gaussian densities and their moments

The normal family has a symmetric, bell-shaped density. Its parameters specify the center and spread.

**Definition 8.3 [Standard normal distribution].** A standard normal random variable $Z$, written $Z \sim N(0,1)$, has density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad z \in \mathbb{R}.$$

Its CDF is denoted by

$$\Phi(z) = \int_{-\infty}^{z} \phi(t)\, dt.$$

The normalization constant makes the total area one. Since $\phi$ is even, $z\phi(z)$ is odd and its integral is zero. Therefore $\mathbb{E}[Z] = 0$. For the second moment, use $\phi'(z) = -z\phi(z)$ and integration by parts:

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \phi(z)\, dz = \Big[-z\phi(z)\Big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z)\, dz = 1.$$

The boundary term vanishes because $ze^{-z^2/2}$ tends to zero at both ends. Hence $\mathrm{Var}(Z) = 1$.

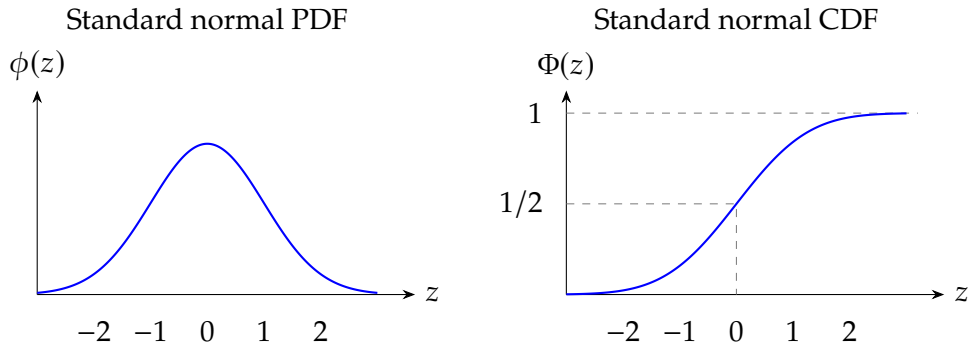



**Figure 25:** The standard normal density is symmetric about zero. Its CDF is smooth and increasing, with $\Phi(0) = 1/2$. Both curves are drawn from their defining formulas.

**Definition 8.4 [General normal distribution].** A normal, or Gaussian, random variable $X \sim N(\mu, \sigma^2)$, with $\sigma > 0$, has density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The substitution $z = (x - \mu)/\sigma$ changes $f_X(x)\,dx$ into $\phi(z)\,dz$. Thus

$$\mathbb{E}[X] = \mu + \sigma\mathbb{E}[Z] = \mu, \qquad \operatorname{Var}(X) = \sigma^2\mathbb{E}[Z^2] = \sigma^2.$$

The second parameter of $N(\mu, \sigma^2)$ is the variance; $\sigma$ is the standard deviation.

For $Y = aX + b$, the expectation and variance rules give

$$\mathbb{E}[Y] = a\mu + b, \qquad \operatorname{Var}(Y) = a^2\sigma^2.$$

When $a \neq 0$, the transformed variable is again normal:

$$\boxed{Y \sim N(a\mu + b, a^2\sigma^2)}.$$

Indeed, after centering and dividing by its positive standard deviation $|a|\sigma$,

$$\frac{Y - (a\mu + b)}{|a|\sigma} = \frac{a}{|a|}\,\frac{X - \mu}{\sigma}.$$

This is either $Z$ or $-Z$. The symmetric standard normal density gives them the same distribution. If $a = 0$, then $Y = b$ is constant with variance zero, so it has no nondegenerate normal density.

## 8.6 Normal probabilities by standardization

The normal CDF has no elementary antiderivative. Numerical values of $\Phi$ let us compute probabilities for every normal distribution. For $X \sim N(\mu, \sigma^2)$, STANDARDIZATION means

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Subtracting the mean centers the variable; dividing by its standard deviation expresses distance from the mean in units of spread. Because $\sigma > 0$, the direction of an inequality is preserved:

$$\boxed{P(X \le x) = \Phi\left(\frac{x - \mu}{\sigma}\right)}.$$

**Example:** *A normal variable with mean 2 and variance 16.* If $X \sim N(2, 16)$, then its standard deviation is 4. Hence

$$P(X \le 3) = P\left(\frac{X - 2}{4} \le \frac{3 - 2}{4}\right) = \Phi(0.25) \approx \boxed{0.5987}.$$

In the table, use row 0.2 and column 0.05: together they give the argument 0.25. The entry is accumulated probability to the left, not a density height.

**Table 1:** Standard normal CDF $\Phi(z)$, rounded to four decimal places. Add the row label and column label to obtain $z$. The bold entry is $\Phi(0.25)$.

| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | **0.5987** | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |

## 8.7 How the descriptions fit together

A PMF assigns mass to values; a PDF assigns density whose integral gives probability. Both lead to a CDF, a mean, and a variance. The notation also distinguishes a single-variable description from a joint or conditional one:

| Description | Discrete | Continuous |
|---|---|---|
| Single variable | $p_X(x)$ | $f_X(x)$ |
| Joint | $p_{X,Y}(x, y)$ | $f_{X,Y}(x, y)$ |
| Conditional | $p_{X\mid Y}(x \mid y)$ | $f_{X\mid Y}(x \mid y)$ |

The CDF notation $F_X(x) = P(X \le x)$ and the meanings of $\mathbb{E}[X]$ and $\mathrm{Var}(X)$ are shared by both columns.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 8, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 9: Multiple Continuous Random Variables

Integrating a joint density gives marginals. Normalizing a slice gives a conditional density.

## 9.1 Review: densities and expectations

The discrete and continuous descriptions use parallel notation:

| Description | Discrete | Continuous |
|---|---|---|
| Single variable | $p_X(x)$ | $f_X(x)$ |
| Joint | $p_{X,Y}(x,y)$ | $f_{X,Y}(x,y)$ |
| Given an event $A$ | $p_{X\|A}(x)$ | $f_{X\|A}(x)$ |
| Given a value $y$ | $p_{X\|Y}(x \mid y)$ | $f_{X\|Y}(x \mid y)$ |

Both descriptions share the CDF $F_X(x) = P(X \le x)$ and the meanings of expectation and variance.

For a continuous random variable $X$, probabilities are areas under a nonnegative density whose total integral is one:

$$P(a \le X \le b) = \int_a^b f_X(x)\,dx, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

The event collects all outcomes whose recorded value lies in $[a, b]$. Its probability becomes area over that interval in the value space.

For a short interval on which the density varies little,

$$P(x \le X \le x + \delta) \approx f_X(x)\delta.$$

Weighting the density by a function gives its expectation:

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx,$$

provided the corresponding absolute-value integral is finite. Taking $g(x) = x$ gives the mean; taking $g(x) = (x - \mathbb{E}[X])^2$ gives the variance when the second moment is finite.

## 9.2 Joint densities, marginals, and independence

For two variables, probability is spread over the plane of pairs $(x, y)$. A double integral collects the probability in a region.

**Definition 9.1 [Joint probability density].** A joint probability density function $f_{X,Y}$ is nonnegative and satisfies

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy\,dx = 1.$$

For a region $S$ over which the integral is defined,

$$\boxed{P((X,Y)\in S) = \iint_S f_{X,Y}(x,y)\,dx\,dy}.$$

The density height is not the probability of the pair. For a small square of side $\delta$, where the density varies little,

$$P(x \le X \le x+\delta,\ y \le Y \le y+\delta) \approx f_{X,Y}(x,y)\delta^2.$$

The factor $\delta^2$ is the square's area. Expectations similarly weight each pair by its joint density:

$$\mathbb{E}[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y)\,dy\,dx,$$

assuming the absolute-value double integral is finite.

To describe $X$ alone, allow every possible value of $Y$. The event $\{a \le X \le b\}$ is a vertical strip in the pair plane:

$$P(a \le X \le b) = \int_a^b \left(\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy\right) dx.$$

The quantity in parentheses therefore serves as the density of $X$. These single-variable densities are the marginal densities:

$$\boxed{f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy}, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx.$$

Marginalizing adds all contributions from the variable being removed. It does not select one value of that variable.

**Definition 9.2 [Independent continuous variables].** Variables $X, Y$ with a joint density are independent if their joint density factors as

$$\boxed{f_{X,Y}(x,y) = f_X(x) f_Y(y)}.$$

The equality may be taken throughout the plane; changing density values on boundaries of zero area does not change probabilities.

For independent variables, a rectangle probability separates into two interval probabilities:

$$P(a \le X \le b,\ c \le Y \le d) = \int_a^b \int_c^d f_X(x) f_Y(y)\, dy\, dx$$
$$= \left( \int_a^b f_X(x)\, dx \right) \left( \int_c^d f_Y(y)\, dy \right).$$

Thus information restricting one variable to an interval does not change the other's interval probabilities, when the conditioning event has positive probability.

## 9.3 Buffon's needle

**Example:** *A needle crossing parallel lines.* Parallel lines are spaced a distance $d > 0$ apart. A needle has length $\ell$, with $0 < \ell < d$. Find the probability that it intersects a line.

Use two variables to describe its placement. Let $X$ be the distance from its midpoint to the nearest line, so $0 \le X \le d/2$. Let $\Theta$ be its acute angle to the lines, with $0 \le \Theta \le \pi/2$. Model these variables as uniform and independent. Their densities and joint density are

$$f_X(x) = \frac{2}{d}, \qquad f_\Theta(\theta) = \frac{2}{\pi}, \qquad f_{X,\Theta}(x, \theta) = \frac{4}{\pi d}$$

on the rectangle $0 \le x \le d/2, 0 \le \theta \le \pi/2$, and zero outside it.

Half the needle has length $\ell/2$. Its component perpendicular to the lines is $(\ell/2) \sin \Theta$. The needle reaches the nearest line exactly when

$$\boxed{X \le \frac{\ell}{2} \sin \Theta}.$$

Since $\ell < d$, this threshold is always below $d/2$. If the needle reaches any line, it reaches a nearest one.

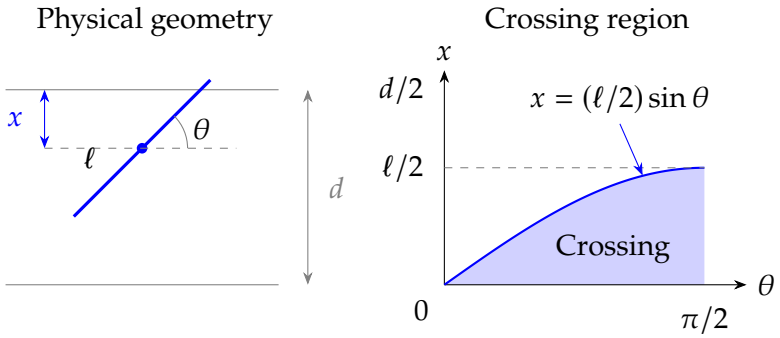



**Figure 26:** The distance $x$ is measured perpendicular to a nearest line; $\theta$ is measured from a parallel direction. In the value plane, crossing occurs below $x = (\ell/2) \sin \theta$. The geometry is schematic.

For each angle, integrate over the distances that permit crossing, then integrate over all angles:

$$\begin{aligned} P(\text{crossing}) &= \frac{4}{\pi d}\int_0^{\pi/2}\int_0^{(\ell/2)\sin\theta} 1\,dx\,d\theta \\ &= \frac{4}{\pi d}\int_0^{\pi/2}\frac{\ell}{2}\sin\theta\,d\theta \\ &= \frac{2\ell}{\pi d}\left[-\cos\theta\right]_0^{\pi/2} = \boxed{\frac{2\ell}{\pi d}}. \end{aligned}$$

The joint density is constant, so this integral multiplies the crossing region's area by its density.

## 9.4 Conditional densities: normalize a slice

For a continuous variable, $P(Y = y) = 0$. The ordinary event ratio cannot define conditioning on $Y = y$. Instead, use the density near that value.

To see the ratio, restrict $Y$ to a short interval $[y, y + \varepsilon]$. Near a point where the densities are continuous and $f_Y(y) > 0$,

$$\begin{aligned} &P(x \le X \le x + \delta \mid y \le Y \le y + \varepsilon) \\ &\quad\approx \frac{f_{X,Y}(x,y)\delta\varepsilon}{f_Y(y)\varepsilon} = \frac{f_{X,Y}(x,y)}{f_Y(y)}\delta. \end{aligned}$$

This suggests a density obtained by dividing a slice by its total area.

**Definition 9.3 [Conditional probability density].** For a value $y$ with $f_Y(y) > 0$, the conditional probability density function of $X$ given $Y = y$ is

$$\boxed{f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}}.$$

For fixed $y$, the numerator is a slice through the joint density. Its integral over $x$ is $f_Y(y)$, so division normalizes it:

$$\int_{-\infty}^{\infty} f_{X|Y}(x \mid y)\,dx = \frac{1}{f_Y(y)}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx = 1.$$

The conditional interval probability is the area under this density:

$$P(a \le X \le b \mid Y = y) = \int_a^b f_{X|Y}(x \mid y)\,dx.$$

Interchanging the roles of the variables gives $f_{Y|X}(y \mid x) = f_{X,Y}(x,y)/f_X(x)$ when $f_X(x) > 0$.

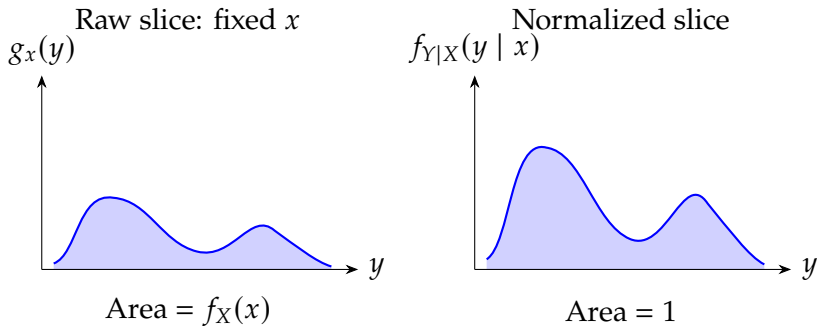



**Figure 27:** For fixed $x$, the slice $g_x(y) = f_{X,Y}(x, y)$ has area $f_X(x)$. Dividing by this area gives the conditional density of $Y$, with area one. Curves and vertical scales are schematic.

For independent variables and $f_Y(y) > 0$,

$$f_{X|Y}(x \mid y) = \frac{f_X(x) f_Y(y)}{f_Y(y)} = f_X(x).$$

The conditional density is unchanged by the observed value. For dependent variables, different slices can give different conditional densities.

## 9.5 Stick breaking: construct the joint density

**Example:** *Two successive breaks.* Start with a stick of length $\ell > 0$. Choose a first break point $X$ uniformly along $[0, \ell]$. Keep the part from 0 to $X$, and choose a second break point $Y$ uniformly along $[0, X]$, conditional on the first break. Both coordinates are measured from the same left endpoint.

The first density is $f_X(x) = 1/\ell$ for $0 < x < \ell$. Given $X = x > 0$, the second break is uniform on an interval of length $x$, so

$$f_{Y|X}(y \mid x) = \begin{cases} 1/x, & 0 < y < x, \\ 0, & \text{otherwise.} \end{cases}$$

The possibility $X = 0$ has probability zero. Density values at individual endpoints do not affect these distributions.

Rearrange the conditional-density definition to obtain the continuous multiplication rule:

$$f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y \mid x).$$

Consequently,

$$\boxed{f_{X,Y}(x, y) = \begin{cases} \dfrac{1}{\ell x}, & 0 < y < x < \ell, \\ 0, & \text{otherwise.} \end{cases}}$$

The SUPPORT, the region where the density can be positive, is the triangle below $y = x$ and to the left of $x = \ell$. The density is not constant on this triangle: smaller retained lengths give larger conditional density heights. Normalization checks the construction:

$$\int_0^\ell \int_0^x \frac{1}{\ell x}\, dy\, dx = \int_0^\ell \frac{1}{\ell}\, dx = 1.$$

For a fixed first break, the conditional mean averages $y$ over $[0, x]$:

$$\mathbb{E}[Y \mid X = x] = \int_0^x y \frac{1}{x}\, dy = \frac{1}{x}\left[\frac{y^2}{2}\right]_0^x = \boxed{\frac{x}{2}}, \qquad x > 0.$$

## 9.6 Stick breaking: marginal density and mean

For fixed $y$, integrate over first breaks $x \in [y, \ell]$.

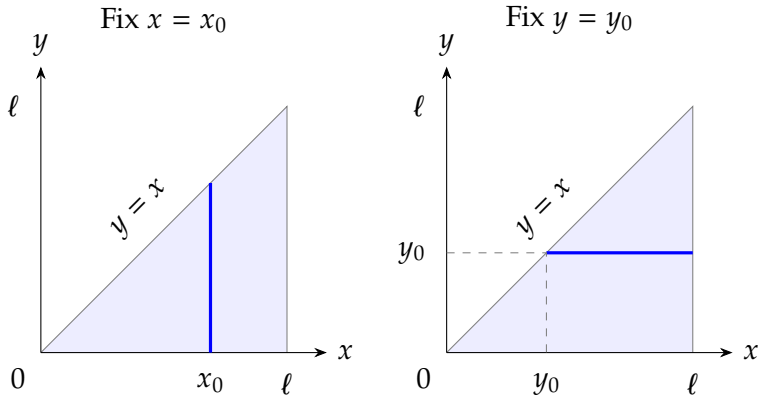



**Figure 28:** A vertical slice at $x_0$ allows $0 < y < x_0$. A horizontal slice at $y_0$ allows $y_0 < x < \ell$. These bounds determine the conditional and marginal integrals. The triangle shows support, not a uniform joint density.

For $0 < y < \ell$,

$$f_Y(y) = \int_y^\ell \frac{1}{\ell x}\, dx = \frac{1}{\ell}\left[\log x\right]_y^\ell$$
$$= \boxed{\frac{1}{\ell} \log \frac{\ell}{y}}.$$

Here log denotes the natural logarithm. Outside $[0, \ell]$, the density is zero. It grows without bound as $y$ approaches zero from the right, but its area is finite:

$$\int_0^\ell \frac{1}{\ell} \log \frac{\ell}{y}\, dy = \frac{1}{\ell}\left[y \log \frac{\ell}{y} + y\right]_0^\ell = 1,$$

using $y \log(\ell/y) \to 0$ as $y \downarrow 0$. This is a continuous distribution; the large density near zero does not create a point mass there.

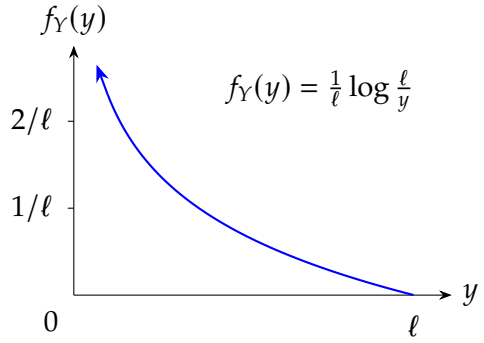



**Figure 29:** The second break's density decreases toward zero at $\ell$ and grows without bound near zero. The plotted portion follows $\ell f_Y(y) = \log(\ell/y)$; the upward arrow indicates the omitted continuation near zero.

Its mean is obtained by weighting this density by $y$:

$$\mathbb{E}[Y] = \frac{1}{\ell} \int_0^\ell y \log \frac{\ell}{y} \, dy.$$

Integration by parts uses $u = \log(\ell/y)$ and $dv = y \, dy$. Then $du = -dy/y$ and $v = y^2/2$, giving

$$\begin{aligned} \mathbb{E}[Y] &= \frac{1}{\ell} \left[ \frac{y^2}{2} \log \frac{\ell}{y} + \frac{y^2}{4} \right]_0^\ell \\ &= \frac{1}{\ell} \frac{\ell^2}{4} = \boxed{\frac{\ell}{4}}. \end{aligned}$$

The lower endpoint is a limit, with $y^2 \log(\ell/y) \to 0$ as $y \downarrow 0$. Although the second break is uniform given the first, it is not uniform over the original stick. Its available interval depends on $X$, so $X$ and $Y$ are dependent.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 9, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 10: Continuous Bayes' Rule; Derived Distributions

Bayes' rule updates an unknown's distribution after an observation. A derived distribution describes a function of variables whose law is known.

## 10.1 Review and the discrete Bayes rule

For discrete variables, probabilities are sums of masses. For continuous variables with densities, they are integrals of densities. The corresponding joint, marginal, and conditional formulas are

| | Discrete | Continuous |
|---|---|---|
| Joint | $p_{X,Y}(x,y)$ | $f_{X,Y}(x,y)$ |
| Marginal of $X$ | $\sum_y p_{X,Y}(x,y)$ | $\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$ |
| Conditional on $Y=y$ | $\frac{p_{X,Y}(x,y)}{p_Y(y)}$ | $\frac{f_{X,Y}(x,y)}{f_Y(y)}$ |

Use a conditional ratio only when its denominator is positive. Both descriptions have the CDF $F_X(x) = P(X \le x)$, expectation $\mathbb{E}[X]$, and variance $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$, when these moments are finite.

The prior distribution describes $X$ before observing $Y$. The posterior distribution describes $X$ given the observed value $Y = y$. To obtain it, write the joint law using the prior and the model for the observation, then normalize over $x$.

For discrete $X, Y$ and $p_Y(y) > 0$,

$$\boxed{p_{X|Y}(x \mid y) = \frac{p_X(x)p_{Y|X}(y \mid x)}{p_Y(y)}}, \qquad p_Y(y) = \sum_x p_X(x)p_{Y|X}(y \mid x).$$

The numerator is $p_{X,Y}(x,y)$ by the multiplication rule. The denominator adds this joint probability over every possible hidden value $x$, so the posterior masses sum to one.

**Example:** *Radar.* Let $X = 1$ mean that an airplane is present and $X = 0$ that it is absent. Let $Y = 1$ mean that the radar registers something and $Y = 0$ that it does not. For either observed value $y \in \{0, 1\}$,

$$P(X = 1 \mid Y = y) = \frac{p_X(1)p_{Y|X}(y \mid 1)}{p_X(0)p_{Y|X}(y \mid 0) + p_X(1)p_{Y|X}(y \mid 1)}.$$

The two denominator terms account for the same observation with an airplane absent or present.

## 10.2 Continuous and mixed Bayes rules

When both variables are continuous, the same calculation uses densities and replaces the sum over hidden values by an integral. For $f_Y(y) > 0$,

$$\boxed{f_{X|Y}(x \mid y) = \frac{f_X(x) f_{Y|X}(y \mid x)}{f_Y(y)}}, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_X(u) f_{Y|X}(y \mid u)\, du.$$

Here $u$ is an integration variable. The numerator is the joint density $f_{X,Y}(x, y)$; integrating it over $x$ gives the denominator. The resulting posterior density therefore integrates to one. The denominator is a density at $y$, rather than $P(Y = y)$.

**Example:** *A noisy continuous signal.* The unknown signal $X$ has prior density $f_X$. The observation $Y$ is a noisy version of it, with conditional density $f_{Y|X}(y \mid x)$ specifying the noise model. For a fixed observation $y$, multiply the prior by this conditional density and divide by its integral over $x$. This gives the distribution of the signal after the observation.

The unknown $X$ determines whether the posterior is a PMF or PDF and whether its normalizer is a sum or integral.

**Discrete $X$, continuous $Y$.** Each possible $x$ has mass $p_X(x)$, but the observation has a conditional density. For $f_Y(y) > 0$,

$$\boxed{p_{X|Y}(x \mid y) = \frac{p_X(x) f_{Y|X}(y \mid x)}{f_Y(y)}}, \qquad f_Y(y) = \sum_u p_X(u) f_{Y|X}(y \mid u).$$

For a discrete signal observed with continuous noise, the posterior assigns a probability mass to each possible signal value. The denominator adds the contributions of all possible signals to the observed density at $y$.

**Continuous $X$, discrete $Y$.** Now the prior is a density, while the observation model gives a conditional probability mass. For $p_Y(y) > 0$,

$$\boxed{f_{X|Y}(x \mid y) = \frac{f_X(x) p_{Y|X}(y \mid x)}{p_Y(y)}}, \qquad p_Y(y) = \int_{-\infty}^{\infty} f_X(u) p_{Y|X}(y \mid u)\, du.$$

A light beam's intensity $X$ can be continuous, while an observed photon count $Y$ is discrete. The model $p_{Y|X}(y \mid x)$ gives the probability of that count at intensity $x$. Multiplying by $f_X(x)$ and normalizing gives a posterior density for intensity. In each mixed case, summing or integrating the numerator over $x$ produces exactly the stated denominator.

## 10.3 Derived distributions and discrete transformations

**Definition 10.1 [Derived distribution].** A derived distribution is the probability law of a function of one or more random variables whose law is known. For a discrete output, describe this law by a PMF; for an output with a density, describe it by a PDF.

For example, suppose $(X, Y)$ has joint density one on the unit square and zero outside it. The ratio $g(X, Y) = Y/X$ is itself a random variable. Finding its PDF means deriving a distribution from the known joint law. Since $P(X = 0) = 0$, the value assigned to the ratio on that boundary does not affect its distribution.

If only an expectation is needed, a derived PDF is unnecessary:

$$\mathbb{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y)\, dy\, dx,$$

when the absolute-value integral is finite. This computes the expectation directly from the joint law.

For a discrete transformation $Z = g(X)$, collect every input value that produces the same output value $z$:

$$\boxed{p_Z(z) = P(g(X) = z) = \sum_{x:\, g(x)=z} p_X(x)}.$$

The events $\{X = x\}$ in this sum are disjoint. If several inputs map to one output, their masses must be added.

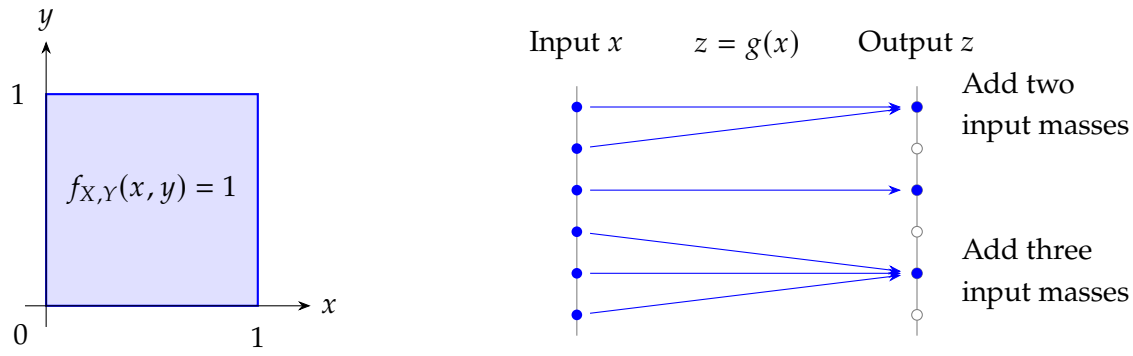



**Figure 30:** Left: a known joint law from which a function such as $Y/X$ derives a new distribution. Right: a schematic discrete map; inputs with the same output contribute to the same mass. Empty output points have zero mass.

## 10.4 Continuous transformations: start with the CDF

For $Z = g(X)$, first find the input values that make $g(X) \le z$. Their probability gives the CDF; differentiating gives a density.

$$\boxed{F_Z(z) = P(g(X) \le z)}, \qquad \boxed{f_Z(z) = \frac{d}{dz} F_Z(z)}.$$

If the output has a density, this procedure gives it, as in the examples below. Below the range of possible outputs, the CDF is zero; above that range, it is one.

**Example:** *Cubing a uniform variable.* Let $X$ be uniform on $[0, 2]$ and let $Y = X^3$. Find the density of $Y$.

Cubing is increasing on $[0, 2]$, with output support $[0, 8]$. For $0 \le y \le 8$, take a cube root without reversing the inequality:

$$F_Y(y) = P(X^3 \le y) = P(X \le y^{1/3})$$
$$= \int_0^{y^{1/3}} \frac{1}{2}\, dx = \frac{y^{1/3}}{2}.$$

Complete the CDF outside the support and differentiate on its interior:

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y^{1/3}/2, & 0 \le y \le 8, \\ 1, & y > 8. \end{cases} \qquad \boxed{f_Y(y) = \begin{cases} \dfrac{1}{6y^{2/3}}, & 0 < y < 8, \\ 0, & \text{otherwise.} \end{cases}}$$

The density grows without bound near zero, but its area is $\int_0^8 (6y^{2/3})^{-1}\, dy = [y^{1/3}/2]_0^8 = 1$. The continuous CDF has no jump at zero, so there is no point mass there. Individual endpoint density values do not affect the law.

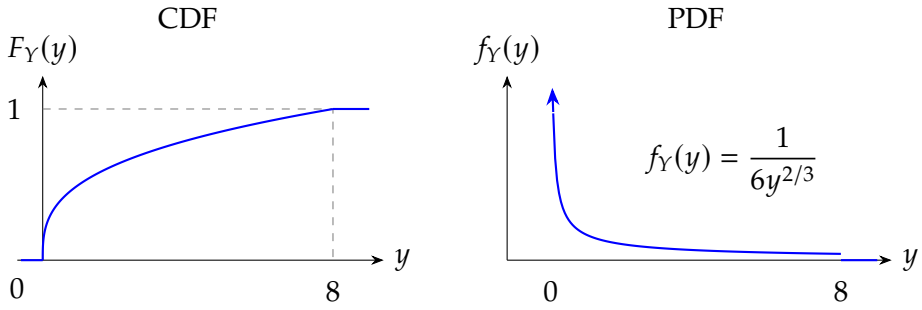



**Figure 31:** For $Y = X^3$, the CDF reaches one at 8. Its density decreases on $(0, 8)$ and grows without bound as $y \downarrow 0$. The upward arrow indicates the omitted continuation of the PDF.

## 10.5 Driving time: a decreasing transformation

**Example:** *Boston to New York.* Joan drives a distance of 200 miles at a random speed $V$ that is uniform between 30 and 60 miles per hour. Find the distribution of the trip duration $T = 200/V$.

The speed density is $f_V(v) = 1/30$ for $30 < v < 60$, and zero elsewhere. A greater speed gives a shorter trip. Thus the possible times, in hours, run from

$$t_{\min} = \frac{200}{60} = \frac{10}{3} \quad \text{to} \quad t_{\max} = \frac{200}{30} = \frac{20}{3}.$$

For $10/3 \le t \le 20/3$, the event $T \le t$ requires a speed *at least* $200/t$. Since $V, t$ are positive,

$$\begin{aligned} F_T(t) &= P(200/V \le t) = P(V \ge 200/t) \\ &= \int_{200/t}^{60} \frac{1}{30}\, dv = \frac{60 - 200/t}{30} = 2 - \frac{20}{3t}. \end{aligned}$$

The full CDF and its density are therefore

$$F_T(t) = \begin{cases} 0, & t < 10/3, \\ 2 - \dfrac{20}{3t}, & 10/3 \le t \le 20/3, \\ 1, & t > 20/3, \end{cases} \qquad \boxed{f_T(t) = \begin{cases} \dfrac{20}{3t^2}, & 10/3 < t < 20/3, \\ 0, & \text{otherwise.} \end{cases}}$$

The CDF equals zero and one at the two support endpoints. The density is positive on the interior, and it integrates to one:

$$\int_{10/3}^{20/3} \frac{20}{3t^2}\, dt = \left[ -\frac{20}{3t} \right]_{10/3}^{20/3} = 1.$$

Uniform speed does not give uniform time. The reciprocal map changes interval widths, yielding the decreasing density above.

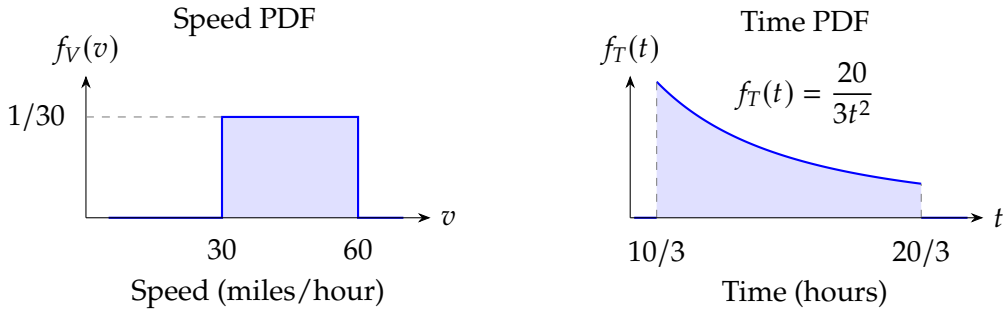



**Figure 32:** The speed PDF is flat, while the time PDF decreases across its support. The horizontal axes use different units; each density has total area one.

## 10.6 Affine transformations and normal variables

Scaling by $a$ changes interval widths by $|a|$; adding $b$ shifts their locations. The density must adjust its height to preserve probability.

**Proposition 10.2 [Density under an affine map].** *Let $X$ have a density and let $Y = aX + b$, where $a \neq 0$. Then*

$$\boxed{f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right).}$$

*Proof.* Invert the inequality in the CDF, treating the sign of $a$ explicitly. If $a > 0$,

$$F_Y(y) = P\left(X \leq \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right).$$

Differentiating gives $f_Y(y) = f_X((y-b)/a)/a$. If $a < 0$, the inequality reverses. Since $X$ has no point masses,

$$F_Y(y) = P\left(X \geq \frac{y-b}{a}\right) = 1 - F_X\left(\frac{y-b}{a}\right).$$

Differentiating gives $f_Y(y) = -f_X((y-b)/a)/a$. In either case, the positive scaling factor is $1/|a|$. □

For $Y = 2X + 5$, first stretch the horizontal axis by two and halve the density height; then shift the stretched density five units to the right. In particular,

$$f_Y(y) = \frac{1}{2} f_X\left(\frac{y-5}{2}\right).$$

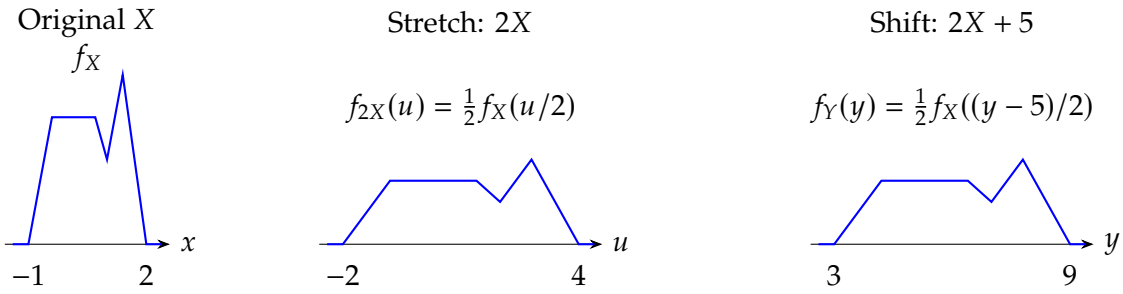



**Figure 33:** A schematic density with support $[-1, 2]$ is stretched to support $[-2, 4]$, then shifted to $[3, 9]$. The last two panels have half the original height. All panels use the same horizontal and vertical scales.

**Example:** *A transformed normal variable.* Let $X$ be normal with mean $\mu$ and variance $\sigma^2 > 0$. For $a \neq 0$, substituting its density into the affine formula gives

$$\begin{aligned} f_Y(y) &= \frac{1}{|a|\sigma\sqrt{2\pi}} \exp\left(-\frac{((y-b)/a-\mu)^2}{2\sigma^2}\right) \\ &= \frac{1}{\sqrt{a^2\sigma^2}\sqrt{2\pi}} \exp\left(-\frac{(y-(a\mu+b))^2}{2a^2\sigma^2}\right). \end{aligned}$$

This is the normal density with mean $a\mu + b$ and variance $a^2\sigma^2$. Thus $Y \sim N(a\mu + b, a^2\sigma^2)$. If $a = 0$, then $Y = b$ with probability one and has no ordinary PDF.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 10, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 11: Derived Distributions; Convolution; Covariance and Correlation

Find a derived law by describing its CDF event. Independent sums use convolution. Covariance and correlation describe how variables vary together.

## 11.1 A ratio from the unit square

**Example:** *Find the density of* $Z = Y/X$. The joint density of $(X, Y)$ is one on the unit square and zero elsewhere.

Ignoring the boundary $X = 0$, which has probability zero, $Z \le z$ means $Y \le zX$. Its probability is the area below $y = zx$ in the square. For $z < 0$, this area is zero.

For $0 \le z \le 1$, the line stays below the square's top edge:

$$F_Z(z) = \int_0^1 \int_0^{zx} 1 \, dy \, dx = \int_0^1 zx \, dx = \frac{z}{2}.$$

For $z \ge 1$, the line reaches the top at $x = 1/z$. To its right, every $y$ in the square is allowed:

$$\begin{aligned} F_Z(z) &= \int_0^{1/z} zx \, dx + \int_{1/z}^1 1 \, dx \\ &= \frac{1}{2z} + 1 - \frac{1}{z} = 1 - \frac{1}{2z}. \end{aligned}$$

Complete the CDF and differentiate on its two positive intervals:

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ z/2, & 0 \le z \le 1, \\ 1 - 1/(2z), & z > 1, \end{cases} \qquad \boxed{f_Z(z) = \begin{cases} 1/2, & 0 < z \le 1, \\ 1/(2z^2), & z > 1, \\ 0, & \text{otherwise.} \end{cases}}$$

Both CDF formulas give $1/2$ at $z = 1$, and $F_Z(z) \to 1$ as $z \to \infty$. The density integrates to $\int_0^1 (1/2)\, dz + \int_1^\infty (1/(2z^2))\, dz = 1$. Values assigned to the PDF at 0 and 1 do not change the law.

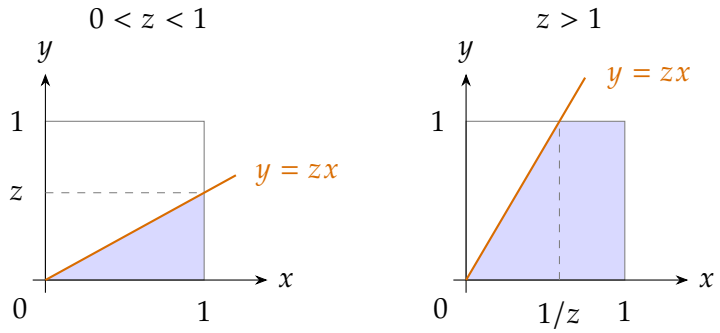



**Figure 34:** The shaded area is $P(Y/X \le z)$. For $0 < z < 1$, it is a triangle. For $z > 1$, it includes a full strip to the right of $x = 1/z$. Slopes represent the two regimes schematically.

## 11.2 A density formula for a monotone transformation

A STRICTLY MONOTONE function is either strictly increasing or strictly decreasing, so each output has at most one input. For $Y = g(X)$, this lets us convert a density by accounting for the change in interval width.

For a short input interval of length $\delta$ near $x$, the output interval has length approximately $|g'(x)|\delta$. Both intervals describe the same event, hence have the same probability:

$$f_X(x)\delta \approx f_Y(g(x))\,|g'(x)|\delta.$$

An increasing map preserves the order of endpoints. A decreasing map reverses them, but its output interval still has positive width.

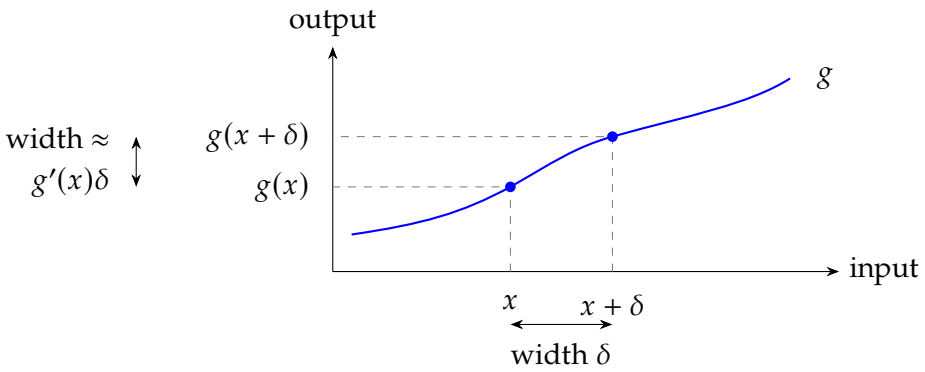



**Figure 35:** An increasing map sends an input interval of width $\delta$ to an output interval of width approximately $g'(x)\delta$. The corresponding probabilities agree. The curve and widths are schematic.

**Proposition 11.1 [MONOTONE CHANGE OF VARIABLE].** *Let $X$ have a density on an interval $I$. Suppose $g$ is continuously differentiable and strictly monotone there, with $g'(x) \neq 0$ in its interior. Writing $h = g^{-1}$, the*

*density of $Y = g(X)$, for $y$ in the interior of $g(I)$, is*

$$\boxed{f_Y(y) = f_X(h(y))\,|h'(y)| = \frac{f_X(h(y))}{|g'(h(y))|}}.$$

*It is zero outside the output range. Endpoint values are immaterial.*

*Proof.* Express the output CDF through the inverse map. If $g$ is increasing, $F_Y(y) = F_X(h(y))$, and differentiation gives $f_X(h(y))h'(y)$. If $g$ is decreasing, $F_Y(y) = 1 - F_X(h(y))$, because $X$ has no point masses. Differentiation gives $-f_X(h(y))h'(y)$. Thus the positive factor in either case is $|h'(y)|$. Differentiating $g(h(y)) = y$ gives $g'(h(y))h'(y) = 1$, which yields the second formula. □

The Jacobian preserves probability under the change in interval width; for $g(x) = ax + b$, it gives $1/|a|$.

### 11.3 Discrete convolution: the distribution of a sum

Let $W = X + Y$, with $X, Y$ independent and discrete. Each possible value $x$ contributes to $W = w$ exactly when $Y = w - x$. The events for different $x$ are disjoint, so

$$\begin{aligned} p_W(w) &= \sum_x P(X = x, Y = w - x) \\ &= \boxed{\sum_x p_X(x) p_Y(w - x)}. \end{aligned}$$

The second line uses independence. The sum ranges over the possible values of $X$; a term is zero when $w - x$ is not a possible value of $Y$.

**Definition:** The CONVOLUTION of two PMFs is the sum $\sum_x p_X(x)p_Y(w - x)$ as a function of $w$. It is the PMF of the sum of independent variables with those two PMFs.

To compute a convolution graphically, align the two PMFs on a common input axis, then carry out these steps:

1. Flip the PMF of $Y$, giving $p_Y(-x)$.
2. Shift it by $w$, giving $p_Y(w - x)$. A positive $w$ shifts the flipped graph to the right.
3. Multiply the two heights at each $x$.
4. Add all products to obtain $p_W(w)$.

Changing $w$ moves the flipped graph and produces another value of the output PMF. Independence supplies the product of masses; the constraint $x + y = w$ selects which masses to combine.

## 11.4 Continuous convolution

For independent continuous variables, condition on one summand. Given $X = x$, the sum $W = x + Y$ is a shifted version of $Y$. Independence makes the conditional law of $Y$ its original law, so

$$f_{W|X}(w \mid x) = f_Y(w - x).$$

Use the multiplication rule to form a joint density, then integrate over all possible values of $x$:

$$f_{W,X}(w, x) = f_X(x) f_{W|X}(w \mid x) = f_X(x) f_Y(w - x).$$

$$\boxed{f_W(w) = \int_{-\infty}^{\infty} f_X(x) f_Y(w - x)\, dx}.$$

This integral is the continuous convolution. The flip, shift, and multiply steps are the same as for PMFs, with integration replacing addition.

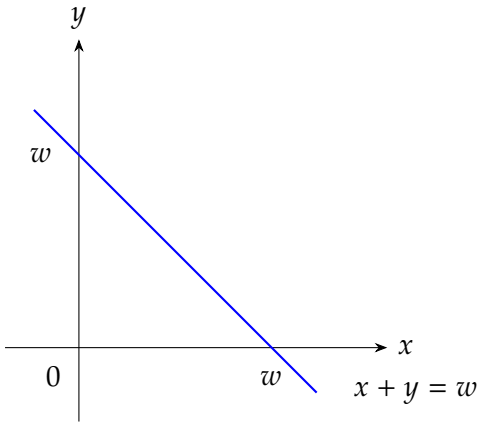



**Figure 36:** For a positive $w$, pairs producing that sum lie on $x + y = w$. The convolution density integrates $f_X(x) f_Y(w - x)$ over $x$. The line itself has probability zero for a joint density; its integral here produces a density in $w$, not a point probability.

The formula automatically respects support: only values of $x$ for which both $f_X(x)$ and $f_Y(w - x)$ can be nonzero contribute. Its total integral is one, since

$$\begin{aligned} \int_{-\infty}^{\infty} f_W(w)\, dw &= \int_{-\infty}^{\infty} f_X(x) \left( \int_{-\infty}^{\infty} f_Y(w - x)\, dw \right) dx \\ &= \int_{-\infty}^{\infty} f_X(x)\, dx = 1. \end{aligned}$$

For each fixed $x$, the inner integral is a translation of the integral of $f_Y$, so it equals one.

## 11.5 The joint density of independent normal variables

Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ be independent, with $\sigma_X, \sigma_Y > 0$. Multiplying their densities gives

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y} \exp\left(-\frac{(x-\mu_X)^2}{2\sigma_X^2} - \frac{(y-\mu_Y)^2}{2\sigma_Y^2}\right).$$

The density has the same height wherever

$$\frac{(x-\mu_X)^2}{2\sigma_X^2} + \frac{(y-\mu_Y)^2}{2\sigma_Y^2} = c$$

for a fixed $c > 0$. These CONTOURS, curves on which the density is constant, are ellipses centered at $(\mu_X, \mu_Y)$. Their horizontal and vertical half-widths are $\sqrt{2c}\,\sigma_X$ and $\sqrt{2c}\,\sigma_Y$. If $\sigma_X = \sigma_Y$, the ellipses are circles.

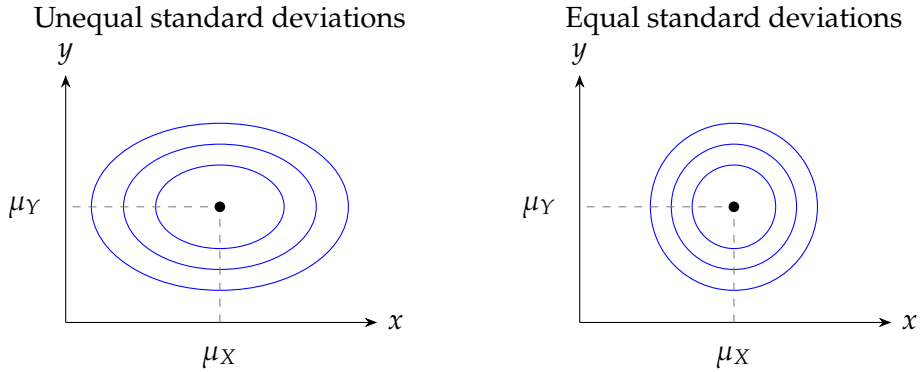



**Figure 37:** Equal-density contours of independent normal variables are centered at their means. Unequal standard deviations give ellipses; equal standard deviations give circles. Larger contours have a larger $c$ and therefore a lower density. Scales are schematic.

## 11.6 The sum of independent normal variables

**Proposition 11.2 [NORMAL CONVOLUTION].** *For the independent normal variables above,*

$$\boxed{X + Y \sim N(\mu_X + \mu_Y,\ \sigma_X^2 + \sigma_Y^2)}.$$

*Proof.* First take both means to be zero. Complete the square in the convolution integral, leaving a normal integral in $x$ and an exponential factor in $w$. Write $A = \sigma_X^2$, $B = \sigma_Y^2$, and

$S = A + B$. Then

$$f_W(w) = \frac{1}{2\pi\sqrt{AB}} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{2A} - \frac{(w-x)^2}{2B}\right) dx,$$

$$\frac{x^2}{A} + \frac{(w-x)^2}{B} = \frac{S}{AB}\left(x - \frac{A}{S}w\right)^2 + \frac{w^2}{S}.$$

The second identity follows by expanding both sides: the coefficient of $x^2$ is $S/(AB)$, the coefficient of $x$ is $-2w/B$, and the constant is $w^2/B$. Substitution gives

$$f_W(w) = \frac{e^{-w^2/(2S)}}{2\pi\sqrt{AB}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x - Aw/S)^2}{2AB/S}\right) dx.$$

The integrand is a normal density without its normalizing factor. Its integral is $\sqrt{2\pi AB/S}$, so

$$f_W(w) = \frac{1}{\sqrt{2\pi S}} \exp\left(-\frac{w^2}{2S}\right).$$

This is the density of $N(0, S)$.
For nonzero means, $X - \mu_X$ and $Y - \mu_Y$ remain independent and are zero-mean normal variables with variances $A, B$. Their sum is $N(0, S)$. Adding $\mu_X + \mu_Y$ shifts its mean and leaves its variance unchanged, proving the stated result. □

Normality of the sum follows from the convolution calculation. Its mean and variance alone would not establish its distribution.

## 11.7 Covariance and the variance of a sum

Subtract each mean to record deviations. Their product is positive for matching signs and negative for opposite signs.

**Definition 11.3 [Covariance].** For variables with finite second moments, their covariance is

$$\boxed{\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}.$$

Covariance averages the products of the centered values. When both means are zero, it reduces to $\mathbb{E}[XY]$. Expanding the centered product gives a useful equivalent formula:

$$\begin{aligned}\mathrm{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y] \\ &= \boxed{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}.\end{aligned}$$

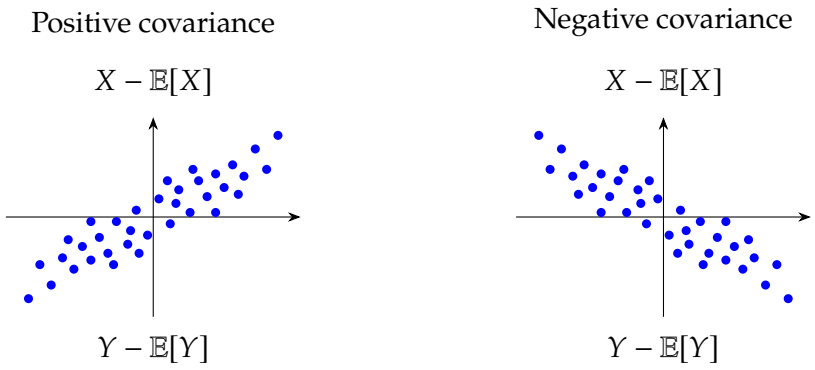



**Figure 38:** Schematic centered clouds show positive and negative covariance. Products of deviations tend to be positive in the left drawing and negative in the right. The dots illustrate the geometry and are not measured data.

To expand a sum's variance, put $D_i = X_i - \mathbb{E}[X_i]$. Linearity of expectation gives

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \mathbb{E}\left[\left(\sum_{i=1}^{n} D_i\right)^2\right]$$
$$= \boxed{\sum_{i=1}^{n} \operatorname{Var}(X_i) + 2 \sum_{1 \le i < j \le n} \operatorname{Cov}(X_i, X_j)}.$$

The factor two appears because each unordered pair occurs twice in the square. With a sum over all ordered pairs $i \neq j$, the same formula uses a factor one instead.

If $X, Y$ are independent, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, so $\operatorname{Cov}(X, Y) = 0$. Consequently, pairwise independent summands have additive variances. Zero covariance does not imply independence: it constrains one expected product, rather than the whole joint law.

## 11.8 Correlation: covariance on a common scale

Covariance carries the product of the variables' units. Divide each centered variable by its standard deviation to compare their variation without units.

**Definition 11.4 [Correlation coefficient].** For finite, positive variances, write $\sigma_X = \sqrt{\operatorname{Var}(X)}$ and $\sigma_Y = \sqrt{\operatorname{Var}(Y)}$. The correlation coefficient is

$$\boxed{\rho = \mathbb{E}\left[\frac{X - \mathbb{E}[X]}{\sigma_X} \frac{Y - \mathbb{E}[Y]}{\sigma_Y}\right] = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}}.$$

Let $U = (X - \mathbb{E}[X])/\sigma_X$ and $V = (Y - \mathbb{E}[Y])/\sigma_Y$. Both have mean zero and second moment one; $\mathbb{E}[UV] = \rho$. The nonnegativity of squared differences and sums gives

$$0 \le \mathbb{E}[(U - V)^2] = 2 - 2\rho, \qquad 0 \le \mathbb{E}[(U + V)^2] = 2 + 2\rho.$$

The first inequality gives $\rho \le 1$ and the second gives $\rho \ge -1$. Therefore

$$\boxed{-1 \le \rho \le 1}.$$

If $\rho = 1$, then $\mathbb{E}[(U - V)^2] = 0$, so $U = V$ with probability one. If $\rho = -1$, the same reasoning gives $U = -V$ with probability one. Rewriting these relations yields

$$X - \mathbb{E}[X] = \pm\frac{\sigma_X}{\sigma_Y}(Y - \mathbb{E}[Y]).$$

Conversely, suppose $X - \mathbb{E}[X] = c(Y - \mathbb{E}[Y])$ with probability one for a constant $c \neq 0$. Then $\sigma_X = |c|\sigma_Y$ and $\mathrm{Cov}(X, Y) = c\sigma_Y^2$, so $\rho = c/|c|$. Thus $|\rho| = 1$ holds exactly when the centered variables are linearly related with probability one.

Variables with zero covariance, or equivalently zero correlation when both variances are positive, are UNCORRELATED. Independence implies $\rho = 0$, but the converse is false. If either variance is zero, the stated correlation coefficient is undefined because its denominator vanishes.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 11, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 12: Iterated Expectations; Sum of a Random Number of Random Variables

Compute conditional moments first, then average over the conditioning variable. This handles both mixtures and sums with a random number of terms.

## 12.1 Conditional expectation as a random variable

For a fixed value $y$, the conditional mean is a number computed from the conditional distribution:

$$m(y) = \mathbb{E}[X \mid Y = y] = \sum_x x\, p_{X|Y}(x \mid y).$$

For a conditional density, replace the sum by $\int_{-\infty}^{\infty} x f_{X|Y}(x \mid y)\, dx$. These formulas apply where the conditional law is defined; assume $\mathbb{E}[|X|] < \infty$.

**Definition:** The conditional expectation as a random variable, $\mathbb{E}[X \mid Y]$, is $m(Y)$. Its value is $m(y)$ whenever $Y = y$. It varies because the value of $Y$ varies.

**Example:** *Break a stick twice.* Start with a stick of length $\ell > 0$. Choose the first break point $Y$ uniformly on $[0, \ell]$. Keep the piece from 0 to $Y$, then choose the second break point $X$ uniformly on that piece.

Given $Y = y > 0$, $X$ is uniform on $[0, y]$, so

$$\mathbb{E}[X \mid Y = y] = \int_0^y x \frac{1}{y}\, dx = \frac{y}{2}, \qquad \boxed{\mathbb{E}[X \mid Y] = \frac{Y}{2}}.$$

The event $Y = 0$ has probability zero, so its conditional value does not affect any average.

## 12.2 The law of iterated expectations

Average the conditional means using the probabilities of the conditions. This recovers the original mean.

**Theorem 12.1 [Law of iterated expectations].** *If $\mathbb{E}[|X|] < \infty$, then*

$$\boxed{\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]}.$$

*For discrete $Y$, the left side is $\sum_y \mathbb{E}[X \mid Y = y] p_Y(y)$. For continuous $Y$ with a density, it is $\int_{-\infty}^{\infty} \mathbb{E}[X \mid Y = y] f_Y(y)\, dy$.*

*Proof.* Multiply each conditional mean by the probability of its condition. In the discrete case, the multiplication rule gives

$$\begin{aligned}\mathbb{E}[\mathbb{E}[X \mid Y]] &= \sum_y \sum_x x\, p_{X|Y}(x \mid y) p_Y(y) \\ &= \sum_x x \sum_y p_{X,Y}(x, y) \\ &= \sum_x x\, p_X(x) = \mathbb{E}[X].\end{aligned}$$

The finite absolute expectation justifies rearranging these sums. For a joint density, the same steps use integrals and $f_{X|Y}(x \mid y) f_Y(y) = f_{X,Y}(x, y)$:

$$\int \int x f_{X,Y}(x, y)\, dx\, dy = \int x f_X(x)\, dx = \mathbb{E}[X].$$

□

For the stick, averaging the random conditional mean gives

$$\boxed{\mathbb{E}[X] = \mathbb{E}[Y/2] = \frac{1}{2}\frac{\ell}{2} = \frac{\ell}{4}}.$$

The first averaging step fixes the retained length; the second averages over that length. No marginal density of $X$ is needed.

## 12.3 Conditional variance and total variance

For a fixed condition, measure deviations from its own conditional mean, rather than from the overall mean.

**Definition 12.2 [Conditional variance].** Assume $\mathbb{E}[X^2] < \infty$. The conditional variance at $Y = y$ is

$$\operatorname{Var}(X \mid Y = y) = \mathbb{E}\big[(X - \mathbb{E}[X \mid Y = y])^2 \mid Y = y\big].$$

The random variable $\operatorname{Var}(X \mid Y)$ takes this value when $Y = y$. Equivalently,

$$\operatorname{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - (\mathbb{E}[X \mid Y])^2.$$

Total variance combines average within-condition variance with the variance of conditional means.

**Theorem 12.3 [Law of total variance].** *If* $\mathbb{E}[X^2] < \infty$, *then*

$$\boxed{\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y])}.$$

*Proof.* Use the second-moment formula on each term. Write $M = \mathbb{E}[X \mid Y]$. Iterated expectations applied to $X^2$ and to $X$ give

$$\begin{aligned}\mathbb{E}[\mathrm{Var}(X \mid Y)] &= \mathbb{E}[\mathbb{E}[X^2 \mid Y]] - \mathbb{E}[M^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[M^2], \\ \mathrm{Var}(M) &= \mathbb{E}[M^2] - (\mathbb{E}[M])^2 \\ &= \mathbb{E}[M^2] - (\mathbb{E}[X])^2.\end{aligned}$$

Adding cancels $\mathbb{E}[M^2]$ and leaves $\mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathrm{Var}(X)$. □

### 12.4 Variability within and between class sections

**Example:** Choose a student uniformly from two class sections. Section 1 has 10 students with mean score 90; section 2 has 20 students with mean score 60. Let $X$ be the chosen student's score and $Y$ the section number.

Let the scores be $x_1, \ldots, x_{30}$, with the first ten in section 1. The given section means are

$$\frac{1}{10}\sum_{i=1}^{10} x_i = 90, \qquad \frac{1}{20}\sum_{i=11}^{30} x_i = 60.$$

The section probabilities are $P(Y = 1) = 1/3$ and $P(Y = 2) = 2/3$. Thus

$$\mathbb{E}[X \mid Y] = \begin{cases} 90, & Y = 1, \\ 60, & Y = 2, \end{cases} \qquad \mathbb{E}[X] = \frac{1}{3}\cdot 90 + \frac{2}{3}\cdot 60 = 70.$$

This equals the direct average $(90 \cdot 10 + 60 \cdot 20)/30$. The variability of the conditional means is

$$\mathrm{Var}(\mathbb{E}[X \mid Y]) = \frac{1}{3}(90 - 70)^2 + \frac{2}{3}(60 - 70)^2 = 200.$$

Now suppose the within-section squared deviations satisfy

$$\frac{1}{10}\sum_{i=1}^{10}(x_i - 90)^2 = 10, \qquad \frac{1}{20}\sum_{i=11}^{30}(x_i - 60)^2 = 20.$$

Because a student is sampled uniformly within a section, these are its conditional variances.

| Section $y$ | $P(Y = y)$ | $\mathbb{E}[X \mid Y = y]$ | $\mathrm{Var}(X \mid Y = y)$ |
|---|---|---|---|
| 1 | 1/3 | 90 | 10 |
| 2 | 2/3 | 60 | 20 |

Average the conditional variances with the same section weights:

$$\mathbb{E}[\mathrm{Var}(X \mid Y)] = \frac{1}{3} \cdot 10 + \frac{2}{3} \cdot 20 = \frac{50}{3}.$$

Therefore

$$\mathrm{Var}(X) = \underbrace{\frac{50}{3}}_{\text{within sections}} + \underbrace{200}_{\text{between section means}} = \frac{650}{3}.$$

The average within-section variance omits the separation between the two means. Total variance restores that contribution.

## 12.5 A two-part density example

**Example:** Let $X$ have density $1/3$ on $(0, 1)$ and $2/3$ on $(1, 2)$, and zero elsewhere. Set $Y = 1$ on the first interval and $Y = 2$ on the second. Compute the conditional moments and total variance.

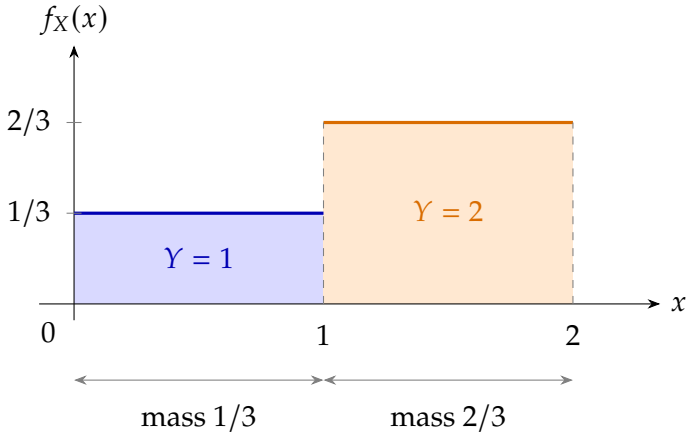



**Figure 39:** Each interval has width one. Its area is the probability of the corresponding value of $Y$: $1/3$ on the left and $2/3$ on the right. Conditioning divides the density by that area.

Integrating on the two intervals gives $P(Y = 1) = 1/3$ and $P(Y = 2) = 2/3$. After dividing by these probabilities, the conditional density is one on the corresponding interval. Thus the conditional laws are uniform, with

$$\mathbb{E}[X \mid Y = 1] = \int_0^1 x\,dx = \frac{1}{2}, \qquad \mathbb{E}[X \mid Y = 2] = \int_1^2 x\,dx = \frac{3}{2},$$

$$\mathrm{Var}(X \mid Y = 1) = \int_0^1 (x - \tfrac{1}{2})^2\,dx = \frac{1}{12}, \quad \mathrm{Var}(X \mid Y = 2) = \int_1^2 (x - \tfrac{3}{2})^2\,dx = \frac{1}{12}.$$

The overall mean and the variance of the conditional means are

$$\mathbb{E}[X] = \frac{1}{3}\frac{1}{2} + \frac{2}{3}\frac{3}{2} = \frac{7}{6},$$
$$\operatorname{Var}(\mathbb{E}[X \mid Y]) = \frac{1}{3}\left(\frac{1}{2} - \frac{7}{6}\right)^2 + \frac{2}{3}\left(\frac{3}{2} - \frac{7}{6}\right)^2 = \frac{2}{9}.$$

Both conditional variances equal $1/12$, so their average is $1/12$. Total variance gives

$$\boxed{\operatorname{Var}(X) = \frac{1}{12} + \frac{2}{9} = \frac{11}{36}}.$$

As a direct check, $\mathbb{E}[X^2] = \frac{1}{3}\int_0^1 x^2\,dx + \frac{2}{3}\int_1^2 x^2\,dx = 5/3$. Subtracting $(7/6)^2$ again gives $11/36$.

## 12.6 The mean of a sum with a random number of terms

**Example:** Visit a random number $N$ of stores, spending $X_i$ in store $i$. Assume $N$ is a nonnegative integer, the $X_i$ are independent and identically distributed, and the whole spending sequence is independent of $N$. The total spending is

$$Y = X_1 + \cdots + X_N = \sum_{i=1}^{N} X_i.$$

When $N = 0$, the empty sum is zero.

Write $\mu = \mathbb{E}[X_1]$ and $\sigma^2 = \operatorname{Var}(X_1)$ for the common mean and variance. Fixing $N = n$ converts the random-length sum into a sum of exactly $n$ terms. Independence from $N$ leaves the summands' joint law unchanged.

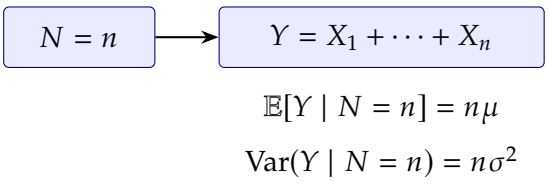



**Figure 40:** Condition on the number of visits first. For each nonnegative integer $n$, the total has mean $n\mu$ and variance $n\sigma^2$. Then average over the law of $N$. The row for $n$ represents any allowed count.

Assume $\mathbb{E}[N] < \infty$ and $\mathbb{E}[|X_1|] < \infty$. Linearity gives

$$\begin{aligned}\mathbb{E}[Y \mid N = n] &= \mathbb{E}[X_1 + \cdots + X_n \mid N = n] \\ &= \mathbb{E}[X_1 + \cdots + X_n] = n\mu.\end{aligned}$$

As a random conditional mean, this is $\mathbb{E}[Y \mid N] = N\mu$. Iterated expectations now yield

$$\boxed{\mathbb{E}[Y] = \mathbb{E}[N\mu] = \mathbb{E}[N]\,\mu}.$$

Expected total spending equals the expected number of visits times the common expected spending per visit.

## 12.7 The variance of a sum with a random number of terms

Assume also $\mathbb{E}[N^2] < \infty$ and $\mathbb{E}[X_1^2] < \infty$. Given $N = n$, independence of the summands makes their variances add:

$$\operatorname{Var}(Y \mid N = n) = \sum_{i=1}^{n} \operatorname{Var}(X_i) = n\sigma^2.$$

Thus $\operatorname{Var}(Y \mid N) = N\sigma^2$, and

$$\mathbb{E}[\operatorname{Var}(Y \mid N)] = \mathbb{E}[N]\,\sigma^2.$$

The conditional mean $N\mu$ varies with the number of visits:

$$\operatorname{Var}(\mathbb{E}[Y \mid N]) = \operatorname{Var}(N\mu) = \mu^2 \operatorname{Var}(N).$$

Apply total variance to combine these contributions:

$$\begin{aligned}\operatorname{Var}(Y) &= \mathbb{E}[\operatorname{Var}(Y \mid N)] + \operatorname{Var}(\mathbb{E}[Y \mid N]) \\ &= \boxed{\mathbb{E}[N]\,\sigma^2 + \mu^2 \operatorname{Var}(N)}.\end{aligned}$$

The first term is the average spending variability for a fixed number of visits. The second comes from changes in that number. If $N$ is fixed, its variance is zero and the formula reduces to the variance of a fixed-length independent sum.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 12, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 13: Bernoulli Process

A Bernoulli process is a sequence of independent successes and failures. Counts and interarrival times give two descriptions of the same sequence.

## 13.1 The Bernoulli process

**Definition 13.1 [Bernoulli process].** A Bernoulli process with parameter $p \in [0, 1]$ is a sequence $X_1, X_2, \ldots$ of independent Bernoulli random variables, each with

$$P(X_i = 1) = p, \qquad P(X_i = 0) = 1 - p.$$

The index $i$ is a trial number or a discrete time slot. A value one is a success, or an arrival; a value zero is a failure, or no arrival.

The success probability is the same at every trial. Independence means that knowing earlier outcomes does not change the law of later trials.

Lottery wins and losses, upward and downward movements of the Dow Jones, and arrivals or no arrivals at a bank or server can all be encoded by zeros and ones. A Bernoulli model for such a sequence assumes independent outcomes with a common success probability. For an arrival model, each slot records whether an arrival occurs.

## 13.2 Two views of a random process

A random process is a collection of random variables indexed by time. One view follows the individual variables. For every fixed time $t$, $X_t$ has a Bernoulli distribution, so

$$\mathbb{E}[X_t] = p, \qquad \mathrm{Var}(X_t) = \mathbb{E}[X_t^2] - (\mathbb{E}[X_t])^2 = p - p^2 = p(1 - p).$$

Here $X_t^2 = X_t$, since both possible values are zero or one.

A second view treats an entire sequence as an outcome. The sample space is

$$\Omega = \{(x_1, x_2, \ldots) : x_i \in \{0, 1\}\}.$$

An individual sequence is a sample path. The random variable $X_t$ reads its $t$th coordinate. Independence specifies the probabilities of finite blocks: a particular block of length $n$ containing $s$ ones has probability $p^s(1 - p)^{n-s}$.

For $0 \le p < 1$, the probability of an all-success infinite path is zero. Indeed, the event of success at every time is contained in the event of success in the first $n$ trials, whose probability is $p^n$. Since this tends to zero,

$$\boxed{P(X_t = 1 \text{ for every } t) = 0 \quad (p < 1)}.$$

If $p = 1$, every trial succeeds and this probability is one. An infinite path is therefore different from a finite block.

Bernoulli processes model memoryless arrivals in discrete time; Poisson processes use continuous time. Markov chains allow dependence across time. The calculations here concern the Bernoulli model.

## 13.3 The number of successes in a fixed interval

Let $S = X_1 + \cdots + X_n$ be the number of successes in $n$ trials. To have $S = k$, choose which $k$ slots succeed. Each choice has probability $p^k(1-p)^{n-k}$, and the choices are disjoint. Consequently,

$$\boxed{P(S = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n}.$$

This is the binomial distribution. Its PMF is zero outside the listed integers. For the endpoint parameters, $p = 0$ gives $S = 0$ and $p = 1$ gives $S = n$ with probability one.

Linearity of expectation and independence give the moments:

$$\mathbb{E}[S] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \boxed{np},$$

$$\operatorname{Var}(S) = \sum_{i=1}^{n} \operatorname{Var}(X_i) = \boxed{np(1-p)}.$$

The mean adds without needing independence. The variance adds here because the trials are independent.

Counting successes fixes the number of trials and lets the count vary. Waiting for a success reverses this: the target count is fixed, while the number of trials varies.

## 13.4 The waiting time to the first arrival

Assume $p > 0$. Let $T_1$ be the number of trials up to and including the first success. Its smallest possible value is one. The event $T_1 = t$ requires $t - 1$ failures followed by a success:

$$\boxed{P(T_1 = t) = (1-p)^{t-1} p, \qquad t = 1, 2, \ldots}.$$

Thus $T_1$ has a geometric distribution on the positive integers. The probability of waiting more than $s$ trials is

$$P(T_1 > s) = (1-p)^s, \qquad s = 0, 1, 2, \ldots.$$

This tends to zero, so the first success occurs in finite time with probability one. If $p = 0$, no success ever occurs.

**Proposition 13.2 [Geometric waiting-time moments].** *For $0 < p \le 1$,*

$$\boxed{\mathbb{E}[T_1] = \frac{1}{p}, \qquad \operatorname{Var}(T_1) = \frac{1-p}{p^2}}.$$

*Proof.* Compute the first two moments by differentiating the geometric series. Put $r = 1 - p$. For $0 < r < 1$,

$$\sum_{t=0}^{\infty} r^t = \frac{1}{1-r}, \qquad \sum_{t=1}^{\infty} t r^{t-1} = \frac{1}{(1-r)^2}, \qquad \sum_{t=2}^{\infty} t(t-1) r^{t-2} = \frac{2}{(1-r)^3}.$$

Multiplying the second series by $p$ gives $\mathbb{E}[T_1] = p/(1-r)^2 = 1/p$. Multiplying the third by $pr$ gives

$$\mathbb{E}[T_1(T_1 - 1)] = \frac{2pr}{(1-r)^3} = \frac{2r}{p^2}.$$

Since $T_1^2 = T_1(T_1 - 1) + T_1$,

$$\operatorname{Var}(T_1) = \frac{2r}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{r}{p^2}.$$

For $p = 1$, $T_1 = 1$ with probability one, giving mean one and variance zero as claimed. □

## 13.5 Memorylessness and a string of lottery losses

A waiting time is memoryless when its remaining wait, conditional on having waited so far, has its original law. For geometric waiting, past failures do not alter the future trials.

**Proposition 13.3 [Memoryless property].** *For $0 < p < 1$ and nonnegative integers $s, t$,*

$$\boxed{P(T_1 > s + t \mid T_1 > s) = P(T_1 > t) = (1-p)^t}.$$

*Equivalently, conditional on $T_1 > s$, the remaining wait $T_1 - s$ has the same distribution as $T_1$.*

*Proof.* The event $T_1 > s + t$ implies $T_1 > s$. Divide their probabilities:

$$P(T_1 > s + t \mid T_1 > s) = \frac{(1-p)^{s+t}}{(1-p)^s} = (1-p)^t.$$

The conditioning probability is positive for $p < 1$. □

**Example:** Buy one lottery ticket every day, with independent wins of probability $p > 0$. Find the length of the initial string of losing days before the first win.

If $L$ is that length, then $L = T_1 - 1$. Unlike $T_1$, it can equal zero: the first ticket may win. For $\ell = 0, 1, 2, \ldots$,

$$\boxed{P(L = \ell) = (1-p)^\ell p}.$$

The power counts the losing tickets; the last factor accounts for the win that ends the string. Counting the final winning day would instead give $T_1$, supported on $1, 2, \ldots$. For $p = 1$, the string has length zero with probability one.

## 13.6 Interarrival times and the time of the kth success

Let $T_2$ be the number of additional trials from the first success to the second, and define subsequent gaps $T_3, T_4, \ldots$ in the same way. The event $T_1 = t$ only fixes the first $t$ trials. Future trials remain independent with success probability $p$. Hence

$$P(T_2 = s \mid T_1 = t) = (1-p)^{s-1}p, \qquad s = 1, 2, \ldots.$$

This conditional law does not depend on $t$, so $T_2$ is independent of $T_1$ and has the same geometric law.

More generally, prescribing gaps $t_1, \ldots, t_k$ prescribes $t_i - 1$ failures and one success in each block. Independence of the underlying trials gives

$$P(T_1 = t_1, \ldots, T_k = t_k) = \prod_{i=1}^{k} (1-p)^{t_i - 1}p.$$

Thus the INTERARRIVAL TIMES $T_1, T_2, \ldots$ are independent and identically distributed geometric variables.

Let $Y_k$ be the trial number of the $k$th success. Then

$$Y_k = T_1 + \cdots + T_k, \qquad \boxed{\mathbb{E}[Y_k] = \frac{k}{p}, \quad \operatorname{Var}(Y_k) = \frac{k(1-p)}{p^2}}.$$

The mean and variance add because the gaps are independent.

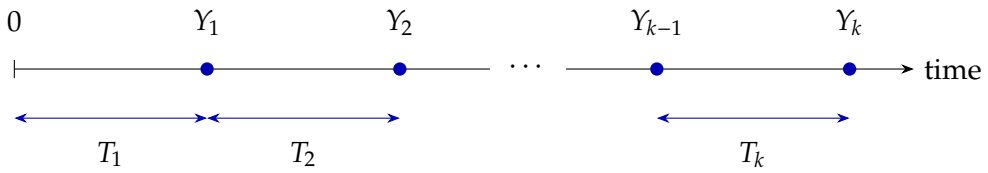



**Figure 41:** Arrival times $Y_i$ accumulate the successive gaps $T_i$. Every gap includes the trial on which its ending success occurs. The spacing is schematic; the dots omit intermediate arrivals.

To have $Y_k = t$, the first $t-1$ trials must contain exactly $k-1$ successes, and trial $t$ must succeed. Therefore

$$\boxed{P(Y_k = t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}, \qquad t = k, k+1, \ldots}.$$

The probability is zero for $t < k$. For $k = 1$, this reduces to the first-arrival PMF. If $p = 1$, the $k$th success occurs at $k$.

## 13.7 Splitting a Bernoulli process

Start with an arrival process of parameter $p$. Whenever an arrival occurs, route it to a kept stream with probability $q$ and to a discarded stream with probability $1-q$. The routing choices are independent of the arrivals and of each other.

Represent the choice in slot $i$ by an independent Bernoulli($q$) variable $C_i$, independent of all $X_j$. The two output indicators are

$$K_i = X_i C_i, \qquad D_i = X_i(1 - C_i).$$

A kept arrival requires both an original arrival and the keep choice. A discarded arrival requires the other choice, so

$$\boxed{P(K_i = 1) = pq, \qquad P(D_i = 1) = p(1-q)}.$$

For different slots, the pairs $(X_i, C_i)$ are independent. Applying the routing rule to each pair preserves independence across slots. Each output is therefore a Bernoulli process, with parameters $pq$ and $p(1-q)$, respectively.

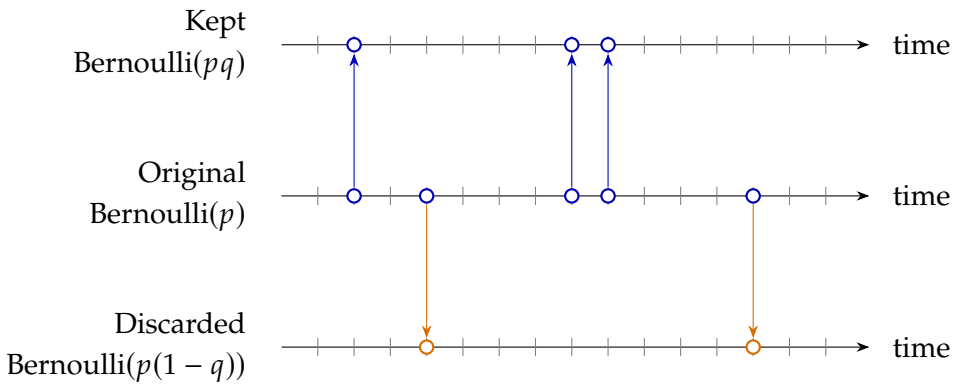



**Figure 42:** Each original arrival enters exactly one output stream, in the same slot. Arrows show the routing choices in a schematic path. Independent routing produces a Bernoulli law for each output stream.

The two output streams are generally dependent: they cannot both have an arrival in the same slot. Independence across time within each stream does not make the two streams independent of one another.

## 13.8 Merging independent Bernoulli processes

Let $A_i$ and $B_i$ be the indicators of two independent Bernoulli processes, with parameters $p$ and $q$. Record one merged arrival if either process has an arrival in that slot. Even when both arrive, the merged slot records just one.

The merged indicator is $M_i = \max\{A_i, B_i\}$. It is zero exactly when both original indicators are zero. Independence gives

$$\begin{aligned} P(M_i = 1) &= 1 - P(A_i = 0, B_i = 0) \\ &= 1 - (1-p)(1-q) = \boxed{p + q - pq}. \end{aligned}$$

Equivalently, add the two arrival probabilities and subtract the overlap $pq$, which was counted twice.

The pairs $(A_i, B_i)$ are independent across slots. The merged indicators depend on separate pairs, so they too are independent. Thus the merged process is Bernoulli with parameter $p + q - pq$.

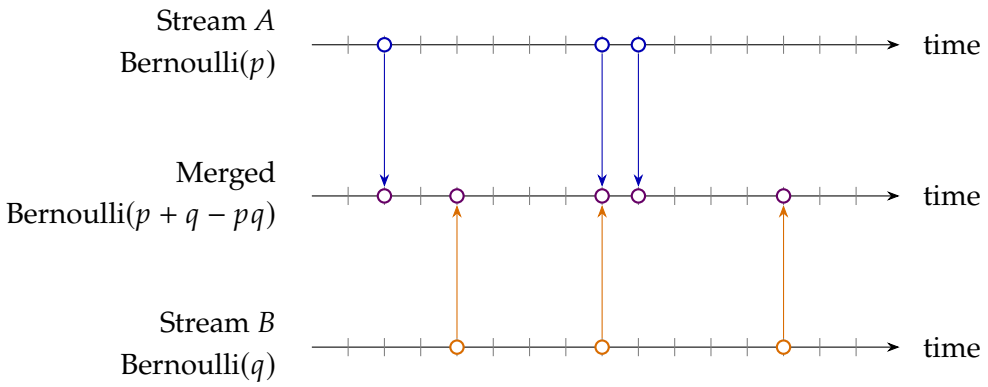



**Figure 43:** The merged stream contains the union of the original arrivals. At the shared slot, two incoming arrows meet one merged marker: a collision counts as one arrival. The paths are schematic.

$A_i + B_i$ counts collisions twice; $M_i$ records whether either stream arrives.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 13, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 14: Poisson Process - I

A Poisson process models continuous-time arrivals with a constant rate and independent counts on disjoint intervals.

## 14.1 From Bernoulli trials to continuous arrivals

For a Bernoulli process with success probability $p$, the number of arrivals in $n$ slots is binomial. The wait to the first arrival is geometric, and the trial number of the $k$th arrival has a Pascal distribution, the law of a sum of $k$ independent geometric waiting times. The relevant PMFs are

$$\begin{aligned} P(S_n = j) &= \binom{n}{j} p^j (1-p)^{n-j}, && j = 0, \dots, n, \\ P(T_1 = m) &= (1-p)^{m-1} p, && m = 1, 2, \dots, \\ P(Y_k = m) &= \binom{m-1}{k-1} p^k (1-p)^{m-k}, && m = k, k+1, \dots. \end{aligned}$$

For $0 < p < 1$, geometric waiting is memoryless: $P(T_1 > s + u \mid T_1 > s) = (1-p)^u$ for nonnegative integers $s, u$. Past failures do not change the remaining wait.

To move to continuous time, divide time into slots of width $\delta$. A rate $\lambda$ suggests a success probability near $\lambda\delta$ in each short slot. As the slots become finer, the arrival rate stays fixed while their individual success probabilities shrink. This leads to Poisson counts and continuous waiting times.

## 14.2 Defining a Poisson process

Let $N_t$ be the number of arrivals in $(0, t]$, with $N_0 = 0$. For $0 \le a < b$, the count in $(a, b]$ is $N_b - N_a$. We describe the process using the laws of these interval counts.

**Definition 14.1 [Poisson process].** A Poisson process of rate $\lambda > 0$ is an arrival process with the following properties.

1. Time homogeneity: the probability of $j$ arrivals in an interval depends only on its duration $\tau$. Write it as $P(j, \tau)$.
2. Independent increments: counts in disjoint time intervals are independent.

3. For a short interval of duration $\delta$,

$$P(0, \delta) = 1 - \lambda\delta + o(\delta),$$
$$P(1, \delta) = \lambda\delta + o(\delta),$$
$$P(\text{at least two arrivals in } \delta) = o(\delta).$$

Here $o(\delta)$ denotes an error whose ratio to $\delta$ tends to zero as $\delta \downarrow 0$.

The ARRIVAL RATE $\lambda$ has units of arrivals per unit time. Thus $\lambda\delta$ is a dimensionless probability approximation. Multiple arrivals in one short interval are possible, but their probability is negligible compared with its duration. The small-interval formulas are approximations, not exact formulas for a long interval.

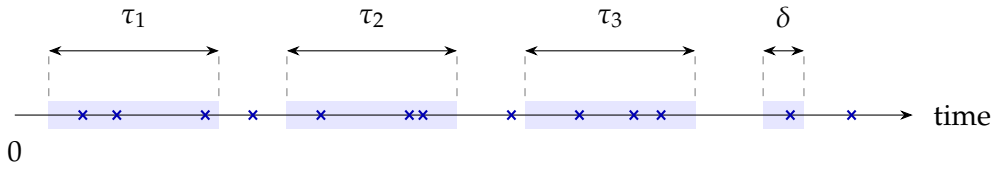



**Figure 44:** The highlighted intervals are disjoint, so their counts are independent. Each count law depends on the interval's duration, not its position. Crosses represent a schematic arrival path; the short interval has duration $\delta$.

## 14.3 The distribution of an arrival count

Fix $t > 0$. Divide $(0, t]$ into $n$ equal slots of width $\delta = t/n$. Let $I_i$ indicate whether slot $i$ contains at least one arrival, and put $B_n = I_1 + \cdots + I_n$. Homogeneity and independent increments make these indicators independent Bernoulli variables, with a common parameter

$$p_n = \lambda\frac{t}{n} + o(1/n), \qquad np_n \longrightarrow \lambda t.$$

Consequently, $B_n$ is binomial. It counts occupied slots, which agrees with $N_t$ unless some slot has multiple arrivals. By the union bound,

$$P(B_n \neq N_t) \leq n\, P(\text{at least two arrivals in } t/n) = n\, o(1/n) \longrightarrow 0.$$

Thus the slot approximation becomes exact in probability.

For a fixed nonnegative integer $j$, set $\theta = \lambda t$. The binomial PMF can be written as

$$P(B_n = j) = \frac{(np_n)^j}{j!}\left(\prod_{r=0}^{j-1}\left(1 - \frac{r}{n}\right)\right)(1 - p_n)^n(1 - p_n)^{-j}$$
$$\longrightarrow \frac{\theta^j}{j!}e^{-\theta}.$$

The finite product tends to one, $(1-p_n)^{-j}$ tends to one, and $(1-p_n)^n \to e^{-\theta}$ because $n\log(1-p_n) = -np_n + o(1) \to -\theta$. For $j=0$, the product is empty and equals one. Since the probabilities for $B_n$ and $N_t$ differ by at most $P(B_n \neq N_t)$, the same limit gives the exact count law:

$$\boxed{P(j,t) = P(N_t = j) = e^{-\lambda t}\frac{(\lambda t)^j}{j!}, \qquad j = 0,1,2,\ldots}.$$

Hence $N_t$ is Poisson with parameter $\lambda t$. The same formula applies to any interval of duration $t$.

### 14.4 Count moments and the email example

The count's parameter combines rate and elapsed time. For $\theta = \lambda t$, the exponential series gives

$$\mathbb{E}[N_t] = e^{-\theta}\sum_{j=1}^{\infty} j\frac{\theta^j}{j!} = \theta e^{-\theta}\sum_{r=0}^{\infty}\frac{\theta^r}{r!} = \theta,$$

$$\mathbb{E}[N_t(N_t-1)] = e^{-\theta}\sum_{j=2}^{\infty} j(j-1)\frac{\theta^j}{j!} = \theta^2.$$

Use $N_t^2 = N_t(N_t-1) + N_t$ to obtain

$$\boxed{\mathbb{E}[N_t] = \lambda t, \qquad \mathrm{Var}(N_t) = \theta^2 + \theta - \theta^2 = \lambda t}.$$

These moments follow directly from the exact Poisson PMF.

**Example:** Email arrives according to a Poisson process at a rate of 5 messages per hour. You check every thirty minutes. Find the probabilities of no new messages and one new message between consecutive checks.

Use hours for both rate and duration: $t = 1/2$ hour, so $\lambda t = 5/2$. The number of new messages $N$ is Poisson(5/2), and

$$P(N=0) = e^{-5/2}\frac{(5/2)^0}{0!} = \boxed{e^{-5/2} \approx 0.0821},$$

$$P(N=1) = e^{-5/2}\frac{(5/2)^1}{1!} = \boxed{\frac{5}{2}e^{-5/2} \approx 0.2052}.$$

The interval's starting time does not change these probabilities, and counts between different checks are independent.

## 14.5 The time of the kth arrival

Let $Y_k$ be the elapsed time from zero to the $k$th arrival, where $k$ is a positive integer. By time $y \geq 0$, that arrival has occurred exactly when at least $k$ arrivals have occurred:

$$\{Y_k \leq y\} = \{N_y \geq k\}.$$

This converts a waiting-time question into a count question:

$$F_{Y_k}(y) = 1 - e^{-\lambda y} \sum_{j=0}^{k-1} \frac{(\lambda y)^j}{j!}, \qquad y \geq 0.$$

The CDF is zero for $y < 0$. To differentiate, put $u = \lambda y$. The derivative of $e^{-u} \sum_{j=0}^{k-1} u^j/j!$ cancels all but its last term:

$$e^{-u}\left(\sum_{j=0}^{k-2} \frac{u^j}{j!} - \sum_{j=0}^{k-1} \frac{u^j}{j!}\right) = -e^{-u} \frac{u^{k-1}}{(k-1)!}.$$

For $k = 1$, the first sum is empty. Multiplying by the chain-rule factor $\lambda$ gives the Erlang density:

$$\boxed{f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}, \qquad y \geq 0}.$$

It is zero for $y < 0$; its value at the endpoint does not affect the law. The CDF tends to one, since an exponential dominates each of the finitely many polynomial terms in its tail.

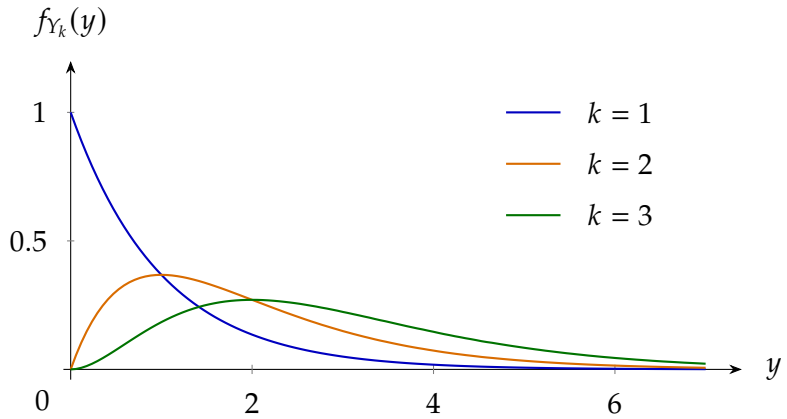



**Figure 45:** Erlang densities for $k = 1, 2, 3$ at rate $\lambda = 1$. The first-arrival density decreases from one. Later-arrival densities begin at zero and rise before decaying. These are formula plots, not measured arrival data.

## 14.6 Exponential waiting and memorylessness

For $k = 1$, the Erlang law becomes exponential:

$$\boxed{f_{Y_1}(y) = \lambda e^{-\lambda y}, \qquad y \geq 0}.$$

The wait exceeds $y$ exactly when there are no arrivals in $(0, y]$. Thus

$$P(Y_1 > y) = P(N_y = 0) = e^{-\lambda y}.$$

For $s, u \geq 0$, the survival probabilities give

$$\begin{aligned} P(Y_1 > s + u \mid Y_1 > s) &= \frac{e^{-\lambda(s+u)}}{e^{-\lambda s}} \\ &= \boxed{e^{-\lambda u} = P(Y_1 > u)}. \end{aligned}$$

The remaining wait after no arrival in the first $s$ units has the original exponential law. This is the continuous-time memoryless property.

More generally, observe the process at a fixed time $a$. The time $W_a$ from $a$ to the next arrival satisfies

$$\{W_a > u\} = \{N_{a+u} - N_a = 0\}.$$

Time homogeneity makes this event's probability $e^{-\lambda u}$, and independent increments make the future arrival behavior independent of the past arrival history. Previous arrivals do not change the waiting law. The intervals between successive arrivals likewise have the exponential waiting law of rate $\lambda$.

## 14.7 The Bernoulli–Poisson relation

The discrete approximation keeps the expected arrival count fixed. With $n = t/\delta$ slots and success probability $p = \lambda\delta$, it has

$$np = \lambda t.$$

This choice represents the first-order slot model; the exact occupied-slot probability has the smaller error described above. Taking $\delta \downarrow 0$ changes binomial counts into Poisson counts and discrete waits into continuous waits.

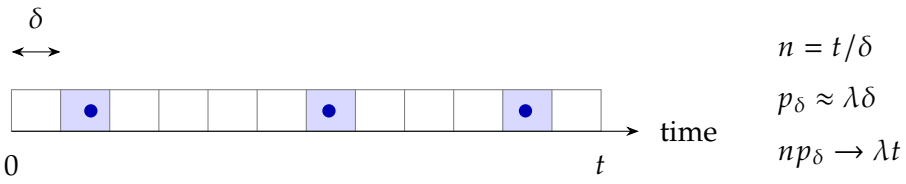



**Figure 46:** A fine time grid converts an arrival path into binary occupied-slot indicators. As the slot width shrinks, multiple arrivals in one slot become negligible. The displayed path and slot width are schematic.

| Quantity | Poisson process | Bernoulli process |
|---|---|---|
| Arrival times | Continuous | Discrete |
| Rate or probability | $\lambda$ per unit time | $p$ per trial |
| Count in an interval | Poisson | Binomial |
| Interarrival time | Exponential | Geometric |
| Time to the $k$th arrival | Erlang | Pascal |

The approximate count laws have the same mean. Bernoulli waiting times count trials; Poisson waiting times measure elapsed time.

## 14.8 Sums and merged Poisson processes

First consider independent Poisson random variables $A$ and $B$ with parameters $\alpha$ and $\beta$. For $m = 0, 1, 2, \ldots$, convolution and the binomial theorem give

$$\begin{aligned} P(A + B = m) &= \sum_{j=0}^{m} e^{-\alpha} \frac{\alpha^j}{j!} e^{-\beta} \frac{\beta^{m-j}}{(m-j)!} \\ &= \frac{e^{-(\alpha+\beta)}}{m!} \sum_{j=0}^{m} \binom{m}{j} \alpha^j \beta^{m-j} \\ &= \frac{e^{-(\alpha+\beta)}(\alpha+\beta)^m}{m!}. \end{aligned}$$

Hence

$$\boxed{A + B \sim \text{Poisson}(\alpha + \beta)}.$$

Now merge two independent Poisson processes with rates $\lambda_1$ and $\lambda_2$. Their count over any interval of duration $t$ is the sum of independent Poisson variables with parameters $\lambda_1 t$ and $\lambda_2 t$. The merged count is therefore Poisson with parameter $(\lambda_1 + \lambda_2)t$. Counts over disjoint intervals remain independent, since they are functions of separate independent pairs of original counts. Thus the merged process is Poisson with rate

$$\boxed{\lambda_1 + \lambda_2}.$$

## 14.9 Which process supplies the next arrival?

**Example:** A red bulb and a green bulb flash according to independent Poisson processes with positive rates $\lambda_1$ and $\lambda_2$. What is the probability that the next flash is red?

From a fixed observation time, let $U$ and $V$ be the remaining waits for a red and green flash. Their densities are exponential, and the waits are independent because the processes are independent. The next flash is red when $U < V$.

Condition on the red waiting time $U = u$. The green wait must exceed it. Integrating gives

$$\begin{aligned} P(U < V) &= \int_0^\infty f_U(u) P(V > u)\, du \\ &= \int_0^\infty \lambda_1 e^{-\lambda_1 u} e^{-\lambda_2 u}\, du \\ &= \boxed{\frac{\lambda_1}{\lambda_1 + \lambda_2}}. \end{aligned}$$

The green probability is $\lambda_2/(\lambda_1 + \lambda_2)$. Since independent continuous waits tie with probability zero, these two probabilities sum to one. The fraction of the total rate contributed by each process gives its probability of supplying the next arrival.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 14, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 15: Poisson Process - II

Poisson counts and exponential waiting times describe the same process. Together they handle stopping rules, routed arrivals, and random observation times.

## 15.1 The count and waiting-time laws

A Poisson process of rate $\lambda > 0$ has time-homogeneous counts and independent increments over disjoint intervals. For a short interval of duration $\delta$,

$$\begin{aligned} P(N_\delta = 0) &= 1 - \lambda\delta + o(\delta), \\ P(N_\delta = 1) &= \lambda\delta + o(\delta), \\ P(N_\delta \geq 2) &= o(\delta). \end{aligned}$$

The error divided by $\delta$ tends to zero. Over an interval of duration $t$, the exact law is

$$P(N_t = k) = e^{-\lambda t}\frac{(\lambda t)^k}{k!}, \qquad k = 0, 1, 2, \ldots, \qquad \mathbb{E}[N_t] = \mathrm{Var}(N_t) = \lambda t.$$

If $T$ is an interarrival time, then

$$f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0, \qquad P(T > t) = e^{-\lambda t}, \qquad \mathbb{E}[T] = \frac{1}{\lambda}.$$

The expectation follows by integrating the survival probability: $\mathbb{E}[T] = \int_0^\infty e^{-\lambda t}\,dt = 1/\lambda$. The time $Y_k$ to the $k$th arrival has the Erlang density

$$f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}, \qquad y \geq 0.$$

Both densities are zero for negative arguments.

Exponential waiting is memoryless. For $s, u \geq 0$,

$$P(T > s + u \mid T > s) = e^{-\lambda u}.$$

At a fixed observation time, the wait to the next arrival is exponential with rate $\lambda$ and independent of the past. This lets us restart a waiting-time calculation after learning that no arrival has yet occurred.

## 15.2 Poisson fishing: the stopping rule and probabilities

**Example:** Fish arrive as a Poisson process of rate $\lambda = 0.6$ per hour. Fish for two hours. If there has been no catch, keep fishing until the first catch.

Let $T$ be the first-catch time, $S$ the total fishing time, and $K$ the total number of fish caught. The rule gives

$$\boxed{S = \max\{2, T\}, \qquad K = \max\{1, N_2\}}.$$

Here $N_2$ counts catches in the first two hours. If it is positive, stop at hour 2 with all those catches. Otherwise, the extension ends with exactly one catch.

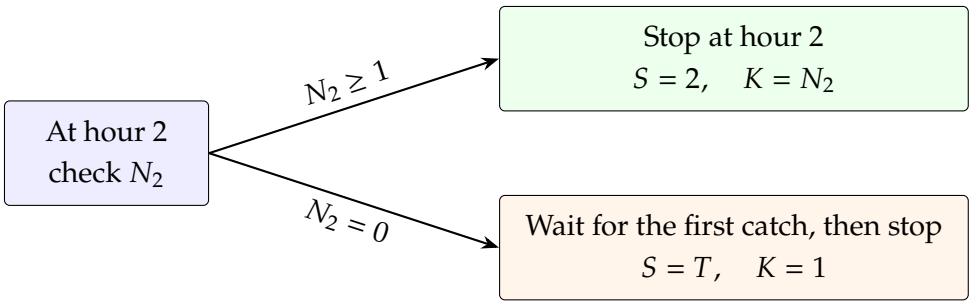



**Figure 47:** The decision at hour 2 determines whether there is an extension. During the extension, the first catch ends the trip.

**(a) More than two hours.** This occurs exactly when the first two hours have no catch:

$$P(S > 2) = P(N_2 = 0) = \boxed{e^{-1.2} \approx 0.3012}.$$

**(b) More than two and less than five hours.** The first catch must occur between hours 2 and 5:

$$P(2 < S < 5) = P(2 < T < 5) = \boxed{e^{-1.2} - e^{-3} \approx 0.2514}.$$

Arrival times have continuous laws, so the endpoints do not change this probability.

**(c) At least two fish.** The extension can produce only one fish. Thus

$$P(K \geq 2) = P(N_2 \geq 2) = \boxed{1 - e^{-1.2}(1 + 1.2) \approx 0.3374}.$$

## 15.3 Poisson fishing: expected catches and time

**(d) Expected number of fish.** Write the catch count as

$$K = N_2 + \mathbf{1}_{\{N_2 = 0\}}.$$

The indicator adds the one catch obtained during an extension. Linearity of expectation gives

$$\boxed{\mathbb{E}[K] = \mathbb{E}[N_2] + P(N_2 = 0) = 1.2 + e^{-1.2} \approx 1.5012}.$$

**(e) Expected future time after four hours of fishing.** Still fishing at hour 4 means $S > 4$, equivalently $T > 4$. There has been no catch, and the remaining trip ends at the first one. Memorylessness gives

$$P(S - 4 > u \mid S > 4) = P(T > 4 + u \mid T > 4) = e^{-0.6u}.$$

Hence

$$\boxed{\mathbb{E}[S - 4 \mid S > 4] = \frac{1}{0.6} = \frac{5}{3} \text{ hours}}.$$

The four hours already spent do not lengthen the expected remaining exponential wait.

**(f) Expected total fishing time.** Every trip lasts at least two hours. An extension occurs with probability $e^{-1.2}$; conditional on an extension, its mean length is $1/0.6$ hour. Therefore

$$\begin{aligned}\mathbb{E}[S] &= 2 + P(T > 2)\mathbb{E}[T - 2 \mid T > 2] \\ &= \boxed{2 + \frac{e^{-1.2}}{0.6} \approx 2.5020 \text{ hours}}.\end{aligned}$$

The unconditional extra time includes trips with zero extension. The conditional future time in part (e) concerns only trips that are still continuing.

## 15.4 Merging processes and identifying the next arrival

Let two independent Poisson processes have positive rates $\lambda_1$ and $\lambda_2$. Combining all their arrivals gives a Poisson process with rate $\lambda_1 + \lambda_2$.

Independent Poisson counts add to Poisson$((\lambda_1 + \lambda_2)t)$, and counts on disjoint intervals remain independent. This establishes the merged process law.

To identify the next arrival, let $U$ and $V$ be the independent exponential waits for the two streams from a fixed observation time. The first stream wins when $U < V$:

$$\begin{aligned}P(U < V) &= \int_0^\infty \lambda_1 e^{-\lambda_1 u} P(V > u)\, du \\ &= \int_0^\infty \lambda_1 e^{-(\lambda_1+\lambda_2)u}\, du = \boxed{\frac{\lambda_1}{\lambda_1 + \lambda_2}}.\end{aligned}$$

The second stream supplies the next arrival with probability $\lambda_2/(\lambda_1 + \lambda_2)$. The waits tie with probability zero. Thus each stream's share of the combined rate is also its probability of supplying the next arrival.

## 15.5 The lifetime of the last of three bulbs

**Example:** Install three bulbs with independent exponential lifetimes of rate $\lambda$. Find the expected time until the last bulb dies.

Let the lifetimes be $X_1, X_2, X_3$ and put $L = \max\{X_1, X_2, X_3\}$. All three have failed by time $t$ exactly when each lifetime is at most $t$. Independence gives

$$P(L \le t) = (1 - e^{-\lambda t})^3, \qquad t \ge 0.$$

Use the survival probability to compute the mean:

$$\begin{aligned}
\mathbb{E}[L] &= \int_0^\infty \left[1 - (1 - e^{-\lambda t})^3\right] dt \\
&= \int_0^\infty \left(3e^{-\lambda t} - 3e^{-2\lambda t} + e^{-3\lambda t}\right) dt \\
&= \frac{3}{\lambda} - \frac{3}{2\lambda} + \frac{1}{3\lambda} = \boxed{\frac{11}{6\lambda}}.
\end{aligned}$$

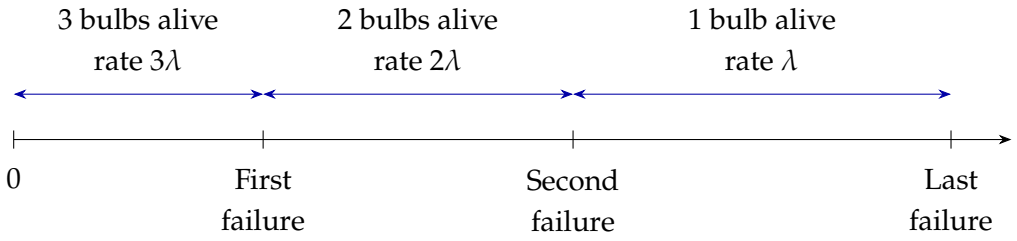



**Figure 48:** Each failure reduces the number of surviving bulbs by one. The indicated rates describe the next failure while that many bulbs remain alive; the drawn gaps are schematic.

The successive waits explain the same answer. With three bulbs alive, the probability of no failure for $u$ time units is $(e^{-\lambda u})^3 = e^{-3\lambda u}$, so the first wait has mean $1/(3\lambda)$. After a failure, memorylessness leaves the surviving bulbs with independent exponential remaining lifetimes. The next two waits have means $1/(2\lambda)$ and $1/\lambda$. Adding their means gives

$$\mathbb{E}[L] = \frac{1}{3\lambda} + \frac{1}{2\lambda} + \frac{1}{\lambda} = \frac{11}{6\lambda}.$$

## 15.6 Splitting a Poisson process

Suppose email reaches a server as a Poisson process of rate $\lambda$. Each message independently goes to the USA with probability $p$ and abroad with probability $1 - p$. Routing choices are also independent of the arrival process. This independent routing is called SPLITTING.

To establish the output laws, fix an interval of duration $t$. Let $A$ and $B$ count its USA and foreign messages. The input count $N = A + B$ is Poisson$(\lambda t)$. Given $N = a + b$, exactly $a$ of these messages choose the USA route with probability $\binom{a+b}{a} p^a (1-p)^b$. For $0 < p < 1$ and nonnegative integers $a, b$,

$$\begin{aligned} P(A = a, B = b) &= e^{-\lambda t} \frac{(\lambda t)^{a+b}}{(a+b)!} \binom{a+b}{a} p^a (1-p)^b \\ &= \left[ e^{-p\lambda t} \frac{(p\lambda t)^a}{a!} \right] \left[ e^{-(1-p)\lambda t} \frac{((1-p)\lambda t)^b}{b!} \right]. \end{aligned}$$

Each bracket is a Poisson PMF. Summing over the other count shows that $A$ is Poisson$(p\lambda t)$ and $B$ is Poisson$((1-p)\lambda t)$. The product factorization also shows that these two counts are independent.

The laws depend only on interval duration. Counts and routing choices in disjoint intervals are independent, so both output streams have independent increments. Therefore the outputs are independent Poisson processes with rates

$$\boxed{p\lambda \quad \text{and} \quad (1-p)\lambda}.$$

At $p = 0$ or $p = 1$, one stream is empty and the other is the entire input. If the total count is fixed by conditioning, the two output counts instead satisfy the constraint $A + B = N$.

## 15.7 Random incidence in a Poisson process

An observer who arrives independently of the process sees the interval containing the observation time. This is called RANDOM INCIDENCE. Consider a homogeneous Poisson process that has been running indefinitely, so arrivals exist on both sides of an arbitrary fixed time $a$.

Let $A$ be the time since the last arrival and $B$ the time to the next one. The containing interval has length $L = A + B$.

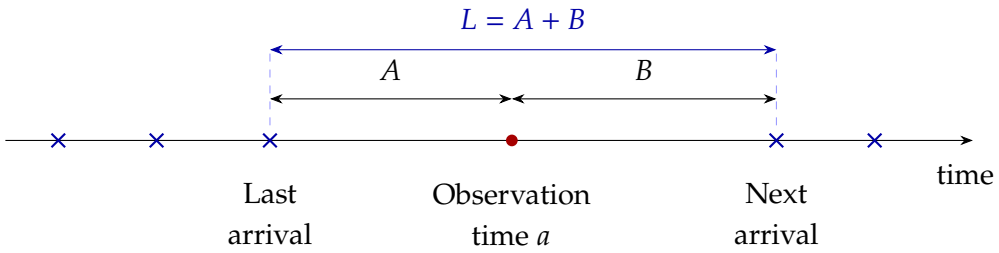



**Figure 49:** The observation time lies inside an interarrival interval. Its total length includes the elapsed part $A$ and the remaining part $B$. Arrival positions are schematic.

For $x, y \geq 0$, the event $A > x$ means no arrival in the preceding $x$ time units, and $B > y$ means no arrival in the following $y$ time units. These intervals are disjoint. Hence

$$P(A > x) = e^{-\lambda x}, \qquad P(B > y) = e^{-\lambda y},$$

$$P(A > x, B > y) = e^{-\lambda x} e^{-\lambda y}.$$

Thus $A$ and $B$ are independent exponential variables of rate $\lambda$. The same conclusion holds for a random observation time chosen independently of the arrivals.

Convolution gives the containing interval's density:

$$\begin{aligned} f_L(\ell) &= \int_0^{\ell} \lambda e^{-\lambda x} \lambda e^{-\lambda(\ell - x)} \, dx \\ &= \boxed{\lambda^2 \ell e^{-\lambda \ell}}, \qquad \ell \geq 0. \end{aligned}$$

It is Erlang with shape 2, and $\mathbb{E}[L] = 2/\lambda$. An interval selected by observing a time has mean length twice the mean $1/\lambda$ of an ordinary interarrival interval. Longer intervals occupy more time and are more likely to contain the observer. The remaining wait $B$ still has mean $1/\lambda$.

## 15.8 Renewal processes and the bus example

A RENEWAL PROCESS has successive arrivals with independent, identically distributed interarrival times. These times need not be exponential.

Suppose bus interarrival times are equally likely to be 5 or 10 minutes. Interpret a random observation time as a time sampled uniformly from a long period, independently of the bus arrivals. About half the intervals have each length, but the 10-minute intervals occupy twice as much time.

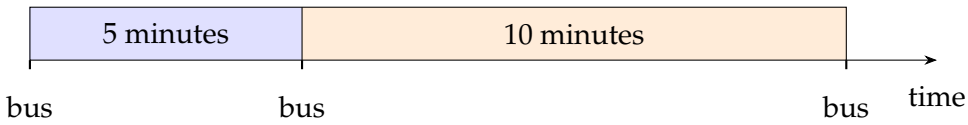



**Figure 50:** A 10-minute interval offers twice as much time in which an observer can arrive as a 5-minute interval. Widths are proportional to duration.

Let $L$ be the containing interval's length. Weighting each length by the time it contributes gives

$$P(L = 5) = \frac{5(1/2)}{5(1/2) + 10(1/2)} = \boxed{\frac{1}{3}}, \qquad P(L = 10) = \frac{2}{3}.$$

Within a selected interval of length $\ell$, the observation point is uniform, so the mean remaining wait is $\ell/2$. For the time $W$ to the next bus,

$$\boxed{\mathbb{E}[W] = \frac{1}{3}\frac{5}{2} + \frac{2}{3}\frac{10}{2} = \frac{25}{6} \text{ minutes} \approx 4.1667 \text{ minutes}}.$$

The observation-time probabilities are $1/3$ and $2/3$, even though the probabilities for a newly generated interval are $1/2$ and $1/2$.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 15, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 16: Markov Chains - I

A Markov state retains the information needed for the next transition: conditional on the present, the earlier history adds nothing. Transition probabilities and recurrent classes determine longer-term behavior.

## 16.1 The checkout counter as a state model

Let $X_n$ be the number of customers at a checkout counter at integer time $n$. Arrivals form a Bernoulli process with probability $p$ per slot. A customer's service time is geometric with parameter $q$: in each slot of service, completion occurs with probability $q$.

Assume independent arrivals and service completions. For the finite queue with states $0, 1, \ldots, 10$, use this slot rule: first attempt to complete service of a customer present at the start, then admit the arrival if space is available. An arrival finding ten customers is turned away.

For $1 \le i \le 9$, the possible changes and probabilities are

$$\begin{aligned} X_{n+1} = i+1 \quad &: \quad u = p(1-q), \\ X_{n+1} = i-1 \quad &: \quad d = (1-p)q, \\ X_{n+1} = i \quad &: \quad h = pq + (1-p)(1-q). \end{aligned}$$

An increase needs an arrival and no completion; a decrease needs a completion and no arrival. Both events or neither leave the count unchanged. Thus $u + d + h = 1$.

At the empty boundary, $p_{01} = p$ and $p_{00} = 1 - p$. At the full boundary, $p_{10,9} = d$ and $p_{10,10} = 1 - d$. There is no service completion when the queue is empty.

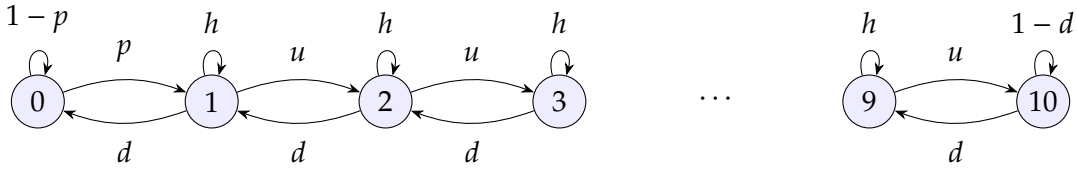



**Figure 51:** The finite queue moves up one, down one, or stays put. Interior rightward arrows have probability $u$, leftward arrows $d$, and loops $h$. The boundary labels implement the stated slot rule.

Geometric service is memoryless. For service time $G$, $P(G = s+1 \mid G > s) = q$ after any $s$ unsuccessful slots. Thus the number of customers is enough to determine the next-step law; elapsed service time need not be added to the state. The state includes the customer being served.

## 16.2 Finite-state Markov chains

**Definition 16.1 [Markov chain].** A finite-state Markov chain is a sequence $X_0, X_1, \dots$ taking values in a finite state set, such as $\{1, \dots, m\}$, with the Markov property

$$P(X_{n+1} = j \mid X_0 = i_0, \dots, X_n = i) = P(X_{n+1} = j \mid X_n = i) = p_{ij}.$$

The equality applies to histories with positive probability. Here the transition probabilities $p_{ij}$ do not depend on $n$, so the chain is time homogeneous.

The transition matrix is $P = (p_{ij})$. Its row $i$ gives the distribution of the state after one step when the current state is $i$. Therefore

$$p_{ij} \geq 0, \qquad \sum_{j=1}^{m} p_{ij} = 1.$$

Rows correspond to starting states and columns to destination states. A directed edge $i \to j$ appears in a state diagram when $p_{ij} > 0$; a self-loop represents staying in state $i$.

Specify the states, transition probabilities, and initial law. The state must retain enough information to make the Markov property valid.

The property also determines path probabilities. For a fixed initial state $i_0$ and any path $i_0, i_1, \dots, i_n$, successive conditioning gives

$$P(X_1 = i_1, \dots, X_n = i_n \mid X_0 = i_0) = \prod_{r=1}^{n} p_{i_{r-1}, i_r}.$$

At every step the next factor depends only on the current state. This describes dependence through the state; it does not make successive states independent.

## 16.3 Transition probabilities over several steps

Define the $n$-step transition probability by

$$r_{ij}(n) = P(X_n = j \mid X_0 = i).$$

At zero steps the starting state has not changed: $r_{ij}(0) = \mathbf{1}_{\{i=j\}}$. At one step, $r_{ij}(1) = p_{ij}$.

To arrive at $j$ after $n$ steps, first arrive at some state $k$ after $n - 1$ steps, then make the last transition $k \to j$.

The law of total probability and the Markov property give

$$r_{ij}(n) = \sum_{k=1}^{m} P(X_{n-1} = k \mid X_0 = i) P(X_n = j \mid X_{n-1} = k, X_0 = i)$$
$$= \boxed{\sum_{k=1}^{m} r_{ik}(n-1) p_{kj}}.$$

Terms with zero intermediate-state probability contribute zero. Writing $R(n) = (r_{ij}(n))$, this recursion is $R(n) = R(n-1)P$. Since $R(0) = I$, induction gives $R(n) = P^n$.

If the initial state is random, let $\alpha_i = P(X_0 = i)$. Conditioning on it yields

$$\boxed{P(X_n = j) = \sum_{i=1}^{m} \alpha_i r_{ij}(n)}.$$

The state distribution is a weighted average of the conditional distributions for the possible starting states. For the row vector $\alpha$, it is $\alpha P^n$.

### 16.4 A two-state calculation

Consider the chain with transition matrix

$$P = \begin{pmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{pmatrix}.$$

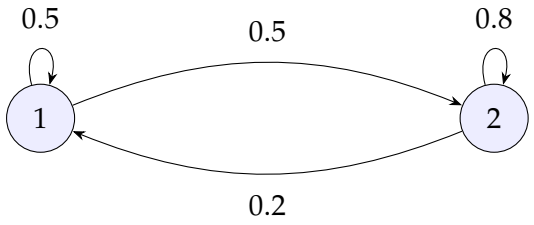



**Figure 52:** From either state, the two outgoing probabilities sum to one. Staying put is one of the possible transitions.

For example, a two-step return from state 1 can pass through either state:

$$r_{11}(2) = p_{11}p_{11} + p_{12}p_{21} = 0.5^2 + 0.5(0.2) = 0.35.$$

Applying the same calculation to each entry gives

$$R(0) = I, \qquad R(1) = P, \qquad R(2) = \begin{pmatrix} 0.35 & 0.65 \\ 0.26 & 0.74 \end{pmatrix}.$$

The requested values are

| | $n = 0$ | $n = 1$ | $n = 2$ | $n = 100$ | $n = 101$ |
|---|---|---|---|---|---|
| $r_{11}(n)$ | 1 | 0.5 | 0.35 | $\approx 2/7$ | $\approx 2/7$ |
| $r_{12}(n)$ | 0 | 0.5 | 0.65 | $\approx 5/7$ | $\approx 5/7$ |
| $r_{21}(n)$ | 0 | 0.2 | 0.26 | $\approx 2/7$ | $\approx 2/7$ |
| $r_{22}(n)$ | 1 | 0.8 | 0.74 | $\approx 5/7$ | $\approx 5/7$ |

The large-$n$ entries are approximations. An exact expression will show why both starting states produce the same limit.

## 16.5 The limit in the two-state example

Fix a starting state $i$ and let $u_n = P(X_n = 1 \mid X_0 = i)$. The other state's probability is $1 - u_n$, so conditioning on the current state gives

$$u_{n+1} = 0.5u_n + 0.2(1 - u_n) = 0.2 + 0.3u_n.$$

The constant solution satisfies $u = 0.2 + 0.3u$, hence $u = 2/7$. Subtracting it from the recursion yields

$$u_{n+1} - \frac{2}{7} = 0.3\left(u_n - \frac{2}{7}\right).$$

Iterating this equality gives

$$u_n = \frac{2}{7} + \left(u_0 - \frac{2}{7}\right)(0.3)^n.$$

Use $u_0 = 1$ for an initial state of 1 and $u_0 = 0$ for an initial state of 2. The exact matrix is

$$\boxed{R(n) = \begin{pmatrix} \frac{2}{7} + \frac{5}{7}(0.3)^n & \frac{5}{7}\left(1 - (0.3)^n\right) \\ \frac{2}{7}\left(1 - (0.3)^n\right) & \frac{5}{7} + \frac{2}{7}(0.3)^n \end{pmatrix}}.$$

This also gives the identity matrix when $n = 0$. Since $(0.3)^n \to 0$,

$$\boxed{R(n) \longrightarrow \begin{pmatrix} 2/7 & 5/7 \\ 2/7 & 5/7 \end{pmatrix}}.$$

At $n = 100$, each displayed approximation has error at most $(5/7)(0.3)^{100} < 4 \times 10^{-53}$; at $n = 101$ it is smaller still. The limiting state probabilities do not depend on the starting state in this example. A random initial state gives the same limit by taking a weighted average of the two rows.

## 16.6 Transition probabilities can oscillate

Convergence is a question about a chain, not an automatic consequence of the Markov property. Consider

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}.$$

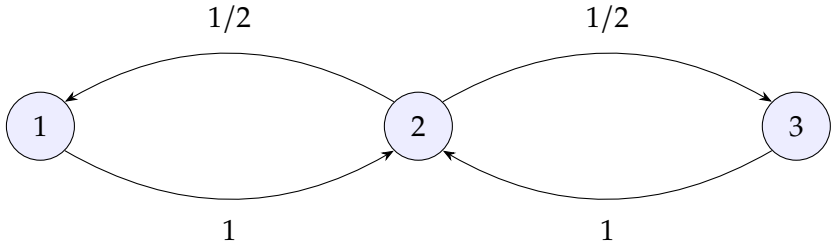



**Figure 53:** State 2 moves to an end state; either end state returns to 2. Starting at 2, the chain alternates between the center and the ends.

Starting at state 2, one step takes the chain to state 1 or 3. The next step returns to state 2 with certainty. Repeating this argument gives

$$\boxed{r_{22}(n) = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}}$$

The even and odd subsequences have different values, so $r_{22}(n)$ has no limit. The possibility of returning to a state does not by itself guarantee convergence of its occupancy probability.

## 16.7 Limits can depend on the initial state

Now consider the following transition structure. State 1 stays at 1. From state 2, the probabilities of moving to 1, staying at 2, and moving to 3 are 0.3, 0.4, and 0.3. States 3 and 4 can move only within their pair.

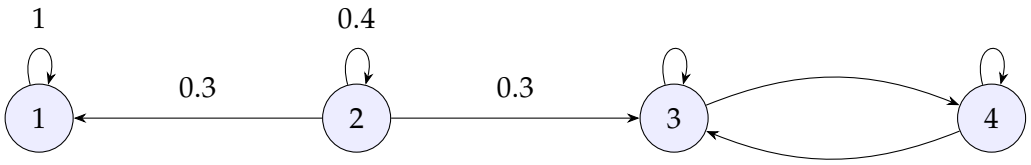



**Figure 54:** Once the chain enters state 1 or the pair $\{3, 4\}$, it cannot leave that set. Unlabeled arrows indicate positive-probability transitions; their values do not affect the probabilities of reaching state 1.

For every $n \geq 0$,

$$\boxed{r_{11}(n) = 1, \qquad r_{31}(n) = 0}.$$

Starting at 1 means staying there. Starting at 3 gives no path to 1.

Starting at 2, arrival at state 1 by step $n$ requires some number $k$ of self-loops at 2, followed by the transition $2 \to 1$. For $n \geq 1$,

$$\begin{aligned} r_{21}(n) &= \sum_{k=0}^{n-1} (0.4)^k (0.3) \\ &= \frac{0.3}{1 - 0.4}\left(1 - (0.4)^n\right) = \boxed{\frac{1}{2}\left(1 - (0.4)^n\right)}. \end{aligned}$$

The final formula also gives $r_{21}(0) = 0$. It tends to $1/2$: eventually leaving 2 for 1 or for 3 is equally likely. Thus the limiting probability of being in state 1 is 1, 0, or $1/2$ for starting states 1, 3, or 2, respectively.

## 16.8 Recurrent states, transient states, and classes

A state $j$ is REACHABLE from $i$ if a path of positive-probability transitions leads from $i$ to $j$. States COMMUNICATE if each is reachable from the other. A COMMUNICATING CLASS groups states that communicate. A class is CLOSED if no transition leads from it to a state outside it.

**Definition 16.2 [RECURRENCE IN A FINITE CHAIN].** A state $i$ is RECURRENT if every state reachable from $i$ has a path back to $i$. Equivalently, its communicating class is closed. A state that fails this condition is TRANSIENT. A closed communicating class is a RECURRENT CLASS.

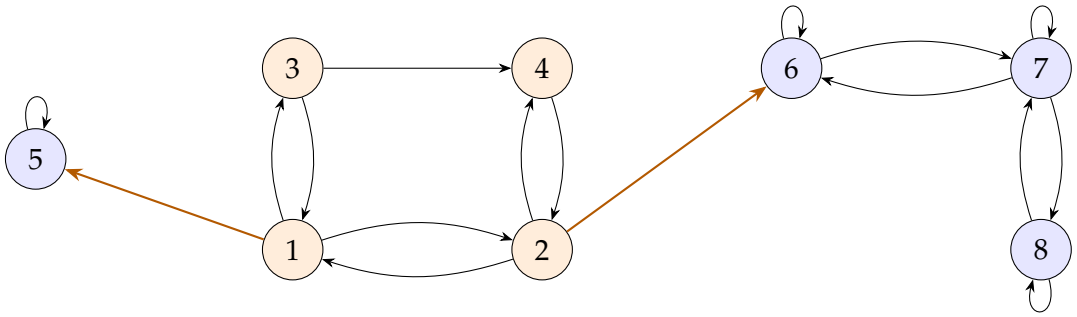



**Figure 55:** The arrows show possible transitions. Blue states form the recurrent classes $\{5\}$ and $\{6,7,8\}$. Orange states $1,2,3,4$ communicate but can escape to a recurrent class, so all four are transient.

States $1,2,3,4$ can reach one another, but $1 \to 5$ and $2 \to 6$ leave their class. There is no path back from 5 or from $6,7,8$. Consequently, none of $1,2,3,4$ satisfies the return-path condition.

State 5 stays at 5, so $\{5\}$ is a recurrent class. The states $6,7,8$ communicate within a closed set and form another recurrent class. An outside state may enter a recurrent class; closedness prohibits leaving it.

## 16.9 Why transient states disappear

In a finite recurrent class, each state has a positive-probability path to $i$, including a positive-length return from $i$ itself. Finiteness gives a common path-length bound $M$ and a positive lower bound $\varepsilon$ on these path probabilities.

The Markov property therefore bounds the chance of avoiding $i$ for $k$ blocks of $M$ steps by $(1-\varepsilon)^k \to 0$. Repeat after each return: $i$ is visited infinitely often almost surely.

For a transient state $i$, choose a reachable state that cannot return to $i$. A shortest path to it does not revisit $i$. Following that path has some probability $\eta > 0$ and prevents any later visit to $i$. Thus at every visit there is at least probability $\eta$ of escaping permanently. Conditional on a first visit,

$$P(\text{at least } k \text{ visits to } i) \le (1-\eta)^{k-1} \longrightarrow 0.$$

So the number of visits is finite almost surely. After the last visit the indicator $\mathbf{1}_{\{X_n=i\}}$ is zero; bounded convergence gives, for any initial distribution,

$$\boxed{P(X_n = i) = \mathbb{E}[\mathbf{1}_{\{X_n=i\}}] \longrightarrow 0}.$$

The probability of returning to a transient state can be positive. Transience means that a permanent escape is possible, and repeated visits eventually end almost surely.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 16, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 17: Markov Chains - II

Recurrent classes and periodicity control long-run behavior. When state probabilities converge, balance equations describe their limiting flow.

## 17.1 Review and the warmup chain

For a time-homogeneous Markov chain, $p_{ij}$ is the probability of the next state being $j$ given that the current state is $i$. The past adds no information once the current state is known. Multi-step probabilities satisfy

$$r_{ij}(n) = P(X_n = j \mid X_0 = i), \qquad r_{ij}(n) = \sum_k r_{ik}(n-1)p_{kj}, \qquad R(0) = I.$$

For a specified path, multiply the transition probabilities. For an endpoint event, sum over all paths of the required length.

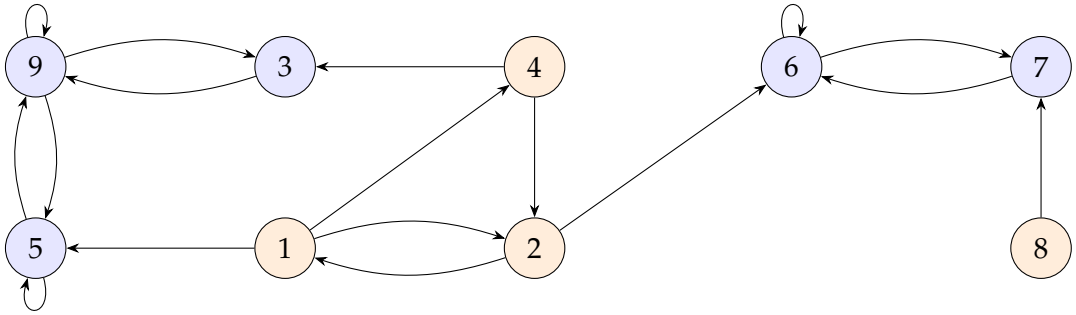



**Figure 56:** Arrows indicate positive-probability transitions. The blue states form the closed classes $\{3, 5, 9\}$ and $\{6, 7\}$. Orange states can leave permanently.

The event specifying the path $1 \to 2 \to 6 \to 7$ has probability

$$\boxed{P(X_1 = 2, X_2 = 6, X_3 = 7 \mid X_0 = 1) = p_{12}p_{26}p_{67}}.$$

There are exactly three four-step paths from 2 to 7:

$$2 \to 1 \to 2 \to 6 \to 7, \qquad 2 \to 6 \to 6 \to 6 \to 7, \qquad 2 \to 6 \to 7 \to 6 \to 7.$$

They are disjoint events, so

$$\boxed{r_{27}(4) = p_{21}p_{12}p_{26}p_{67} + p_{26}p_{66}^2p_{67} + p_{26}p_{67}p_{76}p_{67}}.$$

No numerical transition probabilities have been specified. These expressions retain the probabilities of the displayed edges.

Within a recurrent class every reachable state can return to the starting state. The sets $\{3, 5, 9\}$ and $\{6, 7\}$ are closed and communicating, so they are recurrent classes. States $1, 2, 4$ communicate but can escape to either class; state 8 enters 7 and cannot return. Thus $1, 2, 4, 8$ are transient, and their occupancy probabilities tend to zero.

## 17.2 Periodic recurrent classes

**Definition 17.1 [Periodicity].** A recurrent class is periodic if its states can be partitioned into $d > 1$ nonempty groups $G_0, \ldots, G_{d-1}$ such that every transition from $G_r$ leads to $G_{r+1}$, with indices taken modulo $d$. A class is aperiodic if no such partition exists.

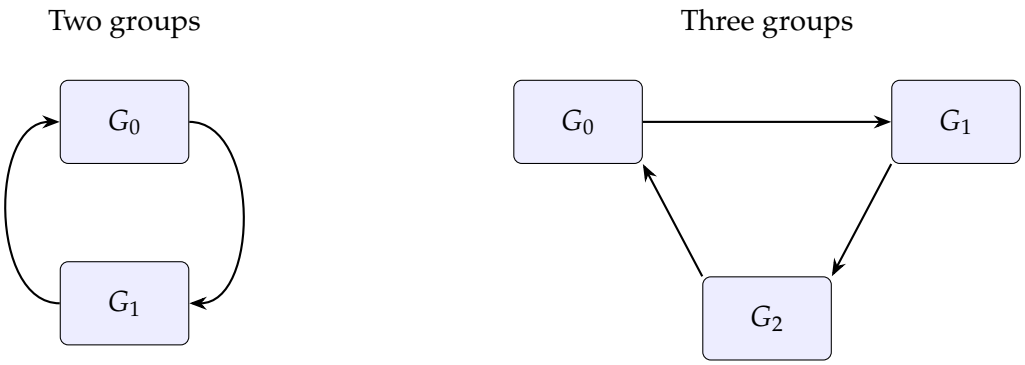



**Figure 57:** A cyclic partition forces the chain to advance one group at each step. The two diagrams illustrate two-group and three-group schedules.

Starting in $G_0$, the chain is in $G_{n \bmod d}$ at time $n$. A return to the starting state is possible only after a number of steps divisible by $d$. This restriction can make state occupancy probabilities oscillate.

A positive-probability self-loop rules out a cyclic partition with $d > 1$: it would stay in the same group instead of moving to the next one. Thus a recurrent class containing a self-loop is aperiodic. The warmup's two recurrent classes are aperiodic, since the first contains loops at 5 and 9 and the second contains a loop at 6. The entire warmup chain still has two recurrent classes.

## 17.3 A periodic class with two return cycles

Consider the following nine-state chain. All displayed arrows have positive probability; at a state with more than one outgoing arrow, their probabilities sum to one.

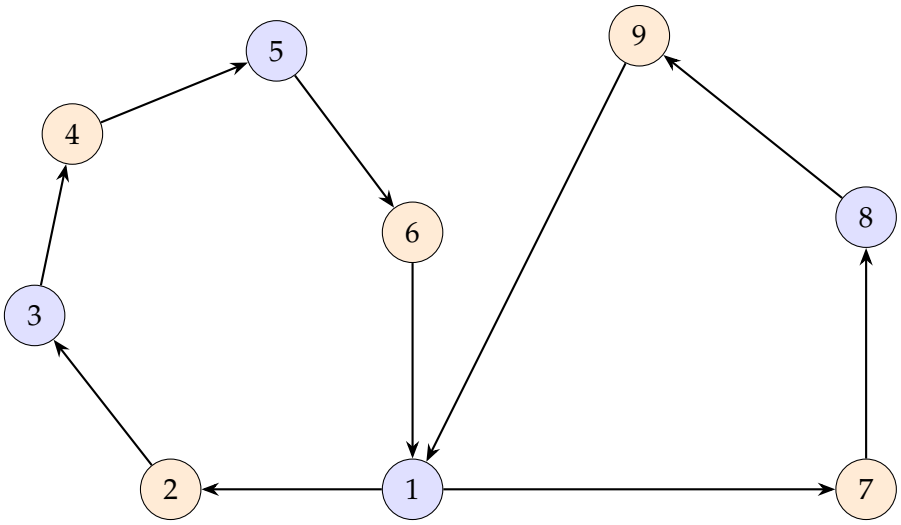



**Figure 58:** The blue group is $G_0 = \{1, 3, 5, 8\}$ and the orange group is $G_1 = \{2, 4, 6, 7, 9\}$. Every arrow changes group. The two directed cycles share state 1.

Every state can reach 1, and 1 can reach every state, so all nine states form one recurrent class. Its cycles are

$$1 \to 2 \to 3 \to 4 \to 5 \to 6 \to 1, \qquad 1 \to 7 \to 8 \to 9 \to 1.$$

Both have even length. The displayed two-group partition forces every return time to be even. Four-step and six-step returns to 1 are both possible. Any cyclic grouping must therefore have a group count dividing both 4 and 6; the only possible count greater than one is 2.

This class has period 2, despite containing cycles of different lengths. Starting in the blue group, the chain is in the blue group with probability one at even times and probability zero at odd times. Consequently, the state probabilities cannot all converge to a common time-independent distribution.

## 17.4 Steady-state convergence and balance equations

**Theorem 17.2 [Steady-state convergence].** *For a finite Markov chain with exactly one recurrent class, if that class is aperiodic, there are probabilities $\pi_j$ such that, for every starting state $i$,*

$$\lim_{n \to \infty} r_{ij}(n) = \pi_j.$$

*The limit is independent of $i$. Transient states have $\pi_j = 0$; states in the recurrent class have $\pi_j > 0$.*

These are the steady-state probabilities. The single-class condition makes the long-run destination common to all starting states. Aperiodicity removes the forced group schedule. Transient starting states are allowed.

To find the probabilities, take limits in the finite sum of the transition recursion:

$$\pi_j = \lim_{n\to\infty} \sum_k r_{ik}(n-1)p_{kj} = \sum_k \pi_k p_{kj}.$$

Taking limits in $\sum_j r_{ij}(n) = 1$ gives normalization. Thus the BALANCE EQUATIONS are

$$\boxed{\pi_j = \sum_k \pi_k p_{kj} \quad \text{for every } j, \qquad \sum_j \pi_j = 1, \qquad \pi_j \geq 0}.$$

For the row vector $\pi$, this is $\pi = \pi P$. A distribution satisfying it is STATIONARY: starting with $X_0 \sim \pi$ gives the same state distribution after every transition.

The convergence theorem supplies the conditions under which this stationary distribution is the common limit. Solving balance equations alone does not establish that arbitrary initial distributions converge. In the warmup, there are two recurrent classes. In the nine-state example, the single recurrent class is periodic. Neither satisfies both convergence conditions.

## 17.5 Solving the two-state balance equations

For the two-state chain

$$P = \begin{pmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{pmatrix},$$

both states communicate and have self-loops. They form one aperiodic recurrent class, so the convergence theorem applies.

The two balance equations and normalization are

$$\pi_1 = 0.5\pi_1 + 0.2\pi_2, \qquad \pi_2 = 0.5\pi_1 + 0.8\pi_2, \qquad \pi_1 + \pi_2 = 1.$$

The first two equations both reduce to $0.5\pi_1 = 0.2\pi_2$. Therefore

$$\pi_2 = \frac{5}{2}\pi_1, \qquad \frac{7}{2}\pi_1 = 1, \qquad \boxed{\pi_1 = \frac{2}{7}, \quad \pi_2 = \frac{5}{7}}.$$

Both starting states yield these limiting probabilities. The balance relation also says that the long-run flows between the two states agree:

$$\pi_1 p_{12} = \frac{2}{7}\frac{1}{2} = \frac{1}{7} = \frac{5}{7}\frac{1}{5} = \pi_2 p_{21}.$$

The chain spends more time in state 2, whose chance of leaving on a given step is smaller.

## 17.6 Visit and transition frequencies

Under the convergence conditions, $\pi_j$ is the long-run fraction of time spent in state $j$. Among steps whose starting state is $k$, a fraction $p_{kj}$ lead to $j$. Thus the long-run frequency of transitions $k \to j$ is

$$\boxed{\pi_k p_{kj}}.$$

This includes self-transitions, whose frequency is $\pi_j p_{jj}$.

For an expected-count calculation, define

$$C_{kj}(n) = \sum_{t=0}^{n-1} \mathbf{1}_{\{X_t=k,\, X_{t+1}=j\}}.$$

Linearity of expectation and one-step conditioning give

$$\frac{\mathbb{E}[C_{kj}(n)]}{n} = \frac{1}{n} \sum_{t=0}^{n-1} P(X_t = k) p_{kj} \longrightarrow \pi_k p_{kj},$$

because averaging a convergent sequence preserves its limit. Similarly, the expected frequency of visits to $j$ tends to $\pi_j$.

Every visit at the end of a step has exactly one incoming transition. Adding the incoming frequencies gives

$$\pi_j = \sum_k \pi_k p_{kj}.$$

This is the probability-flow meaning of the balance equation. The initial endpoint contributes at most one visit, which has no effect after division by a long observation horizon.

## 17.7 Birth-death chains and flow across a cut

A BIRTH-DEATH CHAIN on $\{0, 1, \ldots, m\}$ can move up one state, down one state, or stay put. Write $p_i$ for the upward probability from $i$ and $q_i$ for the downward probability. Here these symbols are transition probabilities themselves.

For $1 \le i \le m-1$, the self-loop probability is $1 - p_i - q_i$. At 0 it is $1 - p_0$; at $m$ it is $1 - q_m$. All probabilities must be nonnegative. Assume the adjacent upward and downward probabilities are positive so that the states communicate.




**Figure 59:** Only neighboring states exchange probability. The end states have one neighbor each; omitted states follow the same interior pattern.

In a stationary distribution, the total mass of $\{0, \dots, i\}$ does not change after a transition. The only transition leaving this set is $i \to i+1$, and the only transition entering it is $i+1 \to i$. Therefore their flows must agree:

$$\boxed{\pi_i p_i = \pi_{i+1} q_{i+1}, \qquad 0 \le i < m}.$$

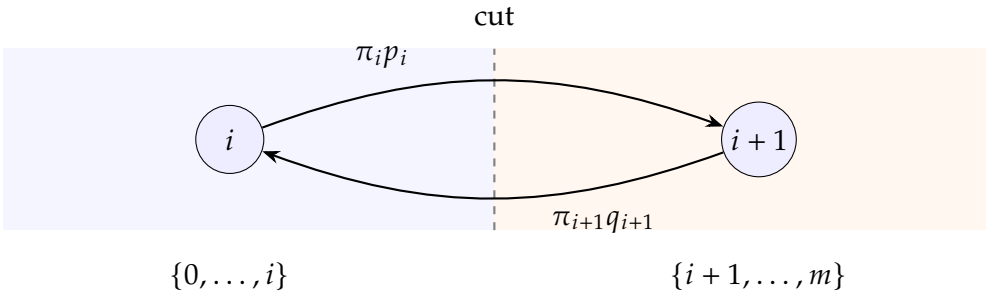



**Figure 60:** The cut separates $\{0, \dots, i\}$ from $\{i+1, \dots, m\}$. In stationarity the two crossing flows balance. All other transitions stay on their own side.

Adjacent balance gives the full state equations: incoming flow replaces the corresponding outward flow, with self-loops accounting for the remainder.

## 17.8 Constant probabilities and the load factor

Suppose every allowed upward transition has probability $p > 0$ and every allowed downward transition has probability $q > 0$, with $p + q \le 1$ for interior states. Define the LOAD FACTOR

$$\rho = \frac{p}{q}.$$

Adjacent balance gives

$$\pi_{i+1} = \frac{p}{q}\pi_i = \rho \pi_i, \qquad \boxed{\pi_i = \pi_0 \rho^i, \quad 0 \le i \le m}.$$

The finite chain's normalization is exact:

$$\pi_0 = \frac{1}{\sum_{i=0}^{m} \rho^i}.$$

In particular, for $p < q$, so $0 < \rho < 1$,

$$\boxed{\pi_0 = \frac{1-\rho}{1-\rho^{m+1}}}.$$

For a large capacity with $\rho^{m+1}$ negligible, this is approximately $1-\rho$. The infinite-capacity geometric law is

$$\boxed{\pi_i = (1-\rho)\rho^i, \qquad i = 0, 1, 2, \ldots}.$$

It requires $\rho < 1$ for the probabilities to sum to one.

In that geometric steady-state law, the expected state is

$$\begin{aligned}\mathbb{E}[X] &= (1-\rho)\sum_{i=0}^{\infty} i\rho^i \\ &= (1-\rho)\rho\frac{d}{d\rho}\left(\frac{1}{1-\rho}\right) = \boxed{\frac{\rho}{1-\rho}}.\end{aligned}$$

Here the geometric series can be differentiated for $0 < \rho < 1$. The mean grows as the load factor approaches one. For a finite capacity, the mean is instead computed from the normalized finite sum $\sum_{i=0}^{m} i\pi_i$; the infinite-capacity formula describes the limiting law.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 17, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 18: Markov Chains - III

Balance equations describe long-run occupancy; first-step conditioning gives hitting probabilities and expected hitting times.

## 18.1 Review of steady-state behavior

For a finite Markov chain with one aperiodic recurrent class, possibly together with transient states,

$$r_{ij}(n) = P(X_n = j \mid X_0 = i) \longrightarrow \pi_j.$$

The limit is independent of the initial state. The steady-state probabilities are the unique solution of

$$\boxed{\pi_j = \sum_k \pi_k p_{kj}, \qquad \sum_j \pi_j = 1, \qquad \pi_j \geq 0}.$$

These equations balance incoming probability at each state and normalize the total mass.

## 18.2 Joint events in the two-state chain

Consider the chain with transition matrix $P = \left(\begin{smallmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{smallmatrix}\right)$ and steady-state probabilities $\pi_1 = 2/7$, $\pi_2 = 5/7$. Assume $X_0 = 1$ throughout this example.

Write $u_n = r_{11}(n)$. One-step conditioning gives $u_{n+1} = 0.5u_n + 0.2(1 - u_n) = 0.2 + 0.3u_n$, with $u_0 = 1$. Subtracting the fixed point $2/7$ and iterating gives

$$u_n - \frac{2}{7} = 0.3^n \left(1 - \frac{2}{7}\right), \qquad u_n = \frac{2}{7} + \frac{5}{7}(0.3)^n.$$

The two joint probabilities are therefore

$$\begin{aligned} P(X_1 = 1, X_{100} = 1 \mid X_0 = 1) &= p_{11} r_{11}(99) \\ &= \frac{1}{2}\left[\frac{2}{7} + \frac{5}{7}(0.3)^{99}\right] \approx \frac{1}{7}, \\ P(X_{100} = 1, X_{101} = 2 \mid X_0 = 1) &= r_{11}(100) p_{12} \\ &= \frac{1}{2}\left[\frac{2}{7} + \frac{5}{7}(0.3)^{100}\right] \approx \frac{1}{7}. \end{aligned}$$

The first event uses 99 transitions after time 1. The second uses 100 transitions to reach state 1, then one transition to state 2. The Markov property supplies the conditional factor; it does not make successive states independent.

## 18.3 The phone company model

Calls arrive according to a Poisson process of rate $\lambda > 0$. Each accepted call occupies one of $B$ available lines for an independent exponential duration with parameter $\mu > 0$, hence mean duration $1/\mu$. Durations are independent of the arrival process. An arriving call is blocked when all $B$ lines are busy.

Let the state be the number of busy lines, in $\{0, 1, \ldots, B\}$. Exponential memorylessness means that the remaining call durations depend only on how many calls are active, so this state description is Markov.

Over a short interval of length $\delta$, an arrival has probability $\lambda\delta + o(\delta)$. In state $i$, each of the $i$ active calls can end, so one departure has probability $i\mu\delta + o(\delta)$. Two or more events have probability $o(\delta)$.

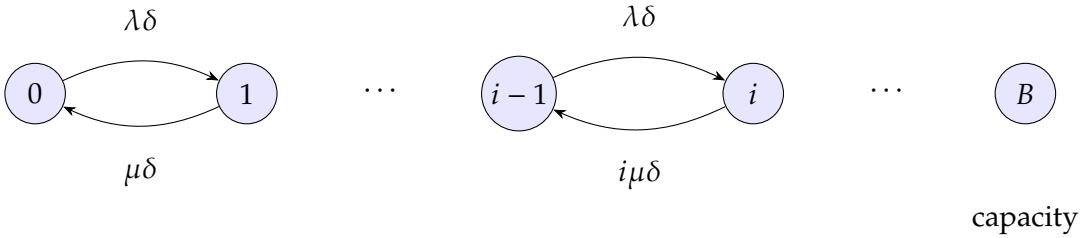



**Figure 61:** First-order transition probabilities over a short interval. Arrivals increase the busy-line count when capacity remains; departures decrease it. Self-loops are omitted for clarity.

For an interior state $0 < i < B$, the probability of no state change is $1 - (\lambda + i\mu)\delta + o(\delta)$. At 0 it is $1 - \lambda\delta + o(\delta)$; at $B$ it is $1 - B\mu\delta + o(\delta)$. A blocked arrival leaves the busy-line count unchanged.

More busy lines give a larger departure rate, unlike a chain with constant downward rate.

## 18.4 Steady-state occupancy and blocking

In stationarity, flows across the cut between $i-1$ and $i$ balance. Using the short-interval probabilities, dividing by $\delta$, and taking $\delta \to 0$ gives

$$\lambda\pi_{i-1} = i\mu\pi_i, \qquad 1 \le i \le B.$$

Set $\rho = \lambda/\mu$. Repeated substitution yields

$$\pi_i = \frac{\rho}{i}\pi_{i-1} = \pi_0 \frac{\rho^i}{i!}.$$

Normalizing over the finite capacity gives

$$\boxed{\pi_0 = \left(\sum_{k=0}^{B} \frac{\rho^k}{k!}\right)^{-1}, \qquad \pi_i = \frac{\rho^i/i!}{\sum_{k=0}^{B} \rho^k/k!}}.$$

The factorial comes from the successive departure rates $\mu, 2\mu, \ldots, i\mu$. The sum stops at $B$ because the system cannot hold more than $B$ simultaneous calls.

A call is blocked exactly when it arrives to state $B$. To see why its blocking probability equals $\pi_B$, consider arrivals in a short interval while the system is stationary. Conditional on the state at its start, the Poisson arrival probability is still $\lambda\delta + o(\delta)$, independent of that state. Thus

$$P(\text{an arrival is blocked during the interval}) = \pi_B \lambda\delta + o(\delta),$$
$$P(\text{an arrival occurs during the interval}) = \lambda\delta + o(\delta).$$

An intervening departure or second arrival changes these expressions only by $o(\delta)$. Dividing and taking the short-interval limit gives

$$\boxed{P(\text{a call is blocked}) = \pi_B = \frac{\rho^B/B!}{\sum_{k=0}^{B} \rho^k/k!}}.$$

This is a steady-state probability. The argument uses the Poisson arrival model, rather than assuming that arbitrary arrival processes observe time averages.

## 18.5 Absorption probabilities

An ABSORBING STATE has $p_{ii} = 1$: once reached, it is never left. For the following chain, let $a_i$ be the probability of eventual absorption in state 4 given $X_0 = i$.

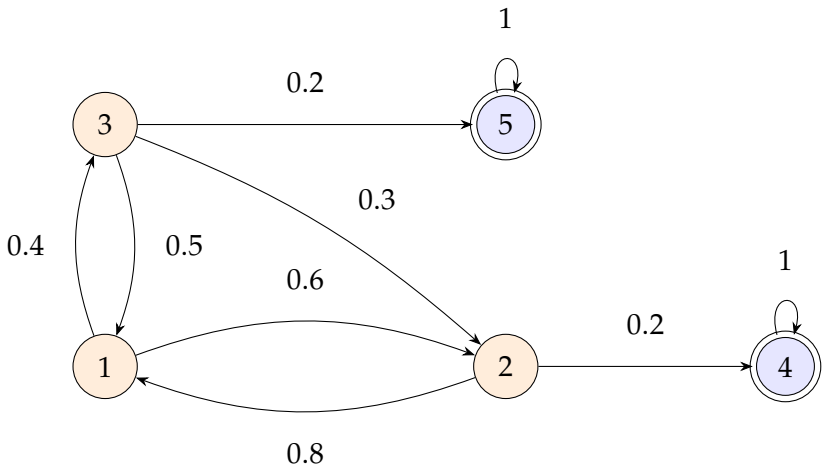



**Figure 62:** States 4 and 5 are absorbing. The transient states $1, 2, 3$ can lead to either outcome. Labels give one-step transition probabilities.

Impose $a_4 = 1$ and $a_5 = 0$, since the desired outcome is absorption in 4. For a transient state, condition on the next state:

$$a_i = \sum_j p_{ij} a_j.$$

The future absorption probability after moving to $j$ is $a_j$. Hence the unknowns satisfy

$$a_1 = \frac{2}{5}a_3 + \frac{3}{5}a_2, \qquad a_2 = \frac{4}{5}a_1 + \frac{1}{5}, \qquad a_3 = \frac{1}{2}a_1 + \frac{3}{10}a_2.$$

Substitute the second equation into the third:

$$a_3 = \frac{37}{50}a_1 + \frac{3}{50}.$$

Then the first equation becomes

$$a_1 = \frac{97}{125}a_1 + \frac{18}{125}, \qquad 28a_1 = 18.$$

Therefore the unique solution is

$$\boxed{a_1 = \frac{9}{14}, \quad a_2 = \frac{5}{7}, \quad a_3 = \frac{15}{28}, \quad a_4 = 1, \quad a_5 = 0}.$$

## 18.6 Expected time to absorption

Now use the following four-state chain, whose only absorbing state is 4. Let $\mu_i$ be the expected number of transitions until reaching 4, given $X_0 = i$. Here $\mu_i$ denotes a mean transition count; it is separate from the call-duration parameter $\mu$.

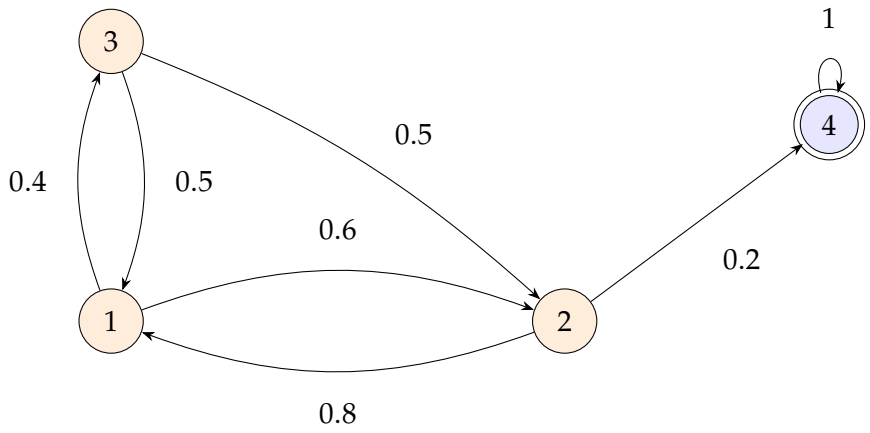



**Figure 63:** State 3 moves to 1 or 2 with equal probability. Every transient state can reach the absorbing state 4.

Starting at the absorbing state requires no transition, so $\mu_4 = 0$. From any other state, count the next transition, then condition on the remaining time:

$$\boxed{\mu_i = 1 + \sum_j p_{ij}\mu_j, \qquad i \neq 4}.$$

For this chain,

$$\mu_1 = 1 + \frac{2}{5}\mu_3 + \frac{3}{5}\mu_2, \qquad \mu_2 = 1 + \frac{4}{5}\mu_1, \qquad \mu_3 = 1 + \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2.$$

The last two equations give

$$\mu_3 = \frac{3}{2} + \frac{9}{10}\mu_1.$$

Substitute into the first equation:

$$\mu_1 = \frac{11}{5} + \frac{21}{25}\mu_1, \qquad \frac{4}{25}\mu_1 = \frac{11}{5}.$$

Thus the unique solution is

$$\boxed{\mu_1 = \frac{55}{4}, \quad \mu_2 = 12, \quad \mu_3 = \frac{111}{8}, \quad \mu_4 = 0}.$$

The 1 in the recursion counts a transition even when it reaches the absorbing state immediately. It is absent from the boundary equation $\mu_4 = 0$.

## 18.7 Mean first passage and recurrence times

Consider a finite chain with one recurrent class and fix a recurrent state $s$. Aperiodicity is not needed for the hitting-time calculations here.

The MEAN FIRST PASSAGE TIME from $i$ to $s$ is

$$t_i = \mathbb{E}[\min\{n \geq 0 : X_n = s\} \mid X_0 = i].$$

Time zero is allowed, so $t_s = 0$. First-step conditioning gives the unique solution of

$$\boxed{t_s = 0, \qquad t_i = 1 + \sum_j p_{ij} t_j \quad (i \neq s)}.$$

To calculate a first passage time, stop counting upon the first visit to $s$; its later transitions do not affect the answer.

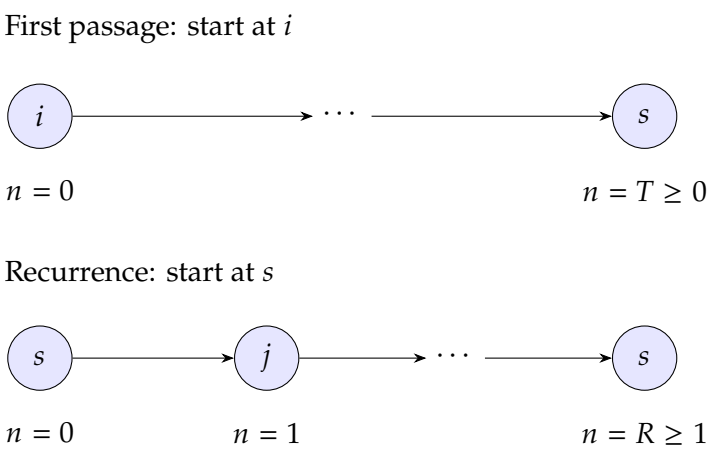



**Figure 64:** First passage permits a hit at time zero. Recurrence starts in $s$ and requires a later visit. If the first transition is a self-loop, the recurrence time is already one. The paths are schematic.

The MEAN RECURRENCE TIME of $s$ is instead

$$t_s^* = \mathbb{E}[\min\{n \geq 1 : X_n = s\} \mid X_0 = s].$$

After one transition from $s$ to $j$, the expected remaining first passage time is $t_j$. Therefore

$$\boxed{t_s^* = 1 + \sum_j p_{sj} t_j}.$$

The sum includes $j = s$, whose contribution is $p_{ss} t_s = 0$. A self-loop returns in one transition, while a move elsewhere requires that first transition plus the time to reach $s$ from the new state. The difference between $n \geq 0$ and $n \geq 1$ prevents confusing $t_s = 0$ with the positive recurrence time.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 18, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 19: Weak Law of Large Numbers

For i.i.d. observations, the sample mean $M_n = (X_1 + \cdots + X_n)/n$ estimates their common mean. The weak law bounds the probability of a fixed estimation error.

## 19.1 Chebyshev's inequality

Large squared deviations consume variance, limiting how often they can occur.

**Theorem 19.1** [Chebyshev's inequality]. *If $X$ has finite mean $\mu$ and variance $\sigma^2$, then for every $c > 0$,*

$$\boxed{P(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}}.$$

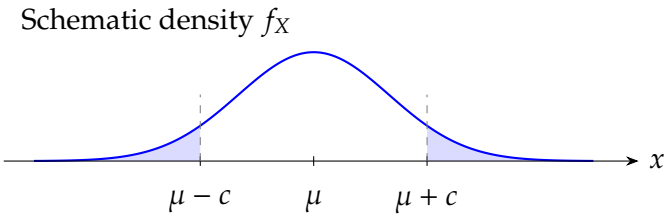



**Figure 65:** The tail event lies outside the open band $(\mu - c, \mu + c)$. On both tails the squared deviation is at least $c^2$. The density is schematic; Chebyshev's inequality does not require a particular shape.

*Proof.* Keep only the contribution from the tail event. Since squared deviations are nonnegative,

$$\sigma^2 = \mathbb{E}[(X - \mu)^2] \geq \mathbb{E}[(X - \mu)^2 \mathbf{1}_{\{|X-\mu| \geq c\}}] \geq c^2 P(|X - \mu| \geq c).$$

Divide by $c^2 > 0$. For a density, the retained contribution is

$$\int_{-\infty}^{\mu - c} (x - \mu)^2 f_X(x)\, dx + \int_{\mu + c}^{\infty} (x - \mu)^2 f_X(x)\, dx.$$

The expectation argument covers both discrete and continuous variables. □

When $\sigma > 0$, taking $c = k\sigma$ gives $P(|X - \mu| \geq k\sigma) \leq 1/k^2$ for $k > 0$. Both forms are upper bounds, not exact tail probabilities.

## 19.2 Deterministic limits and convergence in probability

A deterministic sequence $a_n$ converges to $a$ when its terms eventually stay inside every fixed tolerance band. Precisely, for each $\epsilon > 0$, there exists $n_0$ such that

$$|a_n - a| < \epsilon \qquad \text{for all } n \geq n_0.$$

For a sequence of random variables, each term has a distribution. We can instead ask whether the probability mass outside a fixed band tends to zero.

**Definition 19.2 [Convergence in probability].** The sequence $Y_n$ converges in probability to the number $a$, written $Y_n \xrightarrow{P} a$, if

$$\boxed{\lim_{n\to\infty} P(|Y_n - a| \geq \epsilon) = 0 \quad \text{for every fixed } \epsilon > 0}.$$

The tolerance can be arbitrarily small, but it is held fixed while $n$ increases. This definition concerns probabilities; it does not assert that every possible sequence of realized values eventually stays near $a$.

Consider the two-point distribution

$$P(Y_n = 0) = 1 - \frac{1}{n}, \qquad P(Y_n = n) = \frac{1}{n}, \qquad n \geq 1.$$

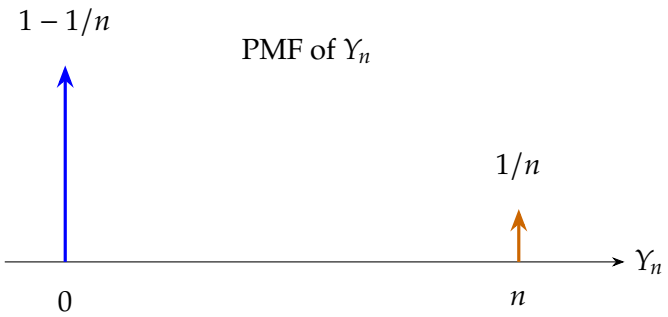



**Figure 66:** A schematic PMF for $n > 2$. The value $n$ moves farther from zero, but its probability $1/n$ tends to zero. Almost all mass is at zero for large $n$.

For any fixed $\epsilon > 0$, once $n \geq \epsilon$ the only value outside the tolerance band around zero is $n$. Thus

$$P(|Y_n| \geq \epsilon) = \frac{1}{n} \longrightarrow 0, \qquad \boxed{Y_n \xrightarrow{P} 0}.$$

The growing size of a rare exceptional value does not prevent convergence in probability.

## 19.3 Convergence of the sample mean

Assume $X_1, X_2, \ldots$ are independent and identically distributed, with finite common mean $\mu$ and variance $\sigma^2$. Averaging preserves the mean and reduces the variance:

$$\mathbb{E}[M_n] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \mu,$$

$$\operatorname{Var}(M_n) = \frac{1}{n^2}\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \operatorname{Var}(X_i) = \frac{\sigma^2}{n}.$$

Independence makes the covariance terms zero, allowing the variances to add.

**Theorem 19.3** [Weak law of large numbers]. *Under these assumptions, $M_n \xrightarrow{P} \mu$. In particular, for every fixed $\epsilon > 0$,*

$$\boxed{P(|M_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \longrightarrow 0}.$$

*Proof.* Apply Chebyshev's inequality to $M_n$, using its mean $\mu$ and variance $\sigma^2/n$. With $\epsilon$ fixed, the bound tends to zero as $n \to \infty$, which is exactly the definition of convergence in probability. □

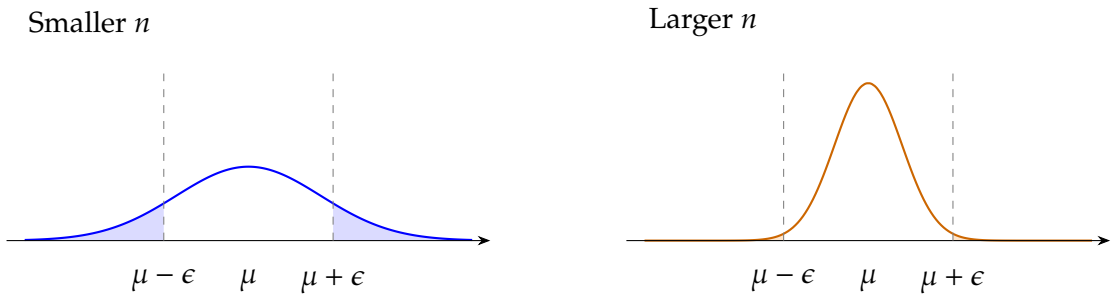



**Figure 67:** Schematic concentration of sample-mean distributions. The tolerance band stays fixed as the tail probability decreases. These curves illustrate the idea; the weak law also applies when the sample mean is discrete.

The weak law of large numbers makes each fixed estimation error unlikely with enough observations. Its moment-based bound can be conservative.

## 19.4 The pollster's problem

Let $f$ be the fraction of a population that would answer yes to a question. Model independent random selections by

$$X_i = \begin{cases} 1, & \text{if person } i \text{ answers yes,} \\ 0, & \text{if person } i \text{ answers no.} \end{cases}$$

Then $P(X_i = 1) = f$, and the observed fraction of yes answers is $M_n = (X_1 + \cdots + X_n)/n$.

Because $X_i^2 = X_i$,

$$\mathbb{E}[X_i] = f, \qquad \operatorname{Var}(X_i) = \mathbb{E}[X_i^2] - f^2 = f(1-f).$$

Although $f$ is unknown, its variance is uniformly bounded:

$$f(1-f) = \frac{1}{4} - \left(f - \frac{1}{2}\right)^2 \le \frac{1}{4}.$$

Thus $\mathbb{E}[M_n] = f$ and $\operatorname{Var}(M_n) = f(1-f)/n \le 1/(4n)$.

We want the probability of an error of at least one percentage point to be at most 0.05:

$$P(|M_n - f| \ge 0.01) \le 0.05.$$

Chebyshev's inequality gives

$$\begin{aligned} P(|M_n - f| \ge 0.01) &\le \frac{f(1-f)}{n(0.01)^2} \\ &\le \frac{1}{4n(0.01)^2}. \end{aligned}$$

To make this bound at most 0.05, choose

$$\frac{1}{4n(0.01)^2} \le 0.05 \quad \Longleftrightarrow \quad \boxed{n \ge \frac{1}{4(0.01)^2(0.05)} = 50{,}000}.$$

This sufficient sample size gives at least a 95% chance of an error smaller than one percentage point under the independent sampling model. It is conservative: 50,000 is what this bound guarantees, not a claim that a smaller sample cannot meet the goal.

## 19.5 Different scalings of the sum

Write $S_n = X_1 + \cdots + X_n$. The same observations produce three quantities with different variance scales:

| Quantity | Mean | Variance |
|---|---|---|
| $S_n$ | $n\mu$ | $n\sigma^2$ |
| $M_n = S_n/n$ | $\mu$ | $\sigma^2/n$ |
| $S_n/\sqrt{n}$ | $\sqrt{n}\,\mu$ | $\sigma^2$ |

The sum accumulates variability. Dividing by $n$ makes the sample mean concentrate. Dividing by $\sqrt{n}$ keeps the variance constant, since

$$\operatorname{Var}\left(\frac{S_n}{\sqrt{n}}\right) = \frac{1}{n}\operatorname{Var}(S_n) = \sigma^2.$$

However, its mean is $\sqrt{n}\,\mu$, so its location still changes when $\mu \neq 0$. To study a stable shape, subtract the mean before scaling:

$$\frac{S_n - n\mu}{\sqrt{n}} \quad \text{has mean } 0 \text{ and variance } \sigma^2.$$

This separates the weak law's concentration question from the question of the shape of the remaining fluctuations.

## 19.6 Standardization and the central limit theorem

Assume additionally $\sigma^2 > 0$. The STANDARDIZED SUM is

$$Z_n = \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\operatorname{Var}(S_n)}} = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\sqrt{n}(M_n - \mu)}{\sigma}.$$

Its mean is zero and its variance is one for every $n$:

$$\mathbb{E}[Z_n] = 0, \qquad \operatorname{Var}(Z_n) = \frac{n\sigma^2}{n\sigma^2} = 1.$$

The normalization fixes both the center and the variance.

**Theorem 19.4 [CENTRAL LIMIT THEOREM].** *For independent, identically distributed $X_i$ with finite mean $\mu$ and variance $0 < \sigma^2 < \infty$, the standardized sum satisfies, for every real $c$,*

$$\boxed{P(Z_n \le c) \longrightarrow P(Z \le c) = \Phi(c)}, \qquad Z \sim N(0,1).$$

Here $\Phi$ is the standard normal cumulative distribution function:

$$\Phi(c) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{c} e^{-z^2/2}\, dz.$$

The statement concerns the probability to the left of each fixed cutoff $c$. The individual observations need not be normally distributed.

The weak law says that the unscaled error $M_n - \mu$ becomes small in probability. The central limit theorem describes its shape after multiplication by $\sqrt{n}/\sigma$, which keeps the variance equal to one.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 19, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 20: Central Limit Theorem

The central limit theorem gives an approximately normal law for a centered, scaled sum of independent observations.

## 20.1 Standardizing the sum

Let $X_1, X_2, \ldots$ be independent and identically distributed, with mean $\mu$ and variance $0 < \sigma^2 < \infty$. For $S_n = X_1 + \cdots + X_n$, independence gives

$$\mathbb{E}[S_n] = n\mu, \qquad \mathrm{Var}(S_n) = n\sigma^2.$$

Subtract the mean and divide by the standard deviation:

$$\boxed{Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}}, \qquad \mathbb{E}[Z_n] = 0, \quad \mathrm{Var}(Z_n) = 1.$$

This fixes the center and spread while $n$ changes.

**Theorem 20.1 [Central limit theorem].** *Under these assumptions, for every fixed real $c$,*

$$\boxed{P(Z_n \le c) \longrightarrow \Phi(c) \quad \text{as } n \to \infty},$$

*where $\Phi(c) = P(Z \le c)$ for $Z \sim N(0,1)$.*

The theorem is useful because the limiting CDF depends on the observations only through the mean and variance used in standardization. It supports normal models for aggregated quantities and supplies a computational shortcut.

For large $n$, the normal approximation treats $Z_n$ as standard normal, or equivalently treats $S_n$ as normal with mean $n\mu$ and variance $n\sigma^2$:

$$P(S_n \le b) \approx \Phi\left(\frac{b - n\mu}{\sigma\sqrt{n}}\right).$$

For an interval, subtract the two approximate CDF values:

$$P(a < S_n \le b) \approx \Phi\left(\frac{b - n\mu}{\sigma\sqrt{n}}\right) - \Phi\left(\frac{a - n\mu}{\sigma\sqrt{n}}\right).$$

These are finite-sample approximations. The limit theorem alone does not give an error bound for a chosen $n$.

## 20.2 What converges, and how quickly?

The CLT states convergence of *CDFs*. It does not state that PDFs or PMFs converge point by point to the normal density. A sum of discrete variables remains discrete, even when its interval probabilities are well approximated by normal areas.

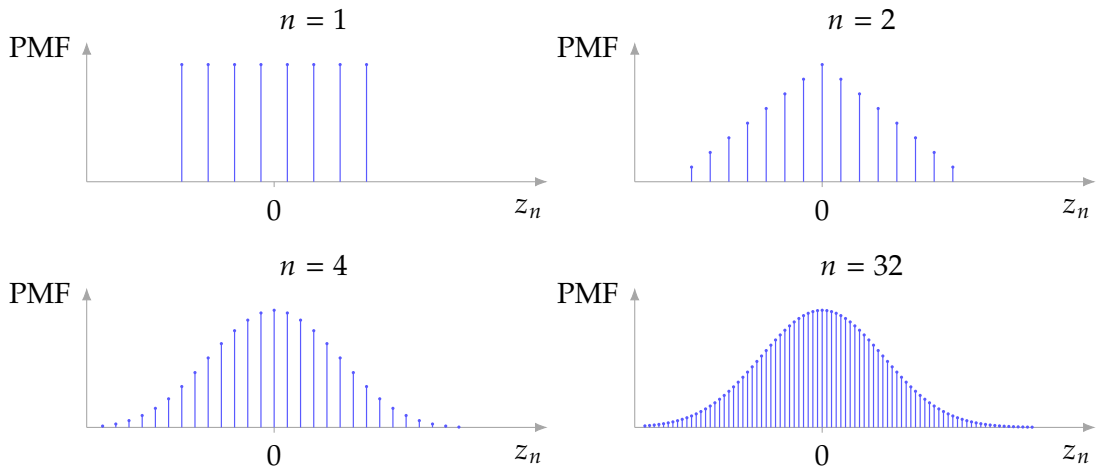



**Figure 68:** Schematic PMFs from a symmetric starting distribution. The sum is centered and scaled in each panel. Its shape becomes smoother and more bell-shaped as $n$ increases. Vertical scales are adjusted to show the shapes.

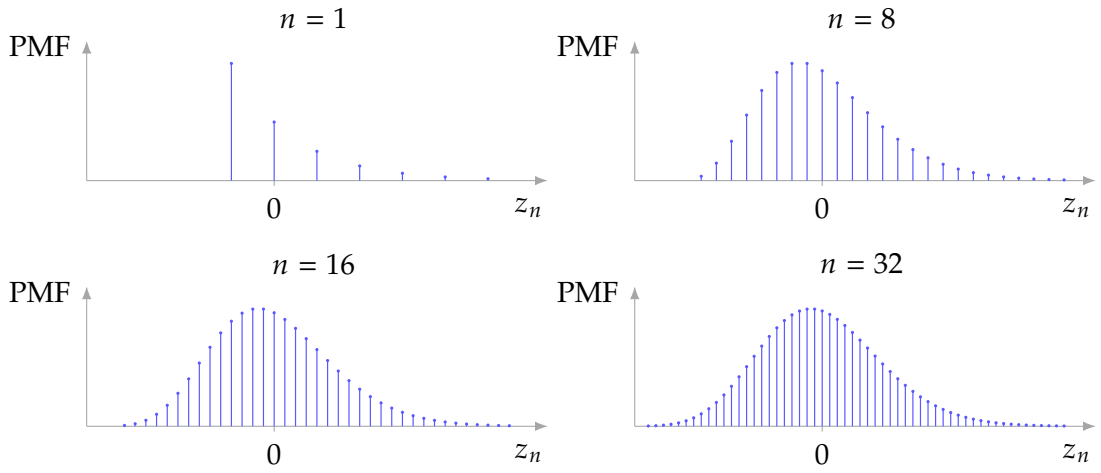



**Figure 69:** Schematic PMFs from a starting distribution with a long right tail. Summation gradually reduces the asymmetry. Centering and scaling keep the mean at zero and the variance at one; vertical scales are adjusted as above.

Approximation quality depends on the starting distribution, especially its skewness. The CLT supplies no universal adequate sample size.

## 20.3 The pollster's problem with the CLT

Let $f$ be the population fraction answering yes. Under independent random sampling, define $X_i = 1$ for a yes answer and $X_i = 0$ otherwise. Then

$$\mathbb{E}[X_i] = f, \qquad \sigma^2 = f(1-f) \le \frac{1}{4}, \qquad M_n = \frac{X_1 + \cdots + X_n}{n}.$$

We want $P(|M_n - f| \ge 0.01) \le 0.05$: at least 95% probability of an error below one percentage point.

For $0 < f < 1$, standardize the sum. The error event is

$$\begin{aligned} |M_n - f| \ge 0.01 \quad &\iff \quad |S_n - nf| \ge 0.01n \\ &\iff \quad \left|\frac{S_n - nf}{\sigma\sqrt{n}}\right| \ge \frac{0.01\sqrt{n}}{\sigma}. \end{aligned}$$

Consequently,

$$P(|M_n - f| \ge 0.01) \approx P\left(|Z| \ge \frac{0.01\sqrt{n}}{\sigma}\right) = 2\left[1 - \Phi\left(\frac{0.01\sqrt{n}}{\sigma}\right)\right].$$

The factor 2 includes both tails of the symmetric normal curve. Because $\sigma \le 1/2$, the standardized cutoff is at least $0.02\sqrt{n}$. Thus the *normal-model* tail is bounded by

$$P\left(|Z| \ge \frac{0.01\sqrt{n}}{\sigma}\right) \le P(|Z| \ge 0.02\sqrt{n}) = 2[1 - \Phi(0.02\sqrt{n})].$$

Using $\Phi(1.96) \approx 0.975$, we choose

$$0.02\sqrt{n} \ge 1.96 \quad \iff \quad \boxed{n \ge \left(\frac{1.96}{0.02}\right)^2 = 9{,}604}.$$

This is an approximate planning calculation. It is much smaller than the sufficient Chebyshev sample size of 50,000, but it does not provide the same finite-sample guarantee. At $f = 0$ or $f = 1$, every response is constant, so $M_n = f$ exactly and no standardization is needed.

## 20.4 Normal approximation to a binomial CDF

Fix $0 < p < 1$, and let the independent $X_i$ be Bernoulli($p$). Their sum is $S_n \sim \text{Binomial}(n, p)$, with

$$\mathbb{E}[S_n] = np, \qquad \text{Var}(S_n) = np(1-p).$$

The CLT says that the CDF of

$$Z_n = \frac{S_n - np}{\sqrt{np(1-p)}}$$

approaches the standard normal CDF as $n \to \infty$.

For $n = 36$ and $p = 1/2$, the mean is 18 and the standard deviation is $\sqrt{36(1/2)(1/2)} = 3$. The exact probability is

$$P(S_{36} \le 21) = \sum_{k=0}^{21} \binom{36}{k} \left(\frac{1}{2}\right)^{36} \approx 0.8785.$$

A direct normal approximation gives $\Phi((21 - 18)/3) = \Phi(1) \approx 0.8413$. The integer cutoff needs more careful treatment.

Since $S_{36}$ is integer-valued, $P(S_{36} \le 21) = P(S_{36} < 22)$. A continuous approximation can put its boundary halfway between 21 and 22. This CONTINUITY CORRECTION uses 21.5:

$$\boxed{P(S_{36} \le 21) \approx \Phi\left(\frac{21.5 - 18}{3}\right) = \Phi(7/6) \approx 0.8783}.$$

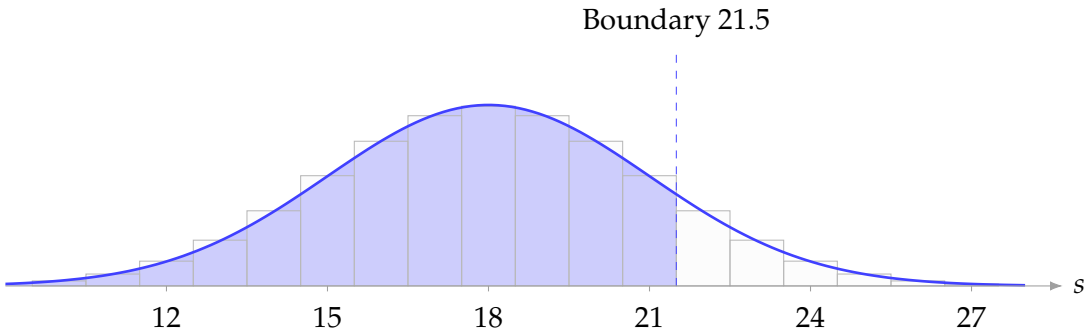



**Figure 70:** Exact binomial masses (gray bars) and the normal density with mean 18 and variance 9 (blue curve). Each integer occupies a width-one cell. The shaded normal area ends at 21.5, the right edge of the cell for 21.

## 20.5 Approximating a binomial probability mass

For a fixed $p$, the binomial form of the CLT is also called the DE MOIVRE–LAPLACE THEOREM. Combining the normal approximation with a half-unit correction lets us estimate a probability mass using a width-one normal area:

$$P(S_n = k) \approx \Phi\left(\frac{k + 1/2 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - 1/2 - np}{\sqrt{np(1-p)}}\right).$$

This is an area approximation, not a claim that a continuous normal variable has positive probability at the point $k$.

Continue with $n = 36$, $p = 1/2$, and find $P(S_{36} = 19)$. Because the sum is integer-valued,

$$\{S_{36} = 19\} = \{18.5 \le S_{36} \le 19.5\}.$$

The standardized endpoints are

$$\frac{18.5 - 18}{3} = \frac{1}{6}, \qquad \frac{19.5 - 18}{3} = \frac{1}{2}.$$

Thus

$$\begin{aligned} P(S_{36} = 19) &\approx P(1/6 \le Z \le 1/2) \\ &= \Phi(1/2) - \Phi(1/6) \approx 0.1253. \end{aligned}$$

Rounding $1/6$ to $0.17$ before using a normal table gives $0.6915 - 0.5675 = 0.1240$. The exact value is

$$P(S_{36} = 19) = \binom{36}{19}\left(\frac{1}{2}\right)^{36} \approx 0.1251.$$

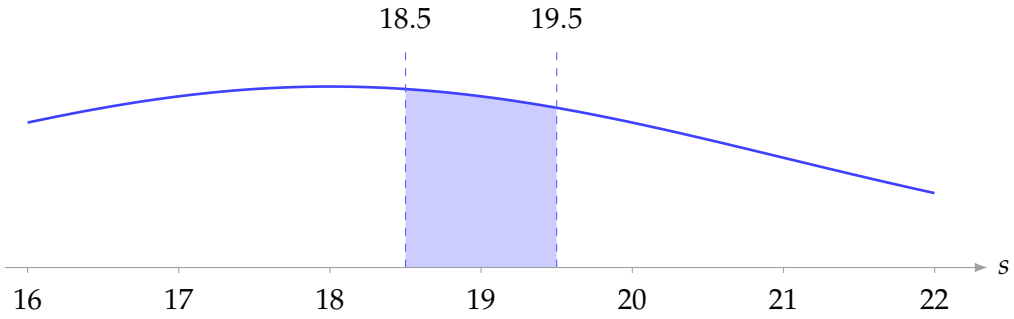



**Figure 71:** The mass at 19 is approximated by the shaded normal area from 18.5 to 19.5. The half-unit endpoints give the whole cell to the integer 19.

## 20.6 Poisson versus normal approximation

The limiting regime matters. For a binomial variable, there are two different ways to let $n$ grow:

| Regime | Approximation |
|---|---|
| $p \in (0,1)$ fixed, $n \to \infty$ | Normal after standardization |
| $np = \lambda$ fixed, $n \to \infty$, $p \to 0$ | Poisson($\lambda$) |

For $n = 100$ and $p = 1/100$, the mean is $np = 1$: rare successes make Poisson(1) the natural approximation. For $n = 500$ and $p = 1/10$, the mean is 50 and the variance is 45; a normal approximation is appropriate.

A tempting but invalid CLT argument divides a Poisson process into ever shorter intervals. If its rate is $\lambda$, write the count during a unit interval as

$$N(1) = Y_{n,1} + \cdots + Y_{n,n}, \qquad Y_{n,i} \sim \text{Poisson}(\lambda/n),$$

with independent interval counts. For each fixed $n$, the summands are identically distributed. However, their distribution changes with $n$: their mean and variance are both $\lambda/n$. The fixed-distribution i.i.d. CLT stated above therefore does not apply to this construction.

The total retains its Poisson($\lambda$) distribution, mean $\lambda$, and variance $\lambda$ regardless of how finely the interval is divided. Increasing the number of pieces alone does not make this fixed count normal.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 20, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 21: Bayesian Statistical Inference - I

Statistical inference uses observations to learn about unknown quantities. A probability model specifies how those observations are generated.

## 21.1 Models and inference questions

Data connect a mathematical model to the system it describes. For example, customer-arrival data can help us choose or assess a Poisson model. Designing an experiment and interpreting its observations are both part of this process.

Even after choosing a model, we must specify which quantity is unknown. In the signal model

$$X = aS + W,$$

$S$ is a signal, $a$ is a coefficient, $W$ is noise, and $X$ is the observation. If $S$ is known, observing $X$ can help infer $a$. If $a$ is known, the same observation can help estimate $S$ in the presence of noise.

Two kinds of inference questions have different goals:

- Hypothesis testing chooses among a few possible values or explanations. The aim is a small probability of making an incorrect decision.
- Estimation assigns a value to an unknown quantity. The aim is a small estimation error, measured by an appropriate criterion.

The model and the quantity being inferred must be specified before either goal can be assessed.

## 21.2 Classical and Bayesian approaches

In classical statistics, an unknown parameter $\theta$ is a fixed number, not a random variable. For example, it might represent the mass of an electron. The distribution $p_X(x;\theta)$ describes how the data would behave at each possible parameter value.

An estimator is a rule $g$ applied to the observations. Before observing $X$, its output $\widehat{\Theta} = g(X)$ is random because the data are random. After observing $X = x$, the estimate $g(x)$ is a number.

In Bayesian inference, the unknown is modeled as a random variable $\Theta$. A prior distribution $p_\Theta$ or $f_\Theta$ describes uncertainty about it before the data are observed. The conditional distribution of $X$ given $\Theta = \theta$ describes the observation model. Bayes' rule then combines the prior and the observed data.

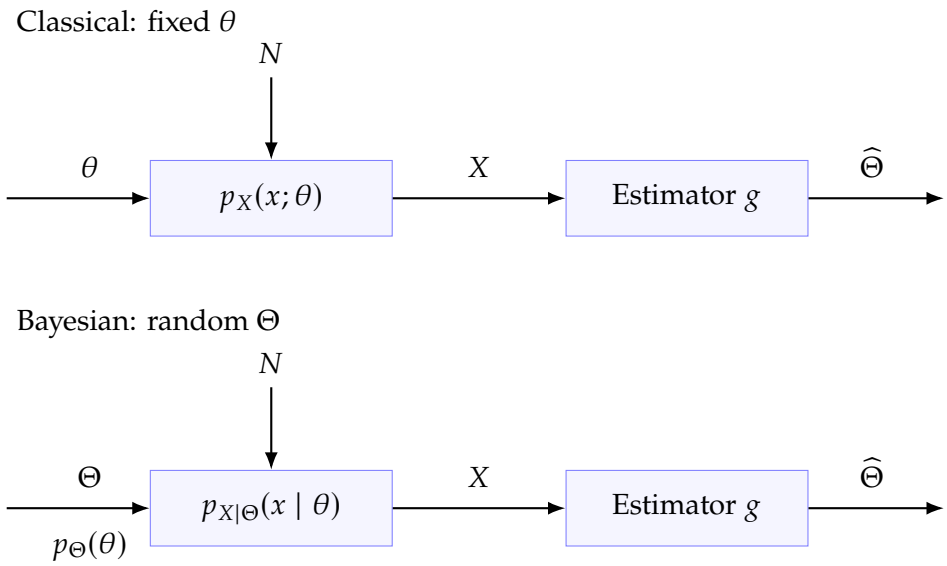



**Figure 72:** Both approaches send observations to an estimator. In the classical model, $\theta$ is fixed and unknown. In the Bayesian model, $\Theta$ has a prior distribution. $N$ represents the randomness or noise in generating $X$.

For a fixed observation $x$, the observation model viewed as a function of $\theta$ is the LIKELIHOOD. It measures how compatible each parameter value is with the data. It is not itself a probability distribution over the parameter.

### 21.3 Bayes' rule for inference

The POSTERIOR DISTRIBUTION describes uncertainty about $\Theta$ after observing $X = x$. Bayes' rule has the form

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{probability or density of the observation}}.$$

The denominator normalizes the posterior. It must be positive at the observation being conditioned on.

For a discrete unknown and discrete data,

$$\boxed{p_{\Theta|X}(\theta \mid x) = \frac{p_\Theta(\theta) p_{X|\Theta}(x \mid \theta)}{p_X(x)}}, \qquad p_X(x) = \sum_\theta p_\Theta(\theta) p_{X|\Theta}(x \mid \theta).$$

For a discrete unknown and continuous data,

$$\boxed{p_{\Theta|X}(\theta \mid x) = \frac{p_\Theta(\theta) f_{X|\Theta}(x \mid \theta)}{f_X(x)}}, \qquad f_X(x) = \sum_\theta p_\Theta(\theta) f_{X|\Theta}(x \mid \theta).$$

For a continuous unknown and continuous data,

$$\boxed{f_{\Theta|X}(\theta \mid x) = \frac{f_\Theta(\theta) f_{X|\Theta}(x \mid \theta)}{f_X(x)}}, \qquad f_X(x) = \int f_\Theta(\theta) f_{X|\Theta}(x \mid \theta)\, d\theta.$$

The posterior is a PMF or a density according to the type of the *unknown*, regardless of the type of the data.

For instance, consider a noisy quadratic signal:

$$Z_t = \Theta_0 + t\Theta_1 + t^2\Theta_2, \qquad X_t = Z_t + W_t, \quad t = 1, \ldots, n.$$

Let $\boldsymbol{\Theta} = (\Theta_0, \Theta_1, \Theta_2)$ and $\mathbf{X} = (X_1, \ldots, X_n)$. A joint prior and observation model give the joint posterior

$$f_{\boldsymbol{\Theta}|\mathbf{X}}(\boldsymbol{\theta} \mid \mathbf{x}) = \frac{f_{\boldsymbol{\Theta}}(\boldsymbol{\theta}) f_{\mathbf{X}|\boldsymbol{\Theta}}(\mathbf{x} \mid \boldsymbol{\theta})}{f_{\mathbf{X}}(\mathbf{x})}.$$

Here $f_{\mathbf{X}}(\mathbf{x})$ is found by integrating the numerator over the three coefficient values. This infers the coefficients together from the noisy measurements.

## 21.4 A continuous unknown with discrete data

Suppose a coin has an unknown probability $\Theta$ of heads. Given $\Theta = \theta$, the $n$ tosses are independent, and $X$ counts the heads. Thus

$$p_{X|\Theta}(x \mid \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}, \qquad x \in \{0, \ldots, n\}.$$

The data are discrete, but the unknown is continuous. Bayes' rule uses a prior density and an observation PMF:

$$\boxed{f_{\Theta|X}(\theta \mid x) = \frac{f_\Theta(\theta) p_{X|\Theta}(x \mid \theta)}{p_X(x)}},$$

$$p_X(x) = \int f_\Theta(\theta) p_{X|\Theta}(x \mid \theta)\, d\theta.$$

Choose a uniform prior on $[0, 1]$, so $f_\Theta(\theta) = 1$ there and zero elsewhere. The posterior on this interval is proportional to $\theta^x(1-\theta)^{n-x}$. To normalize it, repeated integration by parts gives

$$\int_0^1 \theta^x (1-\theta)^{n-x}\, d\theta = \frac{x!(n-x)!}{(n+1)!}.$$

Multiplying by the binomial coefficient yields

$$p_X(x) = \binom{n}{x} \frac{x!(n-x)!}{(n+1)!} = \frac{1}{n+1}.$$

Therefore the complete posterior density is

$$\boxed{f_{\Theta|X}(\theta \mid x) = (n+1)\binom{n}{x}\theta^x(1-\theta)^{n-x}, \quad 0 \le \theta \le 1},$$

and it is zero outside $[0, 1]$. Its integral is one. This distribution, rather than a single fitted number, is the immediate output of the Bayesian calculation.

## 21.5 Posterior distributions and point estimates

A posterior retains the range of plausible values and how much probability each region carries. A single estimate can hide this spread or the presence of several modes.

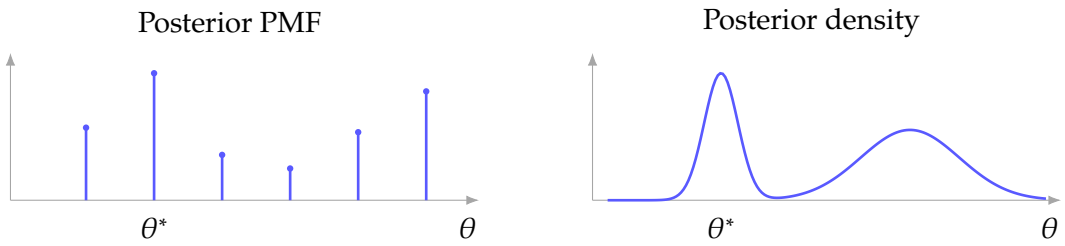



**Figure 73:** Schematic posterior PMF and density after a fixed observation $x$. The highest mass or density identifies a MAP choice $\theta^*$. A point estimate alone does not show the remaining uncertainty or a second mode.

A MAXIMUM A POSTERIORI (MAP) estimate selects a posterior maximizer. For a discrete unknown,

$$\theta^* \in \arg\max_{\theta} p_{\Theta|X}(\theta \mid x).$$

If we decide on a value $c$, the conditional probability of error is $1 - p_{\Theta|X}(c \mid x)$. Maximizing the posterior mass therefore minimizes this error probability, which makes MAP useful in hypothesis testing.

For a continuous unknown, the MAP rule instead maximizes the posterior *density*:

$$\theta^* \in \arg\max_{\theta} f_{\Theta|X}(\theta \mid x).$$

A density value is not a point probability, so the discrete error-probability argument does not apply to this formula.

Another possible point estimate is the posterior mean:

$$\mathbb{E}[\Theta \mid X = x] = \int \theta\, f_{\Theta|X}(\theta \mid x)\, d\theta$$

for a continuous unknown, with a sum replacing the integral for a discrete unknown. The mean and the mode need not coincide. The choice depends on how estimation error is measured.

## 21.6 Least mean squares without observations

Least mean squares (LMS) estimation minimizes expected squared error. Assume $\mathbb{E}[\Theta^2] < \infty$. Without any observations, the estimate is a constant $c$, and the objective is

$$\min_c \mathbb{E}[(\Theta - c)^2].$$

**Proposition 21.1 [Best constant estimate].** *The minimizing constant is $c = \mathbb{E}[\Theta]$, and the minimum mean squared error is* $\mathrm{Var}(\Theta)$.

*Proof.* Center at the mean $m = \mathbb{E}[\Theta]$ before expanding the square:

$$\begin{aligned}\mathbb{E}[(\Theta - c)^2] &= \mathbb{E}[((\Theta - m) + (m - c))^2] \\ &= \mathbb{E}[(\Theta - m)^2] + 2(m - c)\mathbb{E}[\Theta - m] + (m - c)^2 \\ &= \mathrm{Var}(\Theta) + (m - c)^2.\end{aligned}$$

The cross term is zero because $\mathbb{E}[\Theta - m] = 0$. The first term does not depend on $c$, while the second is minimized at $c = m$. □

Without data, the irreducible MSE is $\mathrm{Var}(\Theta)$; shifting from the mean adds $(c - m)^2$.

## 21.7 Least mean squares with observations

Once $X = x$ is observed, the relevant distribution is the posterior. Apply the same squared-error calculation within this conditional distribution. Write $m(x) = \mathbb{E}[\Theta \mid X = x]$. For observations where the conditional second moment is finite,

$$\mathbb{E}[(\Theta - c)^2 \mid X = x] = \mathrm{Var}(\Theta \mid X = x) + (m(x) - c)^2.$$

Thus the best estimate after seeing $x$ is $m(x)$. For any competing rule $g$, this gives

$$\mathbb{E}[(\Theta - m(x))^2 \mid X = x] \le \mathbb{E}[(\Theta - g(x))^2 \mid X = x].$$

**Theorem 21.2 [LMS estimator].** *If $\mathbb{E}[\Theta^2] < \infty$, the estimator*

$$\boxed{\widehat{\Theta} = \mathbb{E}[\Theta \mid X]}$$

*minimizes $\mathbb{E}[(\Theta - g(X))^2]$ over all estimators $g(X)$.*

*Proof.* The conditional inequality holds for almost every $X$:

$$\mathbb{E}[(\Theta - \mathbb{E}[\Theta \mid X])^2 \mid X] \leq \mathbb{E}[(\Theta - g(X))^2 \mid X].$$

Take expectations of both sides. The law of iterated expectations removes the outer conditioning, giving

$$\boxed{\mathbb{E}[(\Theta - \mathbb{E}[\Theta \mid X])^2] \leq \mathbb{E}[(\Theta - g(X))^2]}.$$

Rules with infinite mean squared error cannot improve on the conditional mean, which has finite mean squared error. □

For several observations, condition on all the available data. The best LMS estimator becomes

$$\boxed{\widehat{\Theta} = \mathbb{E}[\Theta \mid X_1, \ldots, X_n]}.$$

The principle is the same, but computing the joint posterior and its mean can require multidimensional integrals and may be difficult to implement.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 21, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 22: Bayesian Statistical Inference - II

The posterior mean minimizes squared error. Restricting to affine estimators gives a simpler problem determined by means, variances, and covariances.

## 22.1 LMS estimation and a uniform-noise model

For a continuous unknown, a MAP estimate maximizes the posterior density. The LMS estimator uses $\widehat{\Theta} = \mathbb{E}[\Theta \mid X]$ and minimizes $\mathbb{E}[(\Theta - g(X))^2]$ over all rules $g$. After observing $X = x$, the number $\mathbb{E}[\Theta \mid X = x]$ minimizes the conditional squared error over constant estimates. Assume $\Theta$ has a finite second moment.

Consider the model

$$\Theta \sim \text{Uniform}[4, 10], \qquad X = \Theta + W, \qquad W \sim \text{Uniform}[-1, 1],$$

where $W$ is independent of $\Theta$. Its prior and observation densities are

$$f_\Theta(\theta) = \frac{1}{6} \quad (4 \le \theta \le 10), \qquad f_{X|\Theta}(x \mid \theta) = \frac{1}{2} \quad (\theta - 1 \le x \le \theta + 1),$$

with zero density outside the stated intervals.

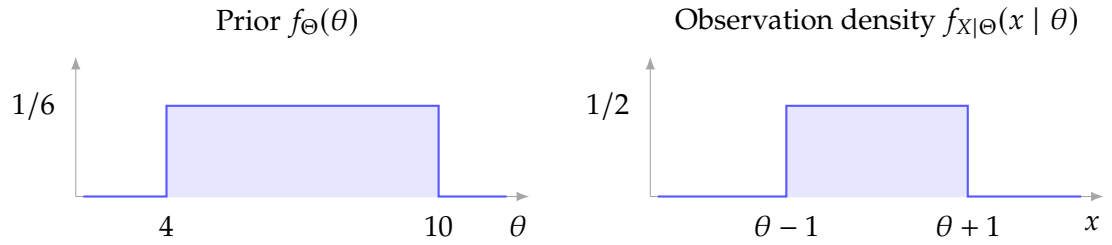



**Figure 74:** The prior spreads $\Theta$ uniformly from 4 to 10. Given $\Theta = \theta$, the observation is uniform from $\theta - 1$ to $\theta + 1$. The height and width of each rectangle give total area one.

Multiplying prior and observation density gives

$$f_{\Theta,X}(\theta, x) = \frac{1}{12} \quad \text{when } 4 \le \theta \le 10 \text{ and } |x - \theta| \le 1,$$

and zero otherwise. The joint support is a diagonal strip clipped at the two ends of the prior interval.

## 22.2 The posterior in the uniform-noise example

For a fixed observation $x$, the possible values of $\Theta$ must satisfy both the prior and the noise constraint:

$$a(x) = \max\{4, x-1\}, \qquad b(x) = \min\{10, x+1\}.$$

For $3 < x < 11$, define the positive interval length $\ell(x) = b(x) - a(x)$. Integrating the constant joint density gives

$$f_X(x) = \frac{\ell(x)}{12}, \qquad \boxed{f_{\Theta|X}(\theta \mid x) = \frac{1}{\ell(x)} \quad \text{for } a(x) \le \theta \le b(x)}.$$

The posterior is uniform on the vertical slice of the joint support, and is zero outside that slice.

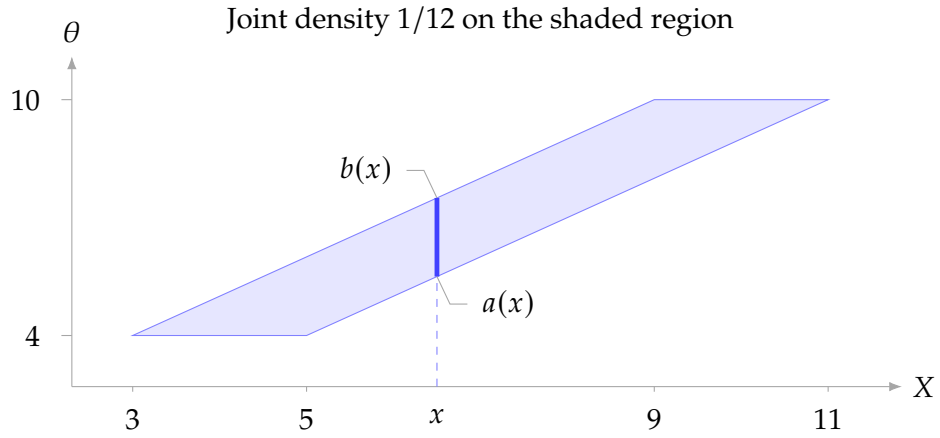



**Figure 75:** The shaded region has joint density $1/12$. Fixing $X = x$ selects a vertical interval from $a(x)$ to $b(x)$. Its midpoint is the LMS estimate; its length controls the conditional uncertainty.

The mean of a uniform interval is its midpoint. Hence

$$\boxed{m(x) = \mathbb{E}[\Theta \mid X = x] = \frac{a(x) + b(x)}{2}}$$

becomes

$$m(x) = \begin{cases} (x+5)/2, & 3 < x < 5, \\ x, & 5 \le x \le 9, \\ (x+9)/2, & 9 < x < 11. \end{cases}$$

In the middle, the noise interval is entirely inside the prior interval. Near an end, the prior clips the interval and moves its midpoint inward. The marginal density is zero at $x = 3$ and $x = 11$; the formulas have continuous limits there, but Bayes' density ratio is not defined at those endpoints.

### 22.3 Conditional mean squared error

For any observation $x$, the minimum conditional MSE is

$$\mathbb{E}[(\Theta - \mathbb{E}[\Theta \mid X])^2 \mid X = x] = \operatorname{Var}(\Theta \mid X = x).$$

This measures how much uncertainty remains after that particular observation, rather than after averaging over all data.

A uniform distribution on an interval of length $\ell$ has variance $\ell^2/12$. In the example,

$$v(x) = \operatorname{Var}(\Theta \mid X = x) = \frac{\ell(x)^2}{12} = \begin{cases} (x-3)^2/12, & 3 < x < 5, \\ 1/3, & 5 \le x \le 9, \\ (11-x)^2/12, & 9 < x < 11. \end{cases}$$

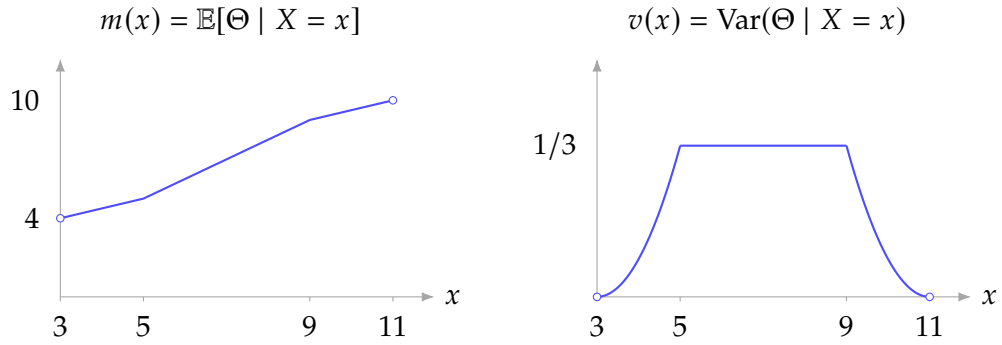



**Figure 76:** Calculated conditional mean and variance for the uniform-noise model. Near the ends, the posterior interval shortens and its variance decreases. Open circles show the limiting values at the zero-density endpoints.

The overall minimum MSE averages this conditional variance using the density of the observations:

$$\mathbb{E}[(\Theta - \widehat{\Theta})^2] = \mathbb{E}[v(X)] = \int_3^{11} v(x) f_X(x)\, dx.$$

Here $\ell(x) = x - 3$, 2, and $11 - x$ in the three regions. Since $v(x) f_X(x) = \ell(x)^3/144$,

$$\begin{aligned} \mathbb{E}[v(X)] &= \frac{1}{144}\left[\int_3^5 (x-3)^3\, dx + \int_5^9 8\, dx + \int_9^{11} (11-x)^3\, dx\right] \\ &= \frac{4 + 32 + 4}{144} = \boxed{\frac{5}{18}}. \end{aligned}$$

The conditional variance varies with $x$ even though its average is a single number.

## 22.4 Properties of the LMS error

Use the error convention

$$\widetilde{\Theta} = \widehat{\Theta} - \Theta, \qquad \widehat{\Theta} = \mathbb{E}[\Theta \mid X].$$

Conditioning on $X$ fixes the estimate, so

$$\mathbb{E}[\widetilde{\Theta} \mid X] = \widehat{\Theta} - \mathbb{E}[\Theta \mid X] = 0.$$

Taking expectations gives $\mathbb{E}[\widetilde{\Theta}] = 0$. Thus the error has zero mean overall and within the conditional distribution for almost every observation.

More generally, it is ORTHOGONAL to functions of the observations: their expected product with the error is zero. For any $h(X)$ with a finite second moment,

$$\begin{aligned}\mathbb{E}[\widetilde{\Theta}h(X)] &= \mathbb{E}\Big[\mathbb{E}[\widetilde{\Theta}h(X) \mid X]\Big] \\ &= \mathbb{E}\Big[h(X)\mathbb{E}[\widetilde{\Theta} \mid X]\Big] = 0.\end{aligned}$$

The condition on $h$ ensures that the product is integrable. Taking $h(X) = \widehat{\Theta}$ yields

$$\operatorname{Cov}(\widetilde{\Theta}, \widehat{\Theta}) = 0,$$

because $\mathbb{E}[\widetilde{\Theta}] = 0$.

Finally, $\Theta = \widehat{\Theta} - \widetilde{\Theta}$. Expanding its variance and using the zero covariance gives

$$\begin{aligned}\operatorname{Var}(\Theta) &= \operatorname{Var}(\widehat{\Theta}) + \operatorname{Var}(\widetilde{\Theta}) - 2\operatorname{Cov}(\widehat{\Theta}, \widetilde{\Theta}) \\ &= \boxed{\operatorname{Var}(\widehat{\Theta}) + \operatorname{Var}(\widetilde{\Theta})}.\end{aligned}$$

The variability of the unknown splits into variability of the estimate and variability of the remaining error. Since the error has mean zero,

$$\operatorname{Var}(\widetilde{\Theta}) = \mathbb{E}[\widetilde{\Theta}^2] = \mathbb{E}[\operatorname{Var}(\Theta \mid X)].$$

These statements describe zero covariance; they do not assert independence between the error and the estimate.

## 22.5 Linear LMS with one observation

LINEAR LMS restricts the estimate to $\widehat{\Theta}_L = aX + b$. The constant term makes this an affine function, though it is conventionally called linear. Assume $X$ also has a finite second moment and $\operatorname{Var}(X) > 0$. Write $\mu_\Theta = \mathbb{E}[\Theta]$, $\mu_X = \mathbb{E}[X]$, and $C = \operatorname{Cov}(X, \Theta)$.

For fixed $a$, the best constant is $b = \mu_\Theta - a\mu_X$. This follows by applying the best-constant rule to $\Theta - aX$. The remaining objective is

$$\begin{aligned}\mathbb{E}[(\Theta - \mu_\Theta - a(X - \mu_X))^2] &= \operatorname{Var}(\Theta) - 2aC + a^2 \operatorname{Var}(X) \\ &= \operatorname{Var}(\Theta) - \frac{C^2}{\operatorname{Var}(X)} + \operatorname{Var}(X)\left(a - \frac{C}{\operatorname{Var}(X)}\right)^2.\end{aligned}$$

The last term is nonnegative and vanishes at $a = C/\operatorname{Var}(X)$. Therefore

$$\boxed{\widehat{\Theta}_L = \mathbb{E}[\Theta] + \frac{\operatorname{Cov}(X, \Theta)}{\operatorname{Var}(X)}(X - \mathbb{E}[X])}.$$

If $\operatorname{Var}(X) = 0$, the observation is constant almost surely and the best affine estimate is simply $\mathbb{E}[\Theta]$.

When $\operatorname{Var}(\Theta) > 0$, let $\rho = C/\sqrt{\operatorname{Var}(X)\operatorname{Var}(\Theta)}$ be the correlation coefficient. The minimum linear MSE is

$$\boxed{\mathbb{E}[(\widehat{\Theta}_L - \Theta)^2] = \operatorname{Var}(\Theta) - \frac{C^2}{\operatorname{Var}(X)} = (1 - \rho^2)\operatorname{Var}(\Theta)}.$$

Only second-moment information is needed to find this estimate. It minimizes MSE within the affine class, while the conditional mean minimizes MSE over all estimators.

**Comparing the estimates in the uniform-noise model.** For this example,

$$\mathbb{E}[\Theta] = \mathbb{E}[X] = 7, \quad \operatorname{Var}(\Theta) = 3, \quad \operatorname{Var}(X) = 3 + \frac{1}{3} = \frac{10}{3}, \quad C = 3.$$

Thus $\widehat{\Theta}_L = 7 + \frac{9}{10}(X - 7) = \frac{9}{10}X + \frac{7}{10}$, with MSE $3 - 9/(10/3) = 3/10$, compared with the unrestricted LMS error $5/18$.

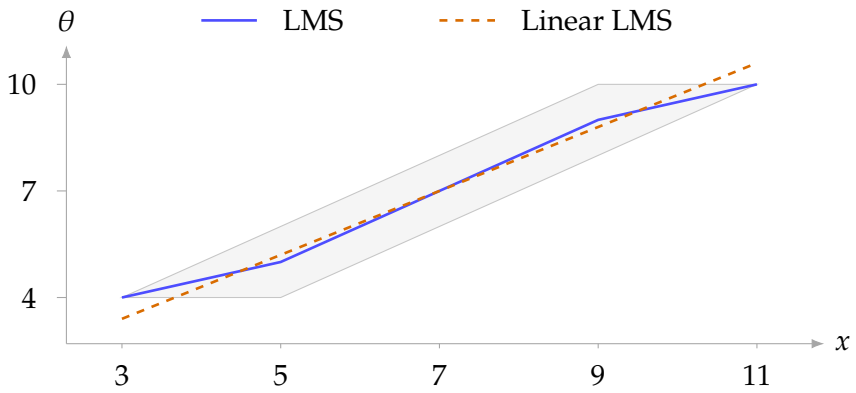



**Figure 77:** The LMS rule follows the posterior midpoint (blue). The best affine rule is a straight line (orange), so it cannot reproduce both changes of slope near the ends. The shaded region is the joint support.

## 22.6 Linear LMS with several observations

For data $X_1, \dots, X_n$, consider

$$\widehat{\Theta}_L = \sum_{j=1}^{n} a_j X_j + b.$$

Assume all variables have finite second moments. We choose the coefficients to minimize $\mathbb{E}[(\sum_j a_j X_j + b - \Theta)^2]$. Differentiating with respect to the constant term gives

$$b = \mathbb{E}[\Theta] - \sum_{j=1}^{n} a_j \mathbb{E}[X_j].$$

Let $U_j = X_j - \mathbb{E}[X_j]$ and $T = \Theta - \mathbb{E}[\Theta]$. The estimate is now $\mathbb{E}[\Theta] + \sum_j a_j U_j$, and its MSE is the quadratic expression

$$\mathbb{E}\left[\left(T - \sum_j a_j U_j\right)^2\right] = \operatorname{Var}(\Theta) - 2\sum_j a_j \operatorname{Cov}(X_j, \Theta) + \sum_{j,k} a_j a_k \operatorname{Cov}(X_j, X_k).$$

Differentiate with respect to each $a_i$ and set the derivative to zero. The resulting NORMAL EQUATIONS are

$$\boxed{\sum_{j=1}^{n} \operatorname{Cov}(X_i, X_j) a_j = \operatorname{Cov}(X_i, \Theta), \qquad i = 1, \dots, n}.$$

These are linear equations for the coefficients. They are sufficient for a minimum: changing a solution by coefficients $d_j$ increases the MSE by $\mathbb{E}[(\sum_j d_j U_j)^2] \geq 0$, because the cross term vanishes by the equations. Independence among the data is not required: the covariances describe their dependence.

The residual is orthogonal to constants and each input, hence to every affine combination of the data.

## 22.7 Independent noisy measurements

A useful model measures the same unknown several times:

$$X_i = \Theta + W_i, \qquad i = 1, \dots, n,$$

where $\Theta, W_1, \dots, W_n$ are independent, with

$$\mathbb{E}[\Theta] = \mu, \quad \operatorname{Var}(\Theta) = \sigma_0^2, \qquad \mathbb{E}[W_i] = 0, \quad \operatorname{Var}(W_i) = \sigma_i^2.$$

Assume $0 < \sigma_i^2 < \infty$ for $i = 0, \ldots, n$. The intercept rule gives $\widehat{\Theta}_L = \mu + \sum_i a_i(X_i - \mu)$. Its error has variance, and hence MSE,

$$R(\mathbf{a}) = \sigma_0^2 \left(1 - \sum_i a_i\right)^2 + \sum_i a_i^2 \sigma_i^2.$$

Independence makes all cross terms between the centered unknown and the noise variables vanish.

Setting the derivative with respect to $a_i$ to zero yields

$$a_i \sigma_i^2 = \sigma_0^2 \left(1 - \sum_j a_j\right).$$

Let $D = \sum_{j=0}^{n} 1/\sigma_j^2$. Solving these equations gives $a_i = (1/\sigma_i^2)/D$ and $1 - \sum_i a_i = (1/\sigma_0^2)/D$. Consequently,

$$\boxed{\widehat{\Theta}_L = \frac{\mu/\sigma_0^2 + \displaystyle\sum_{i=1}^{n} X_i/\sigma_i^2}{\displaystyle\sum_{i=0}^{n} 1/\sigma_i^2}}.$$

The reciprocal variance is called PRECISION. This is a weighted average of the prior mean and the observations: higher precision receives greater weight. Substitution into $R$ gives a minimum MSE of $1/D$.

If the unknown and all noises are normal, this linear rule is also the unrestricted LMS estimator. Indeed, the posterior density is proportional to

$$\exp\left\{-\frac{1}{2}\left[\frac{(\theta - \mu)^2}{\sigma_0^2} + \sum_{i=1}^{n} \frac{(x_i - \theta)^2}{\sigma_i^2}\right]\right\}.$$

Completing the square gives a normal density centered at the displayed weighted average, so $\widehat{\Theta}_L = \mathbb{E}[\Theta \mid X_1, \ldots, X_n]$. Without normality, the formula still gives the best linear estimator.

## 22.8 Choosing inputs and comparing methods

An unrestricted estimator can use all functions of its data. For example, observing $X^3$ provides exactly the same information as observing $X$, because the real cube root recovers $X$. Therefore

$$\mathbb{E}[\Theta \mid X] = \mathbb{E}[\Theta \mid X^3].$$

Linear LMS depends on which inputs we choose. The classes $aX + b$ and $aX^3 + b$ contain different functions of $X$. We can also use several FEATURES, meaning chosen functions of the observation, as

linear inputs:

$$\widehat{\Theta} = a_1 X + a_2 X^2 + a_3 X^3 + b.$$

This is linear in the coefficients and the supplied features, even though it is generally nonlinear in $X$. The chosen features must have finite second moments to apply the covariance equations.

Common observation/prior combinations include uniform observations on $[0, \theta]$ with a uniform prior on the unknown positive endpoint; Bernoulli observations with a uniform or Beta prior on their success probability; and normal observations with unknown mean, known variance, and a normal prior on that mean.

The estimation rules serve different objectives:

- MAP chooses a posterior maximizer. For discrete hypotheses, it minimizes the probability of an incorrect decision.
- LMS uses the posterior mean and minimizes MSE over all estimators based on the observations.
- Linear LMS minimizes MSE over affine functions of the chosen inputs, using their first and second moments.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 22, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 23: Classical Statistical Inference - I

Classical inference studies an estimator's behavior over repeated samples, treating the unknown parameter as fixed.

## 23.1 A fixed unknown parameter

In classical statistics, the unknown parameter $\theta$ is a fixed number. The observations have a model $p_X(x;\theta)$, where $p_X$ denotes a probability mass function or a density, as appropriate. The semicolon separates the data from the parameter: it does not indicate conditioning on a random variable $\Theta$.

Each possible $\theta$ specifies a different probability model. Write $\Pr_\theta$ and $\mathbb{E}_\theta$ for probability and expectation under that model. An estimator is a rule $\widehat{\Theta} = g(X)$. It is random before the data arrive; its observed value $\widehat{\theta} = g(x)$ is an estimate.

The same setup allows vector data and vector parameters:

$$p_{X_1,\dots,X_n}(x_1,\dots,x_n;\theta_1,\dots,\theta_m).$$

Two main questions are:

- Hypothesis testing: decide between parameter claims. For example, $H_0 : \theta = 1/2$ versus $H_1 : \theta = 3/4$. A composite hypothesis allows several parameter values, as in $H_1 : \theta \neq 1/2$.
- Estimation: choose a rule whose error $\widehat{\Theta} - \theta$ is small under each possible model.

The unknown parameter stays fixed in these probability calculations. The observations and the estimator vary.

## 23.2 Maximum likelihood estimation

After observing $x$, ask which parameter value makes that data most plausible under the model. The likelihood is $L(\theta;x) = p_X(x;\theta)$, viewed as a function of $\theta$ with $x$ held fixed. The maximum likelihood (ML) estimate is

$$\boxed{\widehat{\theta}_{\mathrm{ML}} \in \arg\max_\theta p_X(x;\theta)}.$$

For continuous observations, this maximizes a density, not the probability of the single point $X = x$.

Bayesian MAP estimation instead maximizes the posterior:

$$\widehat{\theta}_{\mathrm{MAP}} \in \arg\max_{\theta} \frac{p_{X|\Theta}(x \mid \theta) p_{\Theta}(\theta)}{p_X(x)}.$$

The denominator does not depend on $\theta$. MAP combines the likelihood with a prior; ML uses the sampling model alone.

**Exponential rate.** Let $X_1, \ldots, X_n$ be independent exponential variables with unknown rate $\theta > 0$:

$$p_{X_i}(x_i; \theta) = \theta e^{-\theta x_i} \quad (x_i \geq 0), \qquad \mathbb{E}_\theta[X_i] = \frac{1}{\theta}.$$

Independence gives

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} \theta e^{-\theta x_i} = \theta^n e^{-\theta \sum_i x_i}.$$

Taking a logarithm preserves the maximizer because the logarithm is increasing. Thus maximize

$$\ell(\theta) = n \log \theta - \theta \sum_{i=1}^{n} x_i.$$

When $\sum_i x_i > 0$, differentiate:

$$\ell'(\theta) = \frac{n}{\theta} - \sum_i x_i, \qquad \ell''(\theta) = -\frac{n}{\theta^2} < 0.$$

The unique stationary point is the global maximum, giving

$$\boxed{\widehat{\theta}_{\mathrm{ML}} = \frac{n}{\sum_i x_i}}, \qquad \boxed{\widehat{\Theta}_n = \frac{n}{\sum_i X_i}}.$$

The sum is positive almost surely in this model. The rule estimates the rate by the reciprocal of the sample mean.

### 23.3 Bias, consistency, and mean squared error

An estimator's desirable properties should hold for every allowed value of $\theta$. They concern its sampling distribution, rather than just the one observed estimate.

An estimator is UNBIASED if $\mathbb{E}_\theta[\widehat{\Theta}_n] = \theta$ for every $\theta$. When its expectation is finite, its BIAS is

$$b_n(\theta) = \mathbb{E}_\theta[\widehat{\Theta}_n] - \theta.$$

ML estimation does not automatically give an unbiased rule. For the exponential example with $n = 1$,

$$\mathbb{E}_\theta\left[\frac{1}{X_1}\right] = \int_0^\infty \frac{\theta e^{-\theta x}}{x}\, dx = \infty.$$

On $0 < x < 1/\theta$, the exponential factor is at least $e^{-1}$, and the integral of $1/x$ diverges at zero. This estimator is not unbiased and does not even have a finite mean.

A sequence of estimators is CONSISTENT if, for each $\theta$ and every $\varepsilon > 0$,

$$\Pr_\theta\big(|\widehat{\Theta}_n - \theta| > \varepsilon\big) \longrightarrow 0.$$

This is convergence in probability as the sample grows. In the exponential example, the weak law of large numbers gives

$$M_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\Pr_\theta} \frac{1}{\theta}.$$

The limiting mean is positive, so the reciprocal converges to $\theta$. More explicitly, if $|M_n - 1/\theta| \le 1/(2\theta)$, then

$$\left|\frac{1}{M_n} - \theta\right| = \frac{\theta|M_n - 1/\theta|}{M_n} \le 2\theta^2|M_n - 1/\theta|.$$

The right side tends to zero in probability, and the stated event has probability tending to one. Thus $\widehat{\Theta}_n = 1/M_n$ is consistent.

For an estimator with a finite second moment, its MEAN SQUARED ERROR satisfies

$$\boxed{\mathbb{E}_\theta[(\widehat{\Theta}_n - \theta)^2] = \mathrm{Var}_\theta(\widehat{\Theta}_n) + b_n(\theta)^2}.$$

To verify this, write the error as $(\widehat{\Theta}_n - \mathbb{E}_\theta[\widehat{\Theta}_n]) + b_n(\theta)$. Squaring and averaging removes the cross term because the centered estimator has mean zero. Low MSE requires control of both variability and bias.

## 23.4 Estimating a mean

Suppose $X_1, \dots, X_n$ are i.i.d. with mean $\theta$ and finite variance $\sigma^2$. Equivalently,

$$X_i = \theta + W_i, \qquad \mathbb{E}_\theta[W_i] = 0, \quad \mathrm{Var}_\theta(W_i) = \sigma^2,$$

where the noises $W_i$ are i.i.d. The natural estimator is the SAMPLE MEAN

$$\widehat{\Theta}_n = M_n = \frac{X_1 + \cdots + X_n}{n}.$$

Linearity of expectation gives

$$\mathbb{E}_\theta[M_n] = \frac{1}{n}\sum_i \mathbb{E}_\theta[X_i] = \theta,$$

so it is unbiased. Independence makes the covariance terms vanish, giving

$$\mathrm{Var}_\theta(M_n) = \frac{1}{n^2}\sum_i \mathrm{Var}_\theta(X_i) = \frac{\sigma^2}{n}.$$

Consequently,

$$\boxed{\mathbb{E}_\theta[(M_n - \theta)^2] = \frac{\sigma^2}{n}}.$$

The weak law of large numbers also gives $M_n \xrightarrow{\Pr_\theta} \theta$, so the sample mean is consistent. More observations reduce its standard deviation to $\sigma/\sqrt{n}$.

The sample mean is also the ML estimate in some models. For i.i.d. $X_i \sim \mathcal{N}(\theta, \sigma^2)$ with known $\sigma^2 > 0$, the log likelihood is, apart from a term independent of $\theta$,

$$-\frac{1}{2\sigma^2}\sum_i (x_i - \theta)^2.$$

Let $\overline{x} = n^{-1}\sum_i x_i$. The identity

$$\sum_i (x_i - \theta)^2 = \sum_i (x_i - \overline{x})^2 + n(\overline{x} - \theta)^2$$

follows by expanding around $\overline{x}$ and using $\sum_i (x_i - \overline{x}) = 0$. Only the last term depends on $\theta$, and it is minimized at $\theta = \overline{x}$. Hence the sample mean is the ML rule for the normal mean.

## 23.5 What a confidence interval means

A point estimate alone does not describe its uncertainty. A CONFIDENCE INTERVAL uses the data to produce random endpoints $\widehat{\Theta}_n^-$ and $\widehat{\Theta}_n^+$. It has confidence level $1 - \alpha$ if, for every allowed $\theta$,

$$\boxed{\Pr_\theta\big(\widehat{\Theta}_n^- \le \theta \le \widehat{\Theta}_n^+\big) \ge 1 - \alpha}.$$

Typical choices include $\alpha = 0.05$, 0.25, and 0.01. The probability is over the data that generate the interval.

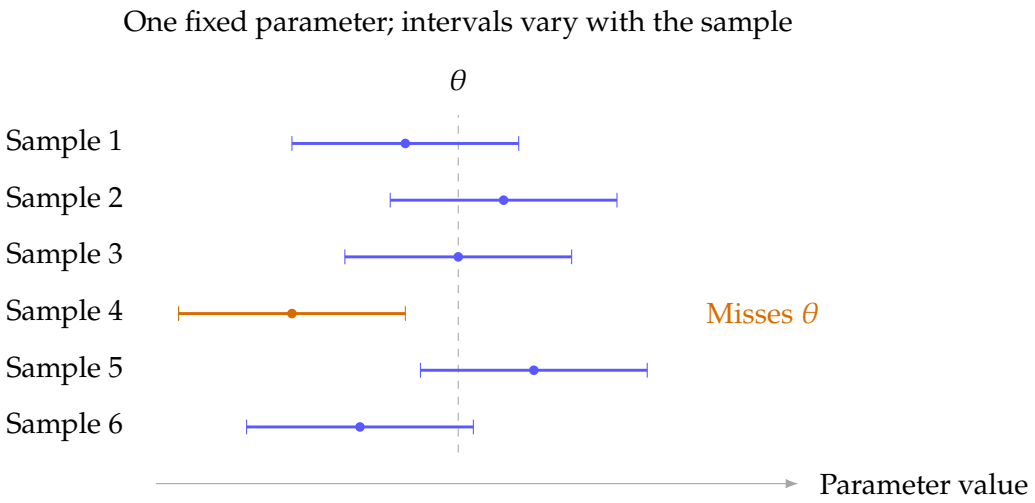



**Figure 78:** Schematic repeated intervals at one fixed value of $\theta$. The interval moves with the sample; the parameter stays in place. Each dot marks an interval's midpoint. Blue intervals cover $\theta$ and an orange interval misses it. These are illustrative intervals, not data or an empirical coverage calculation.

Coverage concerns repeated sampling: the procedure must contain the fixed parameter with probability at least $1 - \alpha$ at every allowed parameter value.

Once the data are observed, the endpoints are numbers. The resulting interval either contains $\theta$ or it does not. Classical confidence does not assign a posterior probability to the fixed parameter inside that observed interval. The probability statement describes the sampling procedure that produced it.

### 23.6 A confidence interval for the mean

For i.i.d. observations with mean $\theta$ and known variance $\sigma^2 > 0$, the central limit theorem gives

$$Z_n = \frac{M_n - \theta}{\sigma/\sqrt{n}} \quad \text{approximately } \mathcal{N}(0,1) \quad \text{for large } n.$$

Let $\Phi$ be the standard normal CDF. Choose $z > 0$ so that $\Phi(z) = 1 - \alpha/2$. Symmetry then gives

$$\Pr(-z \le Z \le z) = 1 - \alpha \quad \text{for } Z \sim \mathcal{N}(0,1).$$

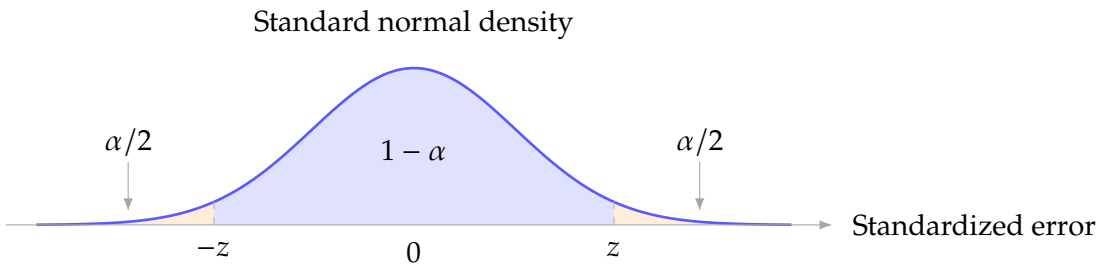



**Figure 79:** The standard normal central area is $1-\alpha$; each tail has area $\alpha/2$. The cutoffs $-z$ and $z$ become the two endpoints of a confidence interval after rearranging the standardized-error inequality.

Applying the normal approximation and rearranging gives

$$\Pr_\theta\left(|M_n-\theta|\le\frac{z\sigma}{\sqrt{n}}\right)=\Pr_\theta\left(M_n-\frac{z\sigma}{\sqrt{n}}\le\theta\le M_n+\frac{z\sigma}{\sqrt{n}}\right)\approx 1-\alpha.$$

Thus the interval is

$$\boxed{\left[M_n-\frac{z\sigma}{\sqrt{n}},\,M_n+\frac{z\sigma}{\sqrt{n}}\right]}.$$

For $\alpha=0.05$, $z\approx 1.96$, giving a 95% interval. The margin of error is $z\sigma/\sqrt{n}$: it grows with noise and shrinks with sample size. Seeking higher confidence increases $z$ and widens the interval.

If the observations themselves are normal, $Z_n$ is standard normal exactly, so the formula has exact coverage with the exact normal quantile. For general i.i.d. data, the CLT supplies a large-sample approximation, rather than an exact finite-sample guarantee.

## 23.7 When the standard deviation is unknown

The normal-based interval needs $\sigma$, but the sampling variance may also be unknown. Two approaches use additional information about the observation model.

**Use a known upper bound.** If $\sigma\le B$, use the wider margin $zB/\sqrt{n}$. For Bernoulli observations with success probability $\theta$, the mean is $\theta$ and

$$\sigma^2=\theta(1-\theta)=\frac{1}{4}-\left(\theta-\frac{1}{2}\right)^2\le\frac{1}{4}.$$

Therefore $B=1/2$ gives

$$\left[M_n-\frac{z}{2\sqrt{n}},\,M_n+\frac{z}{2\sqrt{n}}\right].$$

Within the normal approximation, the bound makes the interval at least as wide as the one using the true $\sigma$. It does not turn the approximation into an exact finite-sample guarantee.

**Estimate the variance from the model.** For Bernoulli data, replace the unknown success probability in $\theta(1-\theta)$ by $M_n$:

$$\widehat{\sigma} = \sqrt{M_n(1-M_n)}.$$

The resulting PLUG-IN interval is

$$\left[ M_n - z\sqrt{\frac{M_n(1-M_n)}{n}},\ M_n + z\sqrt{\frac{M_n(1-M_n)}{n}} \right].$$

For a fixed $0 < \theta < 1$, consistency of $M_n$ makes the estimated standard deviation converge to the positive true value. This supports a large-sample normal approximation. An all-zero or all-one sample makes the estimated margin zero, illustrating why this formula does not promise finite-sample coverage.

A more general approach estimates variance directly from the observed deviations about the sample mean.

## 23.8 Estimating the variance from the sample

Start from $\sigma^2 = \mathbb{E}_\theta[(X_i - \theta)^2]$. If $\theta$ were known, the average

$$\frac{1}{n}\sum_i (X_i - \theta)^2 \xrightarrow{\Pr_\theta} \sigma^2$$

would estimate the variance by the law of large numbers. Because $\theta$ is unknown, center at $M_n$ instead. For $n \geq 2$, define the SAMPLE VARIANCE

$$\boxed{\widehat{S}_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - M_n)^2}.$$

Using the estimated mean reduces the expected sum of squared deviations. The divisor $n-1$ corrects this effect. To see it, expand $X_i - M_n = (X_i - \theta) - (M_n - \theta)$:

$$\sum_i (X_i - M_n)^2 = \sum_i (X_i - \theta)^2 - n(M_n - \theta)^2.$$

Here $\sum_i (X_i - \theta) = n(M_n - \theta)$ combines the cross terms. Taking expectations yields

$$\begin{aligned}\mathbb{E}_\theta\left[\sum_i (X_i - M_n)^2\right] &= n\sigma^2 - n\mathbb{E}_\theta[(M_n - \theta)^2] \\ &= n\sigma^2 - n\frac{\sigma^2}{n} = (n-1)\sigma^2.\end{aligned}$$

Consequently $\mathbb{E}_\theta[\widehat{S}_n^2] = \sigma^2$: the sample variance is unbiased.

The same identity proves consistency:

$$\widehat{S}_n^2 = \frac{n}{n-1}\left[\frac{1}{n}\sum_i (X_i-\theta)^2 - (M_n-\theta)^2\right] \xrightarrow{\Pr_\theta} \sigma^2.$$

The first average converges to $\sigma^2$, the squared mean error tends to zero, and $n/(n-1)$ tends to one. No normality is needed for these variance-estimator properties.

Let $\widehat{S}_n = \sqrt{\widehat{S}_n^2}$. When $\sigma > 0$, replacing $\sigma$ by this consistent estimate gives the large-sample interval

$$\left[M_n - \frac{z\widehat{S}_n}{\sqrt{n}},\ M_n + \frac{z\widehat{S}_n}{\sqrt{n}}\right].$$

The estimated scale converges to the true scale, so the standardized mean retains its limiting normal distribution. Its coverage is approximate at finite $n$.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 23, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 24: Classical Inference - II

Regression fits relationships between observed variables; hypothesis testing chooses between probability models.

## 24.1 Likelihood and confidence: a review

In a classical model $X \sim p_X(x;\theta)$, the unknown parameter $\theta$ is fixed. Maximum likelihood chooses an estimate that maximizes the observed-data likelihood:

$$\widehat{\theta}_{\mathrm{ML}} \in \arg\max_{\theta} p_X(x;\theta).$$

Here $p_X$ is a mass function or a density, as appropriate. Bayesian MAP estimation instead maximizes

$$p_{\Theta|X}(\theta \mid x) = \frac{p_{X|\Theta}(x \mid \theta)p_\Theta(\theta)}{p_X(x)}.$$

The marginal in the denominator is constant with respect to $\theta$; the prior in the numerator can change the maximizer.

For i.i.d. observations with mean $\theta$ and finite variance $\sigma^2$, the sample mean is

$$M_n = \frac{X_1 + \cdots + X_n}{n}.$$

A confidence interval has random endpoints, with the coverage requirement

$$\Pr_\theta(\widehat{\Theta}_n^- \le \theta \le \widehat{\Theta}_n^+) \ge 1-\alpha \quad \text{for every allowed } \theta.$$

This probability describes repeated samples at a fixed parameter value. For known $\sigma > 0$, choosing $\Phi(z) = 1 - \alpha/2$ gives the normal-based interval

$$\left[M_n - \frac{z\sigma}{\sqrt{n}},\ M_n + \frac{z\sigma}{\sqrt{n}}\right].$$

Its coverage is approximately $1-\alpha$ for large $n$ by the CLT. It is exact for normal observations with the exact normal quantile.

## 24.2 Regression, residuals, and least squares

Given pairs $(x_1, y_1), \dots, (x_n, y_n)$, try to describe their relation by $y \approx \theta_0 + \theta_1 x$. Linear regression chooses the intercept and slope of this line from the data. For fitted parameters, the predicted value is

$$\widehat{y}_i = \widehat{\theta}_0 + \widehat{\theta}_1 x_i.$$

The residual is the observed value minus its prediction:

$$r_i = y_i - \widehat{y}_i = y_i - \widehat{\theta}_0 - \widehat{\theta}_1 x_i.$$

Least squares minimizes the sum of squared residuals:

$$(\widehat{\theta}_0, \widehat{\theta}_1) \in \arg\min_{\theta_0, \theta_1} S(\theta_0, \theta_1), \qquad S = \sum_{i=1}^{n} (y_i - \theta_0 - \theta_1 x_i)^2.$$

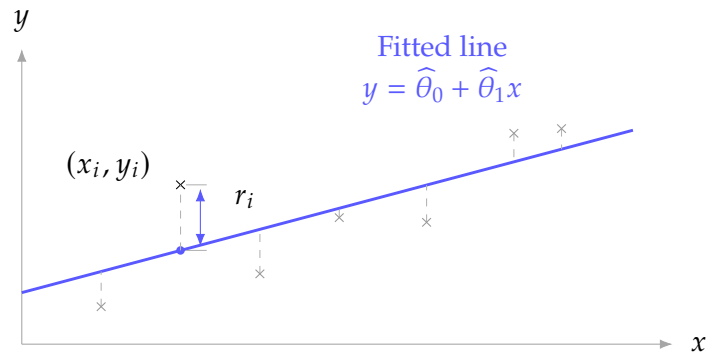



**Figure 80:** Schematic data and a fitted line. The highlighted residual is a vertical difference at the same $x_i$. Least squares minimizes these squared vertical differences, rather than distances perpendicular to the line.

One probabilistic interpretation treats the predictors as fixed and assumes

$$Y_i = \theta_0 + \theta_1 x_i + W_i, \qquad W_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.}$$

For fixed $\sigma^2 > 0$, the likelihood of the observed responses is

$$L(\theta_0, \theta_1) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i} (y_i - \theta_0 - \theta_1 x_i)^2 \right\}.$$

Taking logs leaves a constant minus $S/(2\sigma^2)$. Thus maximizing this likelihood is equivalent to least squares. The least-squares criterion itself can also be used without assuming normal errors.

## 24.3 Deriving the fitted intercept and slope

Define the empirical means

$$\overline{x} = \frac{1}{n}\sum_i x_i, \qquad \overline{y} = \frac{1}{n}\sum_i y_i.$$

Setting the two derivatives of $S$ to zero gives

$$\sum_i (y_i - \theta_0 - \theta_1 x_i) = 0, \qquad \sum_i x_i(y_i - \theta_0 - \theta_1 x_i) = 0.$$

The first equation implies $\theta_0 = \overline{y} - \theta_1 \overline{x}$. Substituting it centers the residuals:

$$y_i - \theta_0 - \theta_1 x_i = (y_i - \overline{y}) - \theta_1(x_i - \overline{x}).$$

Introduce the centered sums

$$S_{xx} = \sum_i (x_i - \overline{x})^2, \qquad S_{xy} = \sum_i (x_i - \overline{x})(y_i - \overline{y}).$$

When $S_{xx} > 0$, the second equation reduces to $S_{xy} - \theta_1 S_{xx} = 0$. Therefore

$$\boxed{\widehat{\theta}_1 = \frac{S_{xy}}{S_{xx}}}, \qquad \boxed{\widehat{\theta}_0 = \overline{y} - \widehat{\theta}_1 \overline{x}}.$$

The fitted line passes through $(\overline{x}, \overline{y})$. The slope measures how centered responses vary with centered predictors.

The derivative equations give the minimum, not just a stationary point. At the fitted parameters, the residuals satisfy $\sum_i r_i = 0$ and $\sum_i x_i r_i = 0$. Perturbing the intercept and slope by $a$ and $b$ changes the objective to

$$\sum_i (r_i - a - b x_i)^2 = \sum_i r_i^2 + \sum_i (a + b x_i)^2 \geq \sum_i r_i^2.$$

The cross term vanishes by those two equations.

If $S_{xx} = 0$, all predictors equal one value $c$. The data then identify only the fitted value $\theta_0 + \theta_1 c = \overline{y}$, rather than a unique intercept and slope. The displayed division by $S_{xx}$ requires variation in the predictors.

## 24.4 The covariance interpretation

Consider the population model

$$Y = \theta_0 + \theta_1 X + W,$$

where $W$ is independent of $X$, $\mathbb{E}[W] = 0$, all variables have finite second moments, and $\operatorname{Var}(X) > 0$. The zero-mean noise gives

$$\mathbb{E}[Y] = \theta_0 + \theta_1 \mathbb{E}[X].$$

Centering the model yields $Y - \mathbb{E}[Y] = \theta_1(X - \mathbb{E}[X]) + W$. Multiply by the centered predictor and take expectations:

$$\begin{aligned}\operatorname{Cov}(X, Y) &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \theta_1 \operatorname{Var}(X) + \mathbb{E}[(X - \mathbb{E}[X])W] \\ &= \theta_1 \operatorname{Var}(X).\end{aligned}$$

Independence and the zero means make the last cross term zero. Hence

$$\boxed{\theta_1 = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(X)}}, \qquad \theta_0 = \mathbb{E}[Y] - \theta_1 \mathbb{E}[X].$$

The fitted formula replaces these population quantities by their sample counterparts. For $n \geq 2$, use

$$\widehat{\operatorname{Cov}}(X, Y) = \frac{S_{xy}}{n-1}, \qquad \widehat{\operatorname{Var}}(X) = \frac{S_{xx}}{n-1}.$$

Their common factor cancels, giving $\widehat{\theta}_1 = S_{xy}/S_{xx}$. The fitted intercept likewise replaces the population means by $\overline{x}$ and $\overline{y}$.

The least-squares minimization itself does not require the independent-noise model.

## 24.5 Several predictors and transformed variables

Multiple linear regression allows several observed predictors in each data row. For data $(x_i, x_i', x_i'', y_i)$, consider

$$y \approx \theta_0 + \theta x + \theta' x' + \theta'' x''.$$

The primes label different predictors, not derivatives. The corresponding least-squares problem is

$$\min_{\theta_0, \theta, \theta', \theta''} \sum_{i=1}^{n} (y_i - \theta_0 - \theta x_i - \theta' x_i' - \theta'' x_i'')^2.$$

Each prediction combines the supplied predictors through unknown coefficients. The intercept is optimized along with all the slopes.

Choosing predictors also includes choosing functions of an original observation. Given a specified function $h$, fit

$$y \approx \theta_0 + \theta_1 h(x)$$

by solving

$$\min_{\theta_0,\theta_1} \sum_{i=1}^{n} (y_i - \theta_0 - \theta_1 h(x_i))^2.$$

This is the same two-parameter fitting problem applied to pairs $(h(x_i), y_i)$. For example, $h(x) = x^2$ gives $y \approx \theta_0 + \theta_1 x^2$. The model is linear in the unknown coefficients even though its graph as a function of $x$ need not be a straight line.

## 24.6 Regression reports and model concerns

A regression report commonly includes uncertainty and fit information alongside the coefficient estimates:

- Confidence intervals for the coefficients describe their sampling uncertainty under the assumed model.
- The residual STANDARD ERROR estimates the noise standard deviation $\sigma$ in the regression model.
- The COEFFICIENT OF DETERMINATION, $R^2$, measures the fraction of observed response variation accounted for by the fitted model.

For a least-squares fit with an intercept and nonconstant responses, the usual definition is

$$R^2 = 1 - \frac{\sum_i (y_i - \widehat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}.$$

The denominator is the squared error from using the same prediction $\overline{y}$ for every response. The numerator is the squared error remaining after the regression fit. If all responses are equal, the denominator is zero and this formula is undefined.

The quality of a fit also depends on the model and the predictors. Two common concerns are:

- HETEROSKEDASTICITY: the noise variance changes with the predictors. This violates the constant variance assumption used in the simple normal-noise interpretation.
- MULTICOLLINEARITY: predictors are close to linear combinations of one another. Their separate coefficients can then be difficult to determine reliably.

Finally, a fitted association alone does not establish a causal relationship. Least squares describes patterns in the supplied observations; a causal conclusion requires justification beyond a good regression fit.

## 24.7 Binary tests, rejection regions, and errors

In binary hypothesis testing, two candidate models are specified. The NULL HYPOTHESIS $H_0$ and the ALTERNATIVE HYPOTHESIS $H_1$ give

$$X \sim p_X(x; H_0), \qquad X \sim p_X(x; H_1),$$

or densities $f_X(x; H_0)$ and $f_X(x; H_1)$ for continuous data. A REJECTION REGION $R$ is a set of possible data: reject $H_0$ exactly when $X \in R$. The complement $R^c$ consists of data for which the test retains $H_0$.

There are two ways to make an error:

- A TYPE I ERROR, or false alarm, rejects a true $H_0$. Its probability is $\alpha(R) = \Pr_{H_0}(X \in R)$.
- A TYPE II ERROR, or missed detection, retains $H_0$ when $H_1$ is true. Its probability is $\beta(R) = \Pr_{H_1}(X \notin R)$.

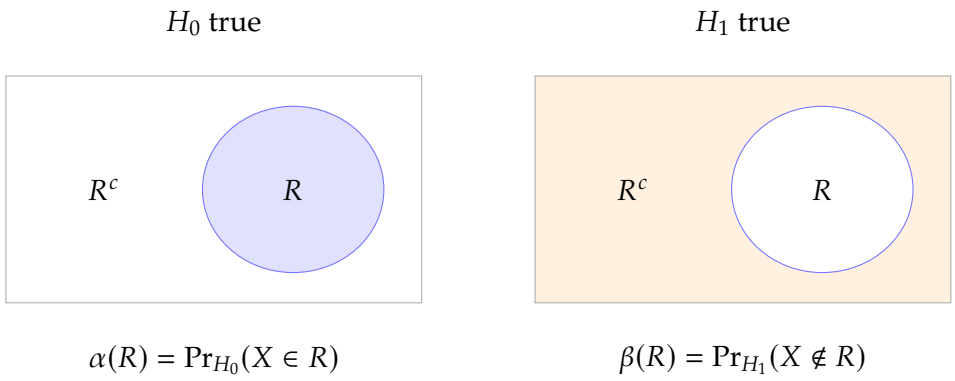



**Figure 81:** Schematic copies of the same data space and rejection region. Under $H_0$, an error occurs inside $R$ (blue). Under $H_1$, an error occurs outside $R$ (orange). Shading identifies events, not numerical probabilities.

| Decision | $H_0$ true | $H_1$ true |
|---|---|---|
| Retain $H_0$ | Correct decision | Type II error |
| Reject $H_0$ | Type I error | Correct decision |

The probabilities use different sampling models: $\alpha(R)$ is evaluated under $H_0$ and $\beta(R)$ under $H_1$. Retaining $H_0$ is a test decision, rather than proof that it is true.

## 24.8 From MAP to a likelihood-ratio test

Write $q_j(x)$ for the data mass function or density under $H_j$. The LIKELIHOOD RATIO compares how the two models explain the same observation:

$$\Lambda(x) = \frac{q_1(x)}{q_0(x)}.$$

Large values favor $H_1$ relative to $H_0$. This is a ratio of likelihoods, not a posterior probability.

In a Bayesian model, let $\pi_j = \Pr(H_j) > 0$, with $\pi_0 + \pi_1 = 1$. The MAP rule chooses $H_1$ when

$$\Pr(H_1 \mid X = x) > \Pr(H_0 \mid X = x).$$

Bayes' formula gives a common marginal denominator, which cancels from the comparison. Thus

$$q_1(x)\pi_1 > q_0(x)\pi_0, \quad \text{equivalently} \quad \boxed{\Lambda(x) > \frac{\pi_0}{\pi_1}}$$

when $q_0(x) > 0$. The prior odds supply the threshold. At equal posterior probabilities, either decision is a MAP choice.

In the classical setting, the hypotheses specify sampling models without prior probabilities. A LIKELIHOOD-RATIO TEST (LRT) instead fixes a threshold $\xi > 0$ and rejects $H_0$ when

$$\boxed{R_\xi = \{x : \Lambda(x) > \xi\}}.$$

For discrete data, $\Lambda(x) = p_X(x; H_1)/p_X(x; H_0)$; for continuous data, it is $f_X(x; H_1)/f_X(x; H_0)$. The strict inequality specifies that ties retain $H_0$.

If $q_0(x) = 0$ but $q_1(x) > 0$, interpret the ratio as infinite and reject $H_0$. If both likelihoods are zero, that observation has zero mass or density under both models and the ratio does not define a comparison there.

## 24.9 Choosing the rejection threshold

The threshold trades false alarms against missed detections. If $\xi_2 > \xi_1$, then

$$R_{\xi_2} \subseteq R_{\xi_1} :$$

data must give stronger relative support for $H_1$ to pass the higher threshold. Taking probabilities of these nested events gives

$$\boxed{\alpha(R_{\xi_2}) \le \alpha(R_{\xi_1}), \qquad \beta(R_{\xi_2}) \ge \beta(R_{\xi_1})}.$$

The first inequality uses the smaller rejection event under $H_0$. The second uses its larger complement under $H_1$.

A common design fixes a tolerated false-alarm probability, such as $\alpha_0 = 0.05$, and chooses $\xi$ using the sampling distribution under $H_0$:

$$\alpha(R_\xi) = \Pr_{H_0}(\Lambda(X) > \xi).$$

When attainable, choose the threshold to make this equal to $\alpha_0$. In a discrete model, rejection probabilities can jump as $\xi$ changes, so exact equality need not be possible. A deterministic threshold can instead enforce $\alpha(R_\xi) \leq \alpha_0$.

The corresponding missed-detection probability is

$$\beta(R_\xi) = \Pr_{H_1}(\Lambda(X) \leq \xi).$$

Classical threshold selection uses an error constraint. Bayesian MAP threshold selection uses the prior odds. Both lead to likelihood comparisons, but their thresholds encode different criteria.

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 24, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

# Lecture 25: Classical Inference - III; Course Overview

A hypothesis test rejects a model when a chosen statistic is sufficiently unusual under its null distribution. The same principle applies to means, variances, and distributional fit.

## 25.1 Simple binary tests: a review

A simple hypothesis specifies one complete probability distribution for the data. In a simple binary test, both $H_0$ and $H_1$ are simple. Write $q_j(x)$ for the probability mass function or density of the observed data under $H_j$.

A rejection region $R$ specifies the decision: reject $H_0$ exactly when $X \in R$. A likelihood-ratio test uses

$$\Lambda(x) = \frac{q_1(x)}{q_0(x)}, \qquad R = \{x : \Lambda(x) > \xi\}, \qquad \xi > 0.$$

Large ratios mean that the observation is more plausible under $H_1$ relative to $H_0$.

The false-rejection probability is computed under the null model:

$$\alpha = \Pr_{H_0}(X \in R) = \Pr_{H_0}(\Lambda(X) > \xi).$$

Choose $\xi$ to attain a prescribed $\alpha$, such as 0.05, when possible. With discrete data, exact equality may be unattainable; a test can instead keep this probability at or below the prescribed level.

For independent observations, the joint likelihood is the product of their individual likelihoods. Taking logarithms turns that product into a sum and preserves the comparison, because the logarithm is increasing. Often the resulting test depends on a single summary of the data.

The threshold controls errors across repeated samples from $H_0$. It does not give the probability that $H_0$ is true after observing a particular sample.

## 25.2 Testing a normal mean

Suppose $X_1, \ldots, X_n$ are i.i.d., with

$$H_0 : X_i \sim \mathcal{N}(0, 1), \qquad H_1 : X_i \sim \mathcal{N}(1, 1).$$

The alternative increases the mean, so large positive sums favor $H_1$.

For observed values $x_1, \dots, x_n$, the common normalizing constants cancel from the likelihood ratio:

$$\Lambda(x) = \frac{(2\pi)^{-n/2} \exp\{-\frac{1}{2}\sum_i (x_i - 1)^2\}}{(2\pi)^{-n/2} \exp\{-\frac{1}{2}\sum_i x_i^2\}}$$
$$\log \Lambda(x) = -\frac{1}{2}\sum_i \big((x_i - 1)^2 - x_i^2\big) = \sum_i x_i - \frac{n}{2}.$$

The expansion $(x_i - 1)^2 - x_i^2 = -2x_i + 1$ gives the last equality. Thus $\Lambda(x) > \xi$ is equivalent to

$$\boxed{\sum_i x_i > c}, \qquad c = \log \xi + \frac{n}{2}.$$

The threshold $c$ is on the sum scale; $\xi$ is on the likelihood-ratio scale.

Under $H_0$, $\sum_i X_i \sim \mathcal{N}(0, n)$, so $Z = \sum_i X_i/\sqrt{n}$ is standard normal. Let $\Phi$ be its CDF and let $z_{1-\alpha} = \Phi^{-1}(1 - \alpha)$. The exact level-$\alpha$ test rejects when

$$\boxed{\sum_i X_i > \sqrt{n}\, z_{1-\alpha}}.$$

Its null rejection probability is $\Pr(Z > z_{1-\alpha}) = \alpha$. This upper-tail rule matches the specified positive mean shift.

## 25.3 Testing a normal variance

Now suppose the observations are i.i.d. with

$$H_0 : X_i \sim \mathcal{N}(0, 1), \qquad H_1 : X_i \sim \mathcal{N}(0, 4).$$

The mean stays zero. The second argument is the variance, so the alternative has standard deviation 2. Large squared observations favor this wider distribution.

Set $Q = \sum_i x_i^2$. Each alternative density has normalizing constant $1/(2\sqrt{2\pi})$, giving

$$\Lambda(x) = \frac{(2\sqrt{2\pi})^{-n} \exp\{-Q/8\}}{(\sqrt{2\pi})^{-n} \exp\{-Q/2\}}$$
$$= 2^{-n} \exp\left\{\frac{3}{8}Q\right\},$$
$$\log \Lambda(x) = -n \log 2 + \frac{3}{8}Q.$$

Therefore the likelihood-ratio test rejects when

$$\boxed{\sum_{i=1}^{n} x_i^2 > c}, \qquad c = \frac{8}{3}\big(\log \xi + n \log 2\big).$$

The statistic is a sum of squares about the known mean zero, rather than about an estimated sample mean.

The CHI-SQUARE DISTRIBUTION with $\nu$ degrees of freedom, denoted $\chi^2_\nu$, is the distribution of the sum of squares of $\nu$ independent standard normal variables. Consequently, under $H_0$,

$$Q = \sum_{i=1}^{n} X_i^2 \sim \chi^2_n.$$

Let $G_n$ be its CDF. Choose

$$\boxed{c = G_n^{-1}(1-\alpha)}$$

so that $\Pr_{H_0}(Q > c) = \alpha$. Chi-square tables give the required upper-tail cutoff. On the likelihood-ratio scale, this corresponds to $\xi = 2^{-n} \exp(3c/8)$.

The chi-square law here is exact: it follows directly from the assumed independent normal observations.

## 25.4 Composite hypotheses: is a coin fair?

A COMPOSITE HYPOTHESIS allows more than one probability distribution. For independent coin tosses with head probability $p$, consider $H_0 : p = 1/2$ versus $H_1 : p \neq 1/2$. The null is simple. The alternative is composite because every allowed $p \neq 1/2$ gives a different distribution.

A STATISTIC is a function of the observed data. Use the number of heads $S$ in $n$ tosses. Departures in either direction matter, so choose a rejection region of the form $R = \{s : |s - n/2| > c\}$. The SIGNIFICANCE LEVEL $\alpha$ is the prescribed false-rejection probability, or an upper bound on it. The CRITICAL VALUE $c$ sets the boundary of $R$. Choose it using the distribution of $S$ under $H_0$.

Under the fair-coin model,

$$S \sim \text{Binomial}(n, 1/2), \qquad \mathbb{E}[S] = \frac{n}{2}, \qquad \text{Var}(S) = \frac{n}{4}.$$

For large $n$, the CLT gives

$$\frac{S - n/2}{\sqrt{n/4}} \approx \mathcal{N}(0,1).$$

Splitting the false-rejection probability equally between the two tails gives the approximate cutoff $c \approx z_{1-\alpha/2}\sqrt{n/4}$.

For $n = 1000$ and $\alpha = 0.05$,

$$\sqrt{n/4} = \sqrt{250} \approx 15.81, \qquad c \approx 1.96\sqrt{250} \approx 31.$$

Thus $\Pr_{H_0}(|S - 500| \le 31) \approx 0.95$. The observed $S = 472$ gives $|472 - 500| = 28 < 31$, so the test does not reject fairness at the approximate 5% level. This does not establish that the coin is fair. The normal calibration is approximate, and the count takes integer values.

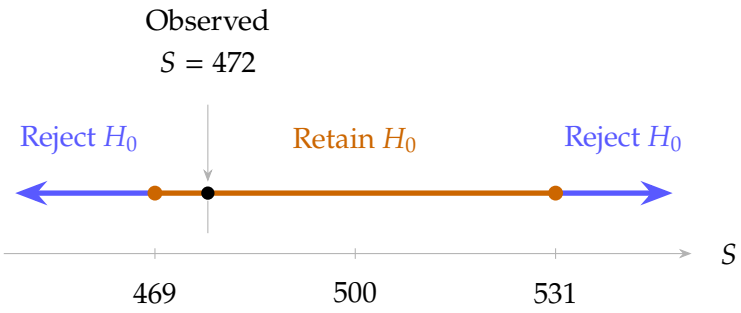



**Figure 82:** With the cutoff $c = 31$, retain $H_0$ for $469 \le S \le 531$ and reject outside this interval. The observed count 472 lies inside. The axis spacing is schematic; counts are discrete.

## 25.5 Is a die fair?

For $n$ independent rolls, the null hypothesis specifies

$$H_0 : \Pr(X = i) = p_i = \frac{1}{6}, \qquad i = 1, \dots, 6.$$

Let $N_i$ count occurrences of face $i$. Then $\sum_i N_i = n$ and the expected count under $H_0$ is $\mathbb{E}_{H_0}[N_i] = np_i = n/6$.

A CHI-SQUARE GOODNESS-OF-FIT TEST measures how far the observed counts are from their expected values:

$$\boxed{T = \sum_{i=1}^{6} \frac{(N_i - np_i)^2}{np_i}}.$$

Each term is nonnegative. The denominator adjusts the squared discrepancy for the expected count in that category. For a fair die this simplifies to

$$T = \frac{6}{n} \sum_{i=1}^{6} \left(N_i - \frac{n}{6}\right)^2.$$

Large values indicate a poor fit, so reject $H_0$ when $T > c$.

For large $n$, under $H_0$,

$$T \approx \chi^2_5.$$

There are five degrees of freedom because the six count deviations sum to zero:

$$\sum_{i=1}^{6}(N_i - np_i) = n - n = 0.$$

Only five deviations can vary freely. The multinomial CLT, followed by the quadratic sum defining $T$, gives the chi-square approximation. It requires sufficiently large expected counts; it is not an exact finite-sample law.

At significance level $\alpha = 0.05$, use the upper-tail cutoff from this null distribution:

$$c = G_5^{-1}(0.95), \qquad \Pr_{H_0}(T > c) \approx 0.05,$$

where $G_5$ is the CDF of $\chi^2_5$. This compares the whole collection of face counts, rather than checking each face separately.

## 25.6 Checking a proposed distribution with bins

To test a proposed density $f_X$, partition its range into disjoint bins $B_1, \ldots, B_k$ that cover the possible observations. Under $H_0$, the probability of bin $i$ is

$$p_i = \Pr_{H_0}(X \in B_i) = \int_{B_i} f_X(x)\,dx.$$

Let $N_i$ be its observed count in an i.i.d. sample of size $n$. The expected count is $np_i$, and

$$\sum_i p_i = 1, \qquad \sum_i N_i = n.$$

These are the same count comparisons used for the die:

$$T = \sum_{i=1}^{k} \frac{(N_i - np_i)^2}{np_i}.$$

Choose bins with positive null probabilities so the displayed denominators are defined. An observation in a bin with null probability zero is incompatible with the proposed model.

For a fully specified null distribution and sufficiently large expected bin counts, the null distribution of $T$ is approximately $\chi^2_{k-1}$. The subtraction of one again comes from the constraint on the total count. Use the corresponding upper-tail critical value to calibrate the test.

The binned test depends on the chosen bins. Comparing CDFs provides an alternative.

## 25.7 Comparing the model and empirical CDFs

For observations $X_1, \dots, X_n$, the EMPIRICAL CUMULATIVE DISTRIBUTION FUNCTION is

$$\widehat{F}_X(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{X_i \le x\}.$$

Here the indicator is 1 when $X_i \le x$ and 0 otherwise. Thus $\widehat{F}_X(x)$ is the fraction of observations at or below $x$. It is a nondecreasing, right-continuous step function, with a jump of $1/n$ at each distinct observation when there are no ties.

The KOLMOGOROV-SMIRNOV STATISTIC measures the largest vertical separation from the proposed CDF:

$$\boxed{D_n = \sup_{x \in \mathbb{R}} |F_X(x) - \widehat{F}_X(x)|}.$$

The supremum includes limiting gaps immediately before an empirical jump as well as gaps at the jump. Both functions are compared at the same value of $x$.

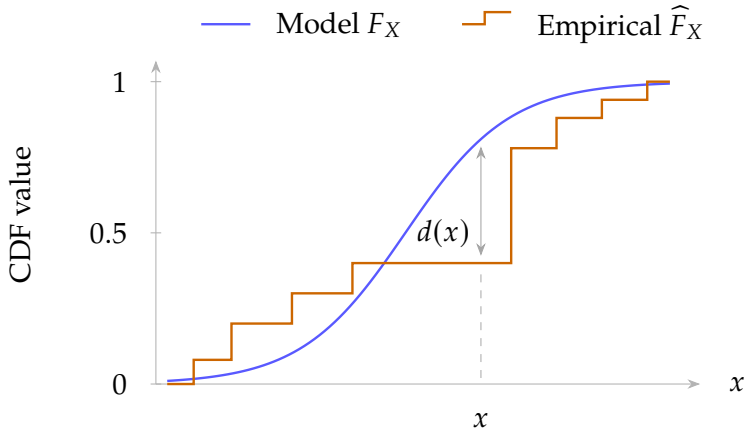



**Figure 83:** Schematic model and empirical CDFs. The marked vertical gap $d(x) = |F_X(x) - \widehat{F}_X(x)|$ compares the two functions at one $x$; $D_n$ takes the supremum of these gaps over all $x$. The curves do not represent a measured sample.

For i.i.d. observations from the fully specified continuous null CDF, the large-sample calibration is

$$\Pr_{H_0}(\sqrt{n}\, D_n \ge 1.36) \approx 0.05.$$

The approximate 5% test rejects when $D_n \ge 1.36/\sqrt{n}$. The number 1.36 is an asymptotic cutoff, rather than an exact finite-sample guarantee.

## 25.8 Further statistical questions

The tests above illustrate a common task: choose a statistic and rejection region, then calibrate the decision using the sampling distribution under a model. Further statistical work addresses several related questions:

- How can rejection regions be chosen systematically?
- How can an unknown density be estimated, for example by constructing a histogram and smoothing it?
- How can signals be processed efficiently as new observations arrive, using recursive calculations?
- How can one choose between simpler and more complex models, including which explanatory variables to use in regression?
- How can inference handle high-dimensional parameter vectors and very large data sets?

**Source.** John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Lecture 25, Fall 2010. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.